

A tall, rectangular stone tower, likely a remnant of a castle or fortress, stands prominently on a rocky mountain peak. The tower is constructed from light-colored stone blocks and has a slightly irregular, weathered appearance. It is surrounded by large, jagged rocks and some sparse vegetation. The sky above is filled with dramatic, white and grey clouds, suggesting a bright but slightly overcast day. The overall scene conveys a sense of historical significance and natural grandeur.

Introduction to Bitcoin

Mohsen Lesani

Original slides adopted from Maurice Herlihy

Abstraction: Distributed Ledger

Cash				
Date	Description	Increase	Decrease	Balance
Jan. 1, 20X3	Balance forward			\$ 50,000
Jan. 2, 20X3	Collected receivable	\$ 10,000		60,000
Jan. 3, 20X3	Cash sale	5,000		65,000
Jan. 5, 20X3	Paid rent		\$ 7,000	58,000
Jan. 7, 20X3	Paid salary		3,000	55,000
Jan. 8, 20X3	Cash sale	4,000		59,000
Jan. 8, 20X3	Paid bills		2,000	57,000
Jan. 10, 20X3	Paid tax		1,000	56,000
Jan. 12, 20X3	Collected receivable	7,000		63,000

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Append-only list of events

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Not just financial

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Everyone agrees on content

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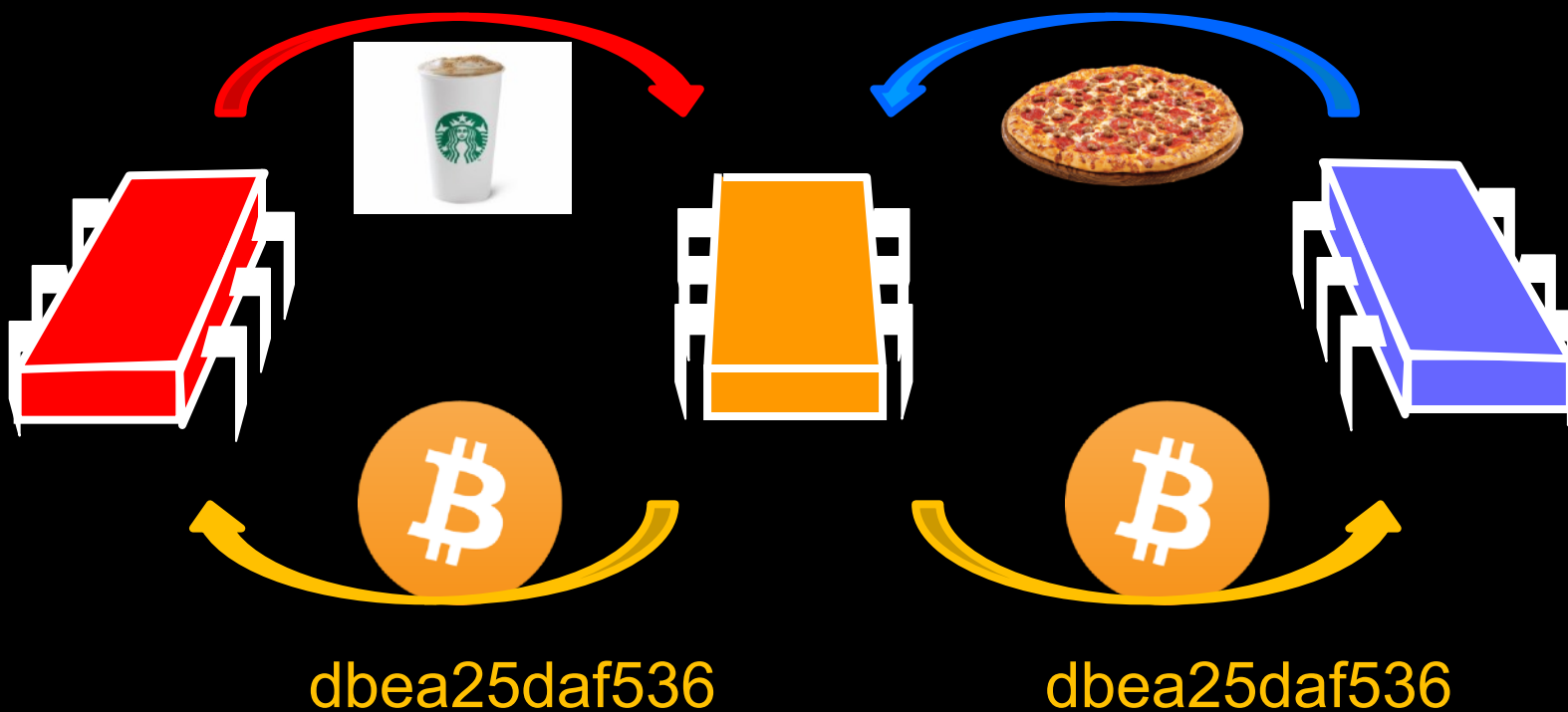
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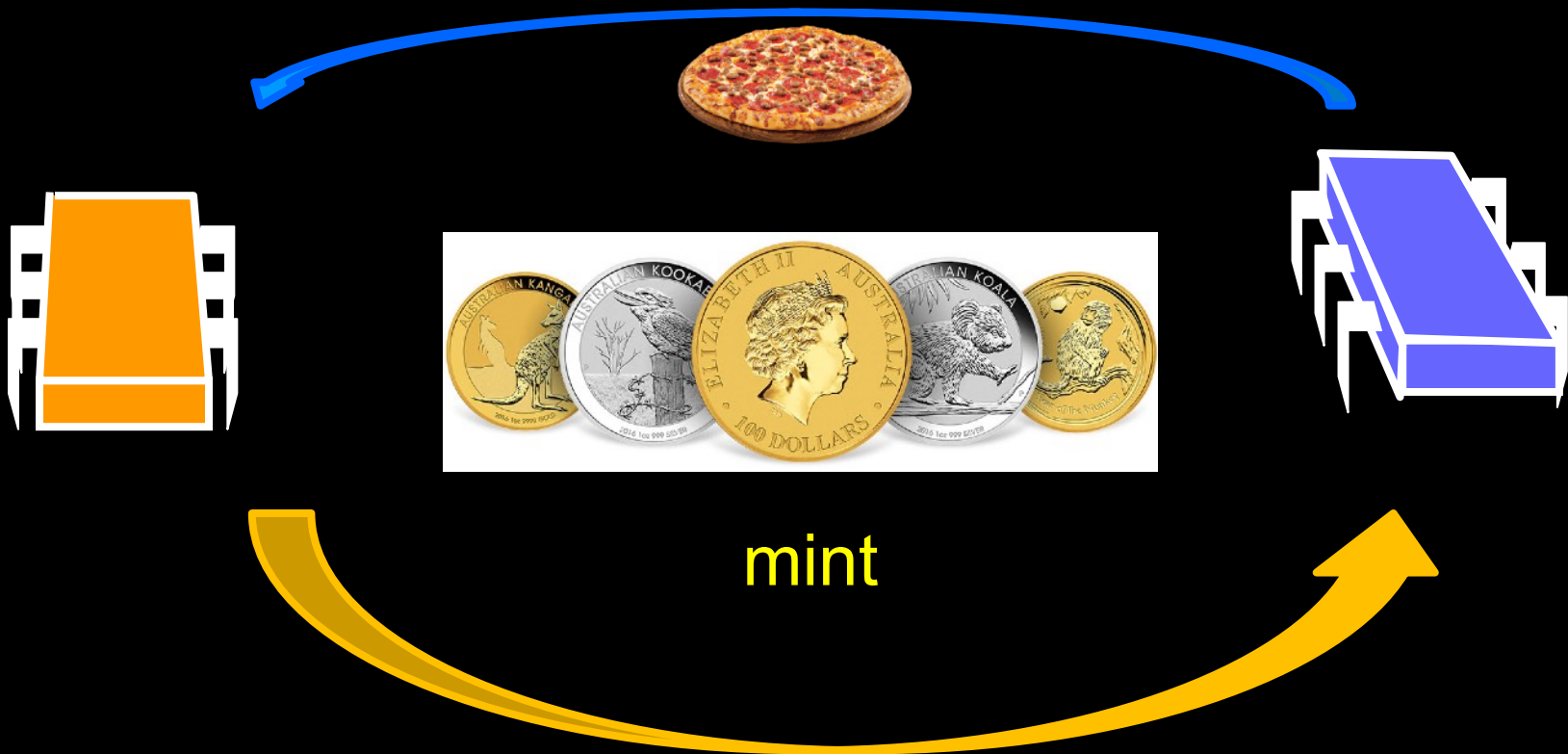
Everyone agrees on content

Tamper-proof!

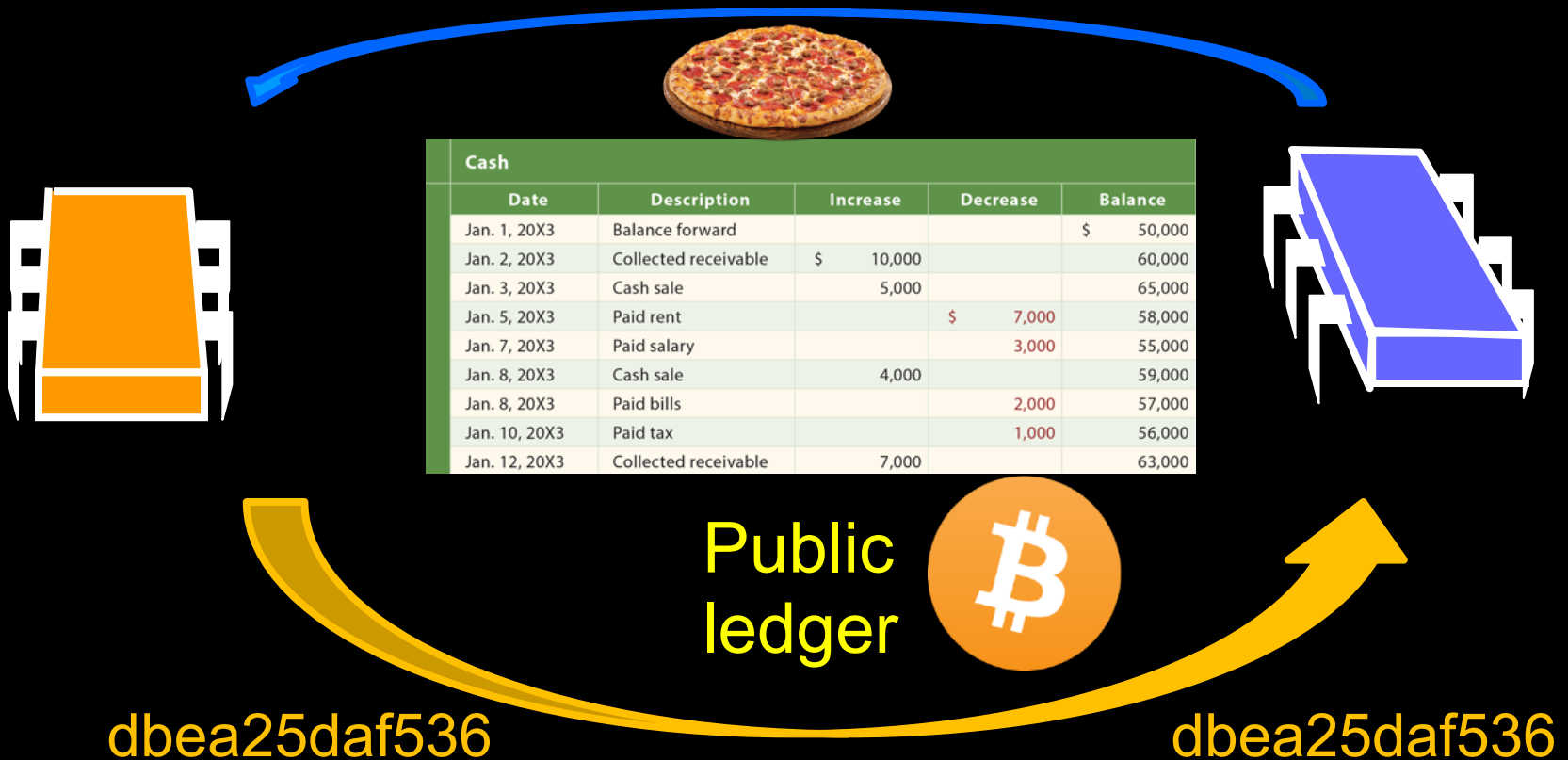
Problem: Double Spending



Old-School Solution

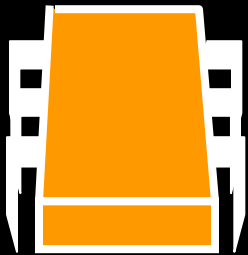


Nakamoto Solution

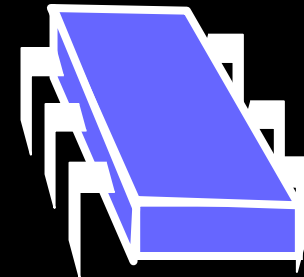


Nakamoto Solution

decided which arrived first. To accomplish this without a trusted party, transactions must be publicly announced [1], and we need a system for participants to agree on a single history of the order in which they were received. The payee needs proof that at the time of each transaction, the



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Public
ledger

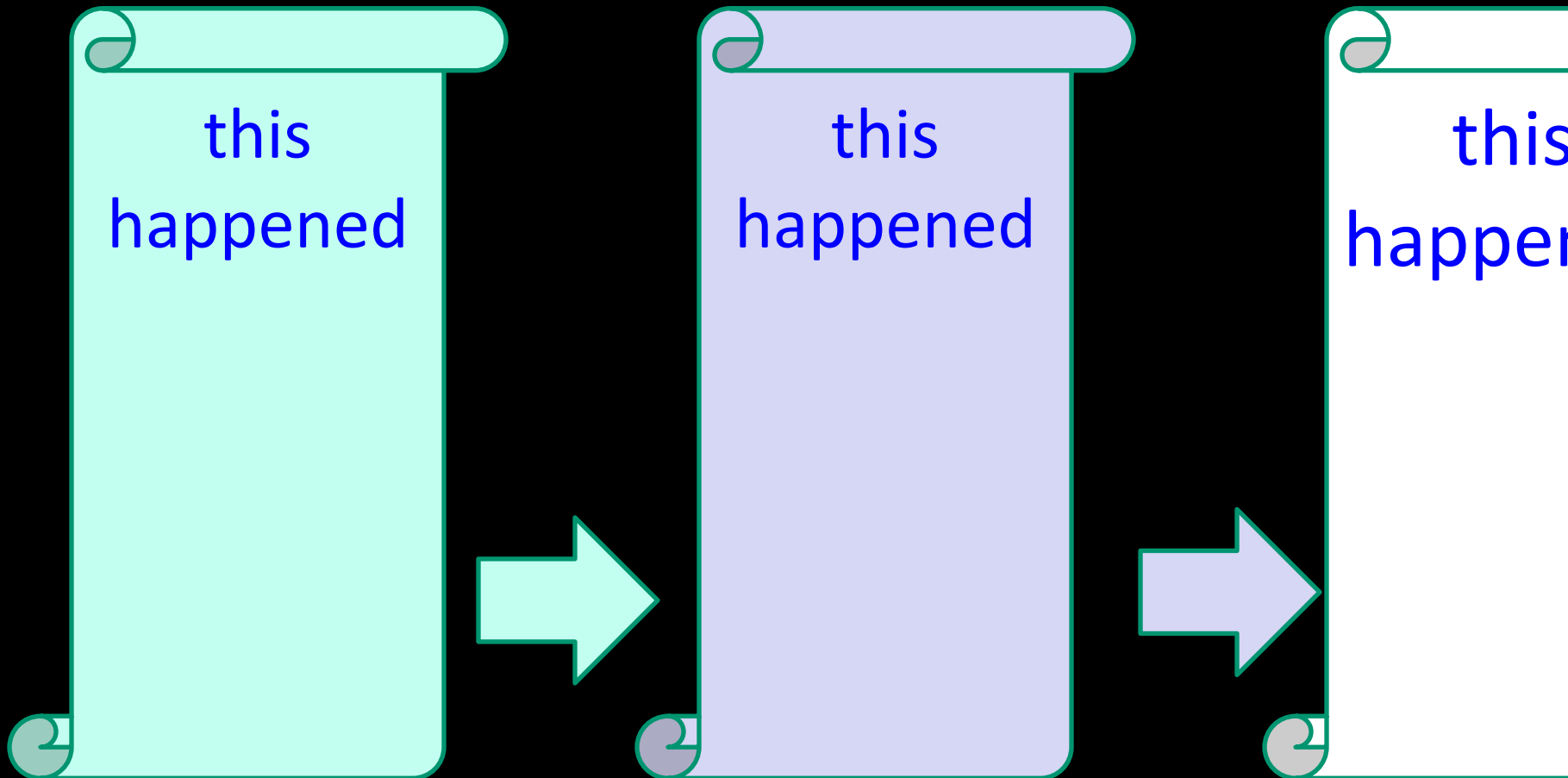


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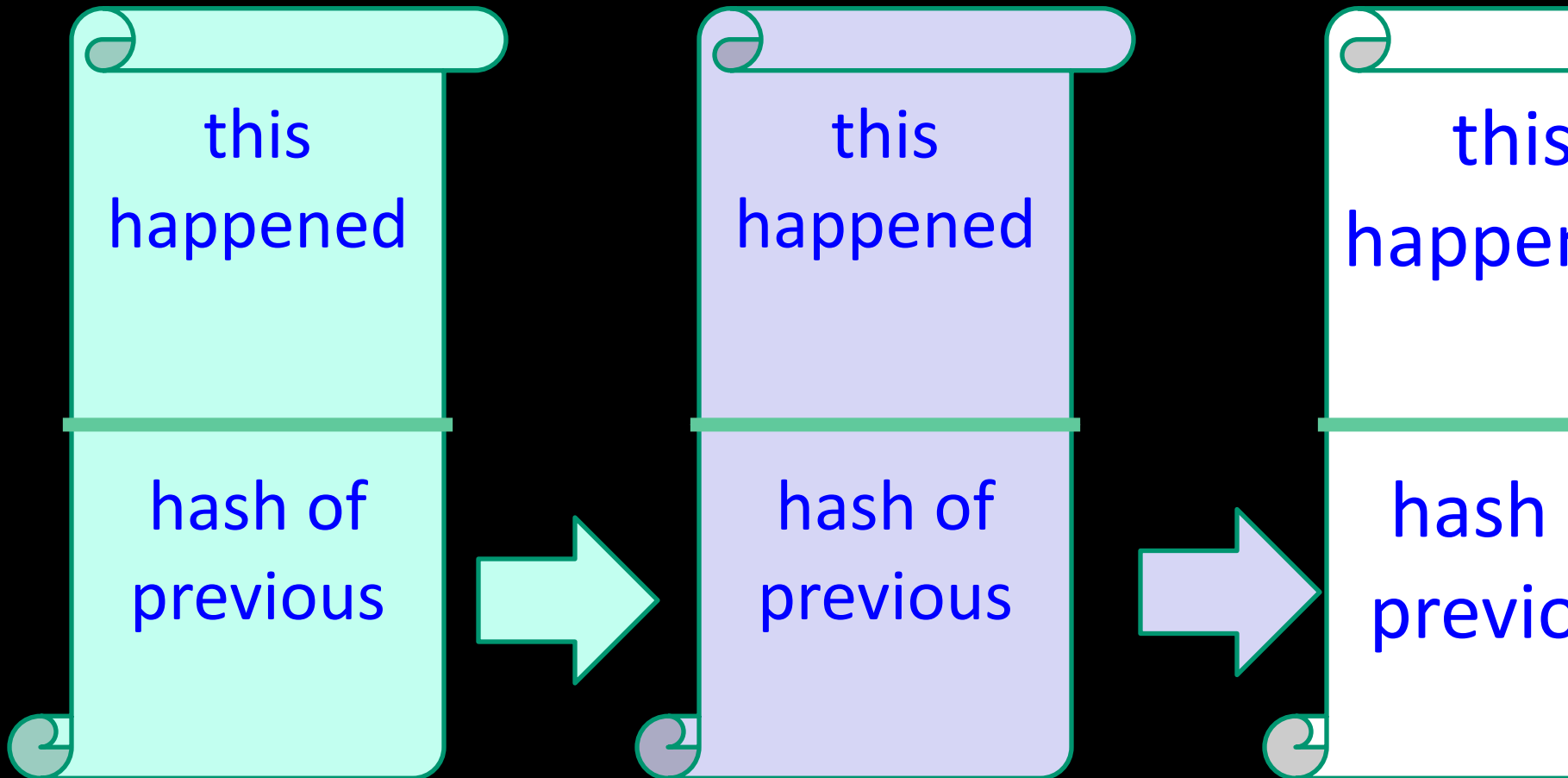
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What is this Blockchain of
which you speak?

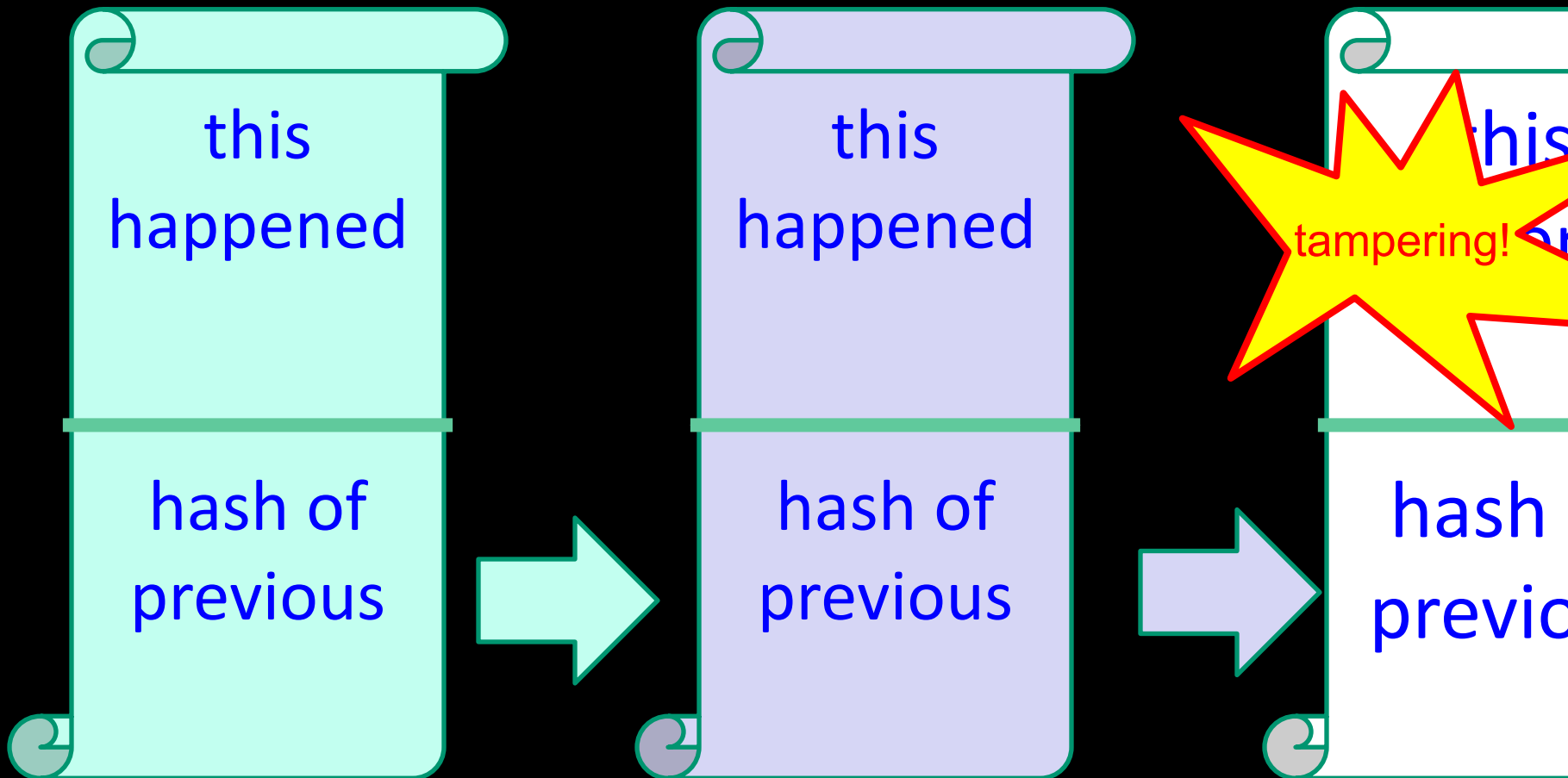
Literally



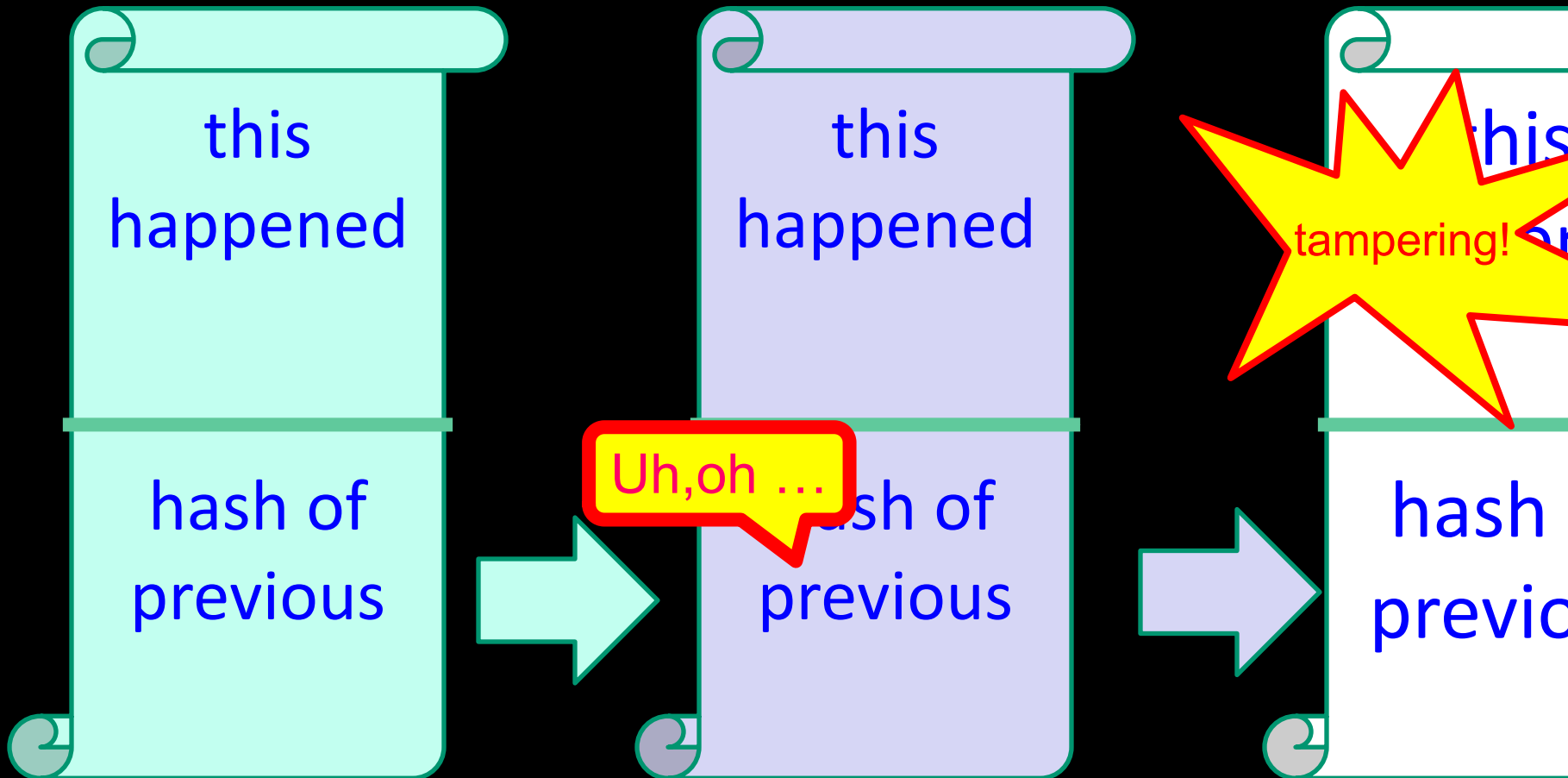
More Literally



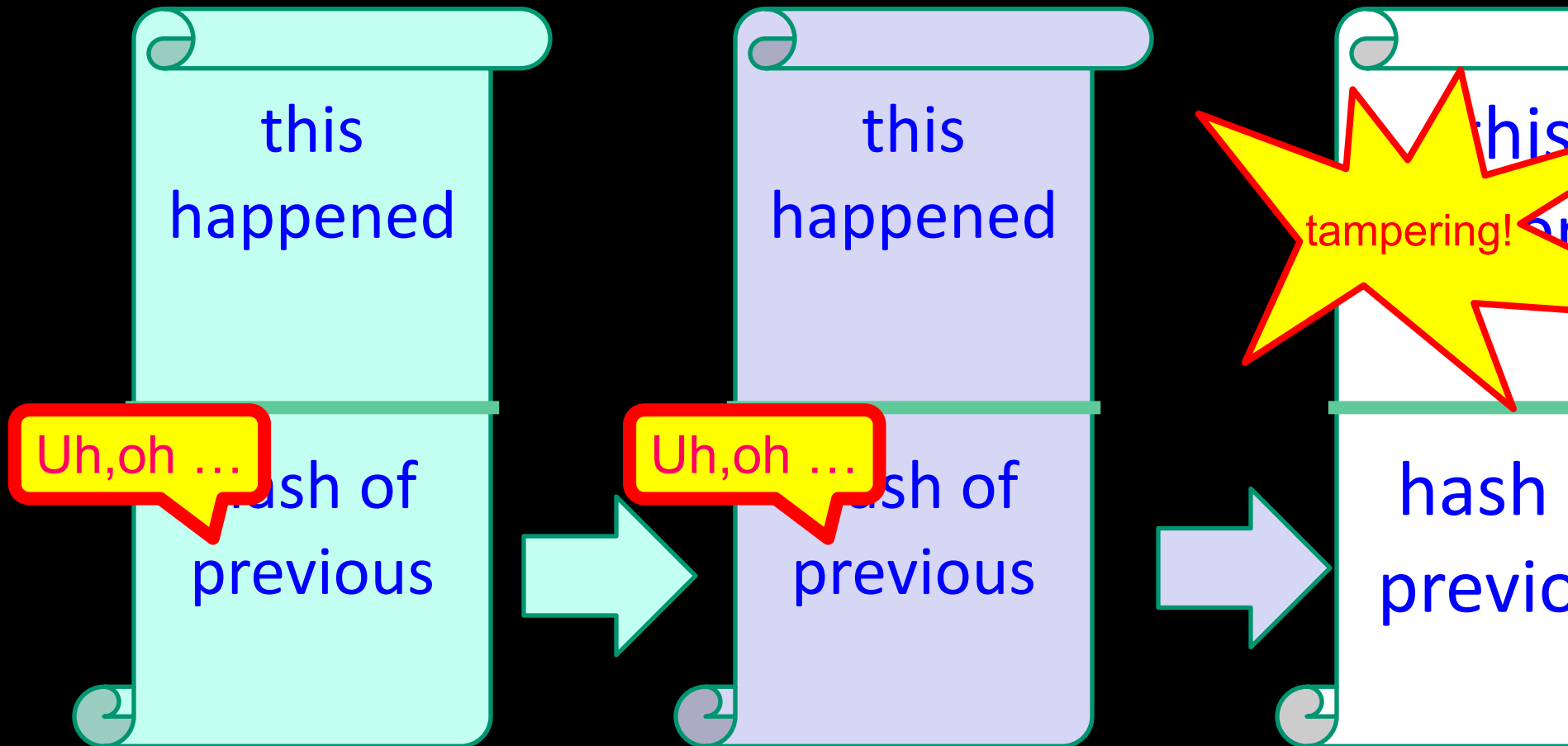
More Literally



More Literally



More Literally



BLOCKCHAINS | By Daniel Oberhaus | Aug 27 2018, 4:19pm

The World's Oldest Blockchain Has Been Hiding in the New York Times Since 1995

This really gives a new meaning to the "paper

How to Time-Stamp a Digital Document*

Stuart Haber
stuart@bellcore.com

W. Scott Stornetta
stornetta@bellcore.com

Bellcore
445 South Street
Morristown, N.J. 07960-1910

Abstract

The prospect of a world in which all text, audio, picture, and video documents are in digital form on easily modifiable media raises the issue of how to certify when a document was created or last changed. The problem is to time-stamp the data, not the medium. We propose computationally practical procedures for digital time-stamping of such documents so that it is infeasible for a user either to back-date or to forward-date his document, even with the collusion of a time-stamping service. Our procedures maintain complete privacy of the documents themselves, and require no record-keeping by the time-stamping service.

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Hash document

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Cryptographic seal from Timestamp & Hash

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Store seals on server ...

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Every day, publish hash of seals to ...

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The New York Times

NOTICES & LOST AND FOUND

(5100-5102)

Universal Registry Entries:
Zone 2 -

dS8492cgVOFAoP9kyE1XzMOrQ
HgEwzkVbVafNylkUz99avq8/ME
p5y9EFSG8XxzMBalGQQ==

Zone 3 -

JnFCg+HCmvhj8GmmUP7VZna71
NgZup/RfuKUQNzCHWXMuqLK
durxHQV5pSHLqBGPRly+mg==

These base64-encoded values represent the combined fingerprints of all digital records notarized by Surety between 2009-06-03Z 2009-06-09Z.

www.surety.com

571-748-5800

The New York Times

NOTICES &
LOST AND
FOUND



Vitalik Non-giver of Ether ✓

@VitalikButerin

Following

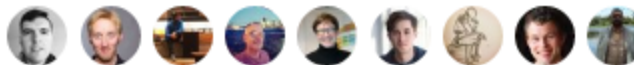


Replying to @ofnumbers @ittaia @ittayeyal

The more realistic attack vector would be to make fake newspapers with a different chain of hashes and circulate them more widely. Still very difficult though :)

4:48 AM - 27 Aug 2018

9 Retweets 55 Likes



8



9



55



The Bitcoin Protocol (simplified)

Bitcoin: A Peer-to-Peer Electronic Cash System

Satoshi Nakamoto
satoshin@gmx.com
www.bitcoin.org

Abstract. A purely peer-to-peer version of electronic cash would allow online payments to be sent directly from one party to another without going through a financial institution. Digital signatures provide part of the solution, but the main benefits are lost if a trusted third party is still required to prevent double-spending. We propose a solution to the double-spending problem using a peer-to-peer network. The network timestamps transactions by hashing them into an ongoing chain of hash-based proof-of-work, forming a record that cannot be changed without redoing the proof-of-work. The longest chain not only serves as proof of the sequence of events witnessed, but proof that it came from the largest pool of CPU power. As long as a majority of CPU power is controlled by nodes that are not cooperating to attack the network, they'll generate the longest chain and outpace attackers. The network itself requires minimal structure. Messages are broadcast on a best effort basis, and nodes can leave and rejoin the network at will, accepting the longest proof-of-work chain as proof of what happened while they were gone.

Introduction

The Internet has come to rely almost exclusively on financial institutions serving as electronic payments. While the system works well enough for the inherent weaknesses of the trust based model. It is not possible, since financial institutions cannot transact without incurring transaction costs, limiting the number of small casual transactions, and thus making it impractical for non-

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Illustrate ideas using the original paper

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Precursors

Pricing via Processing or Combatting Junk Mail*

Cynthia Dwork *

Moni Naor †

Abstract

We present a computational technique for combatting junk mail, in particular, and controlling access to a shared resource, in general. The main idea is to require a user to compute a moderately hard, but not intractable, function in order to gain access to the resource, thus preventing frivolous use. To this end we suggest *pricing functions*, based on, respectively, extracting square roots modulo a prime and the Fiat-Shamir signature scheme, and the One Out of Two Diffie-Hellman scheme.

Precursors

Hashcash - A Denial of Service Counter-Measure

Adam Back

e-mail: adam@cypherspace.org

1st August 2002

Abstract

Hashcash was originally proposed as a mechanism to throttle systematic abuse of un-metered internet resources such as email, and anonymous remailers in May 1997. Five years on, this paper captures in one place the various applications, improvements suggested and related subsequent publications, and describes initial experience from experiments using hashcash.

The *hashcash* CPU cost-function computes a token which can be used as a proof-of-work. Interactive and non-interactive variants of cost-functions can be constructed which can be used in situations where the server can issue a challenge (connection oriented interactive protocol), and where it can not (where the communication is store-and-forward, or packet oriented) respectively.

Key Words: hashcash, cost-functions

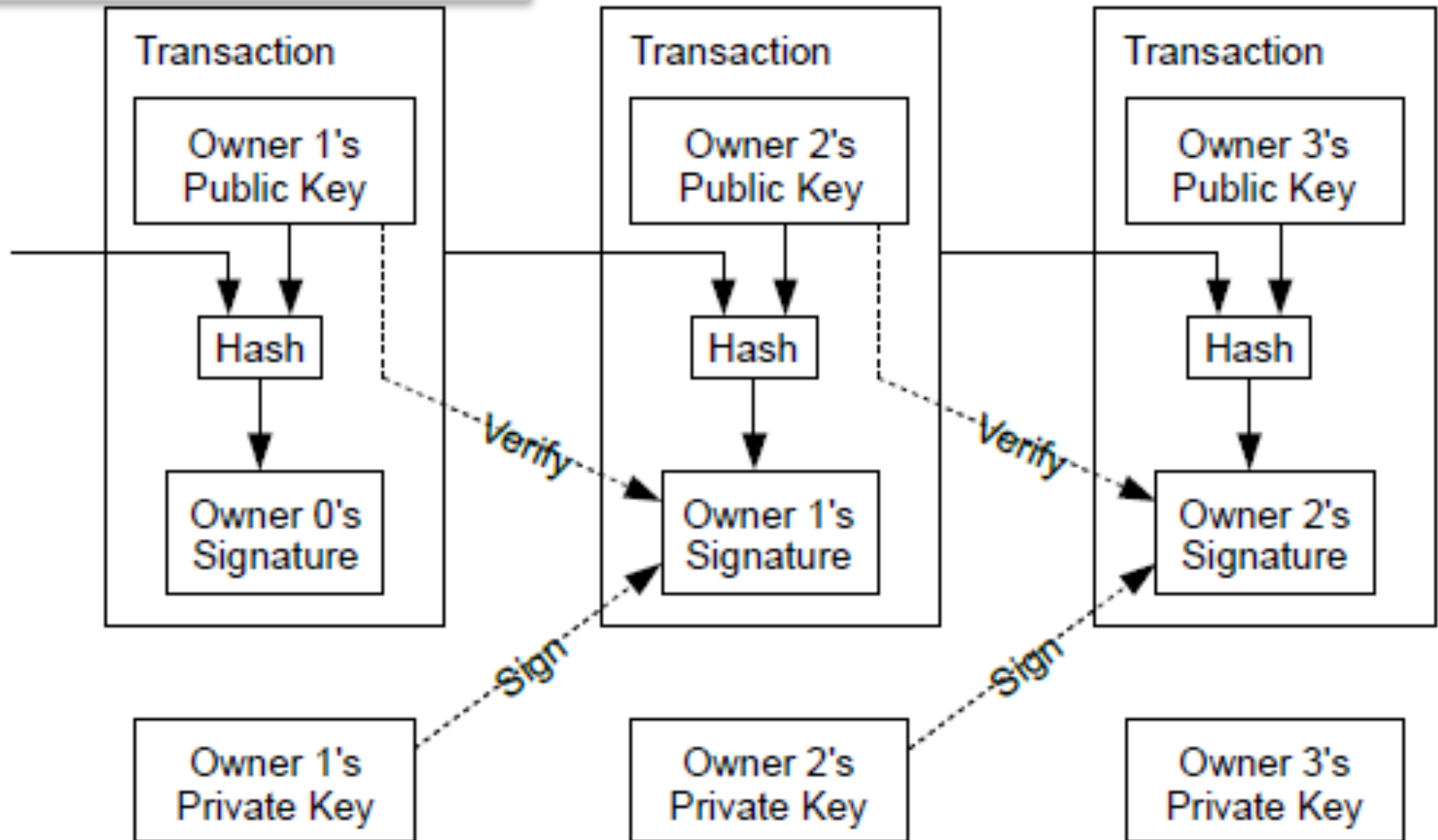
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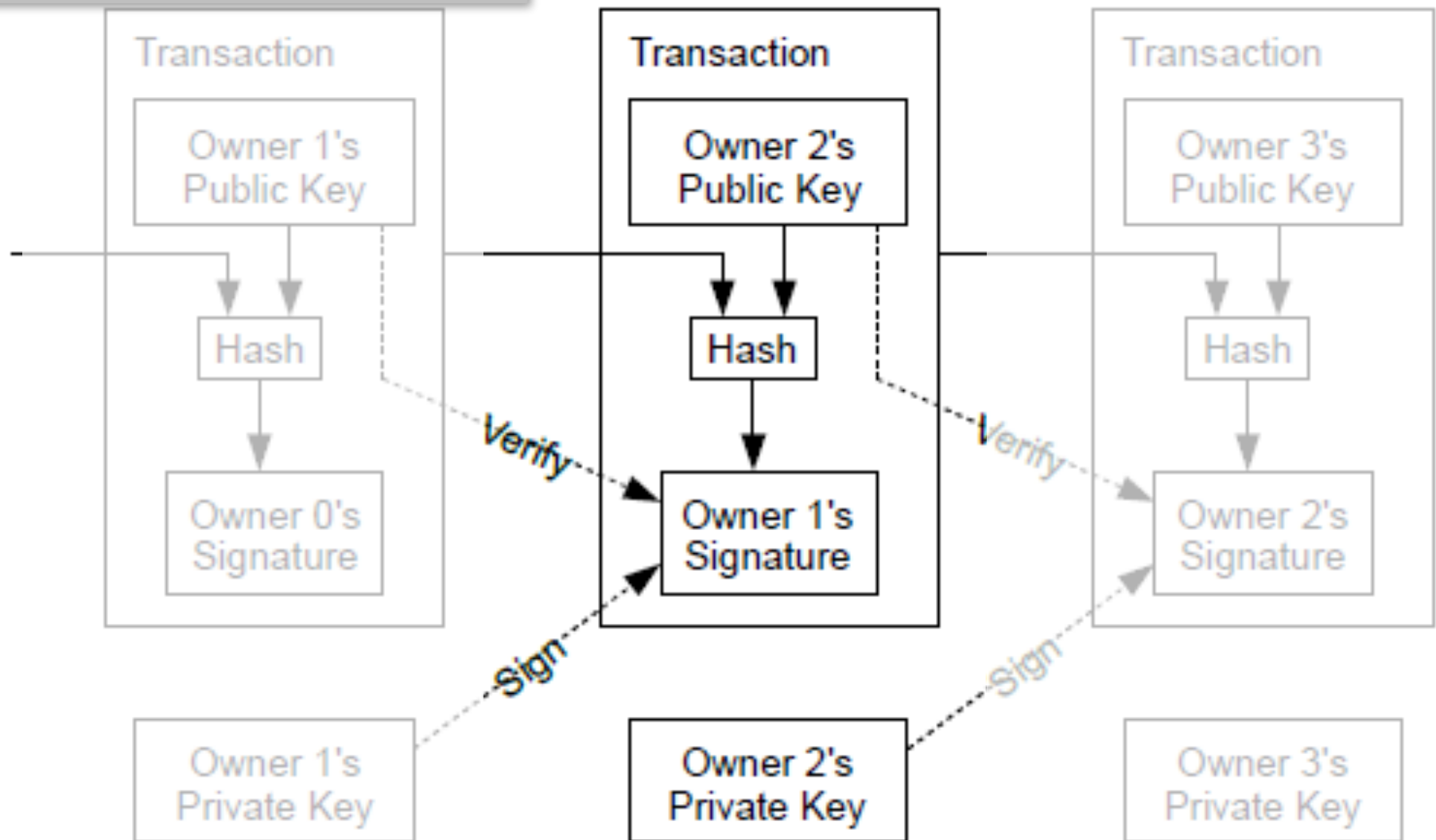
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Naor in [2] w

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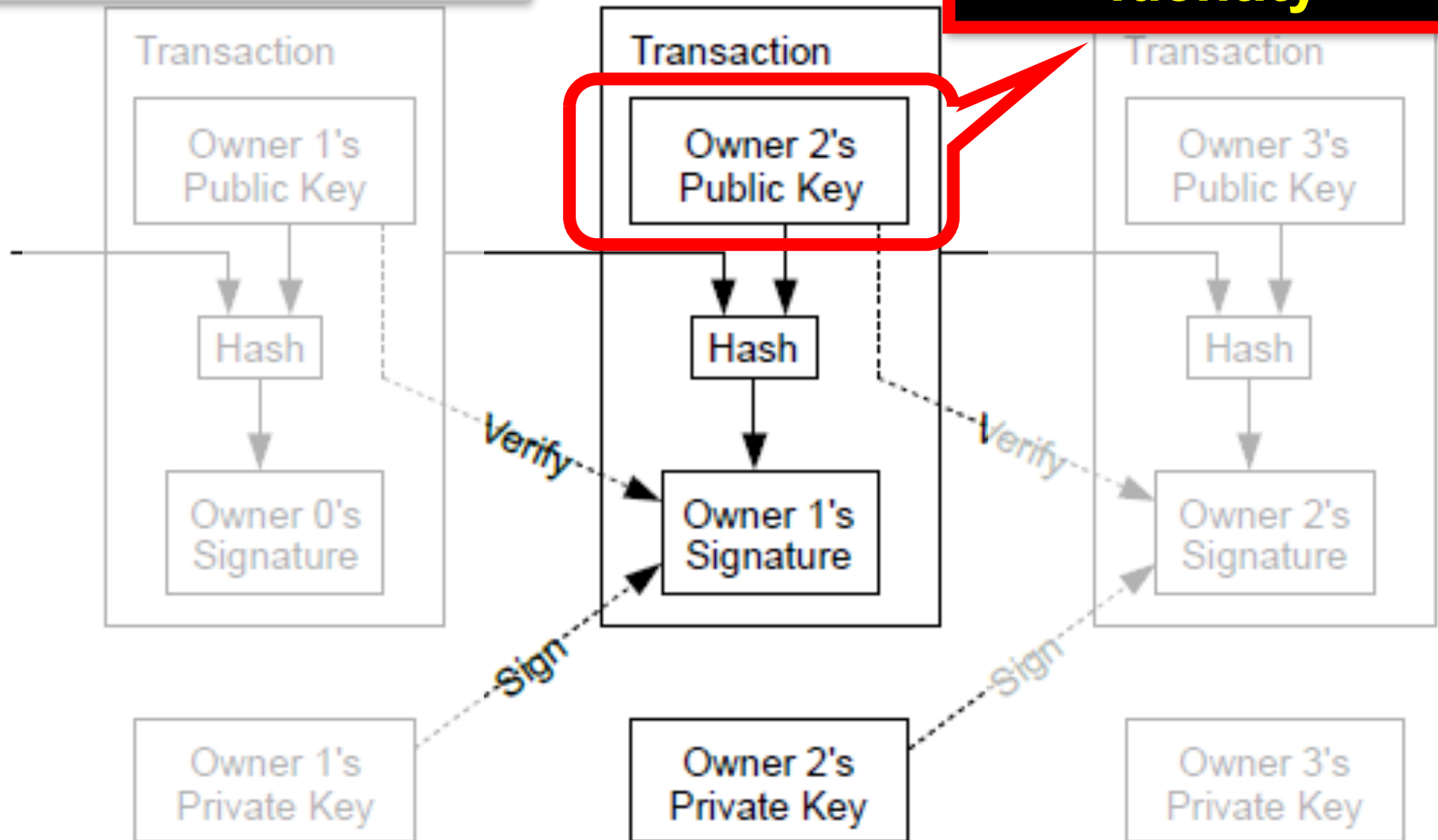


How it works



How it works

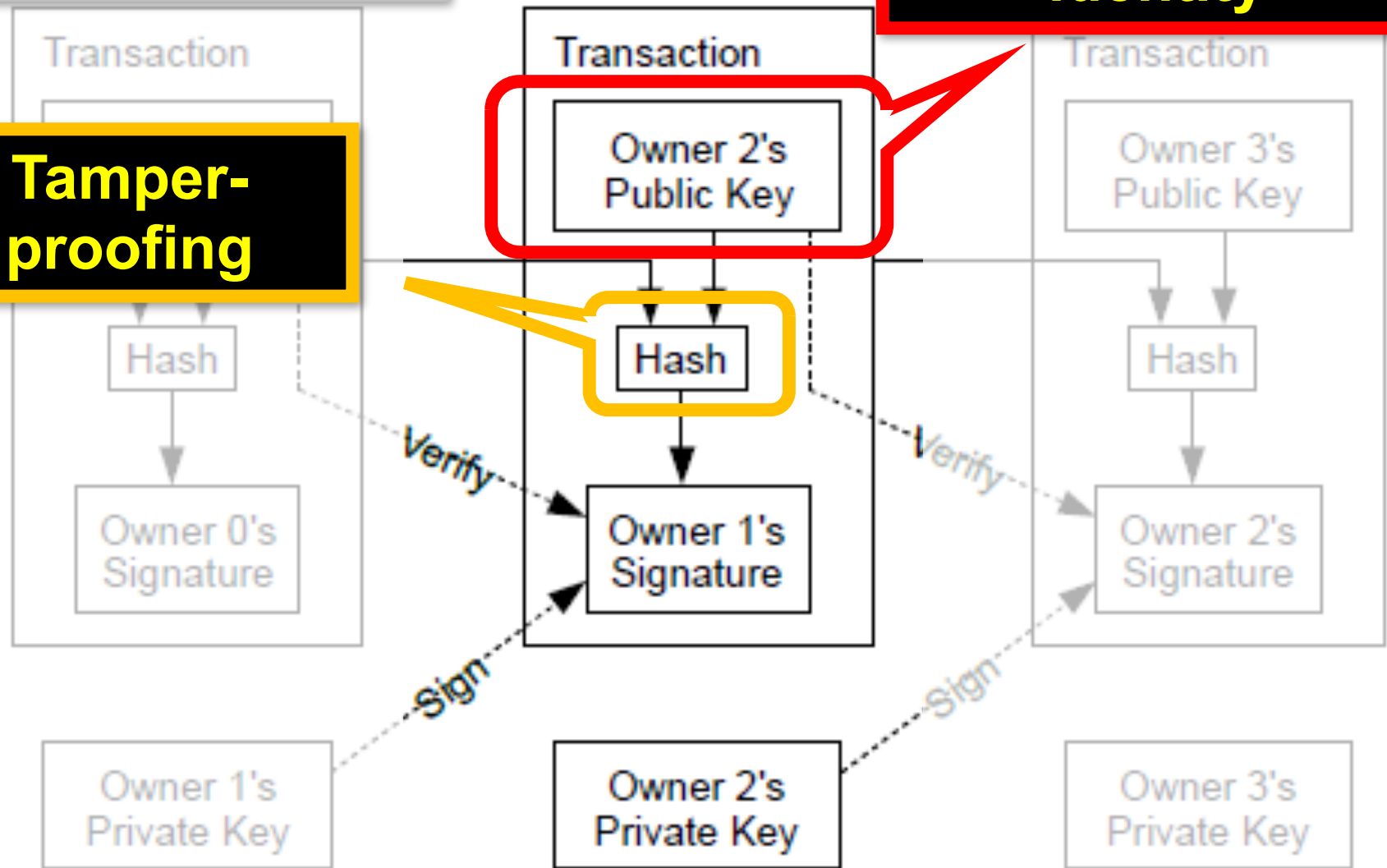
New owner's identity



How it works

Tamper-proofing

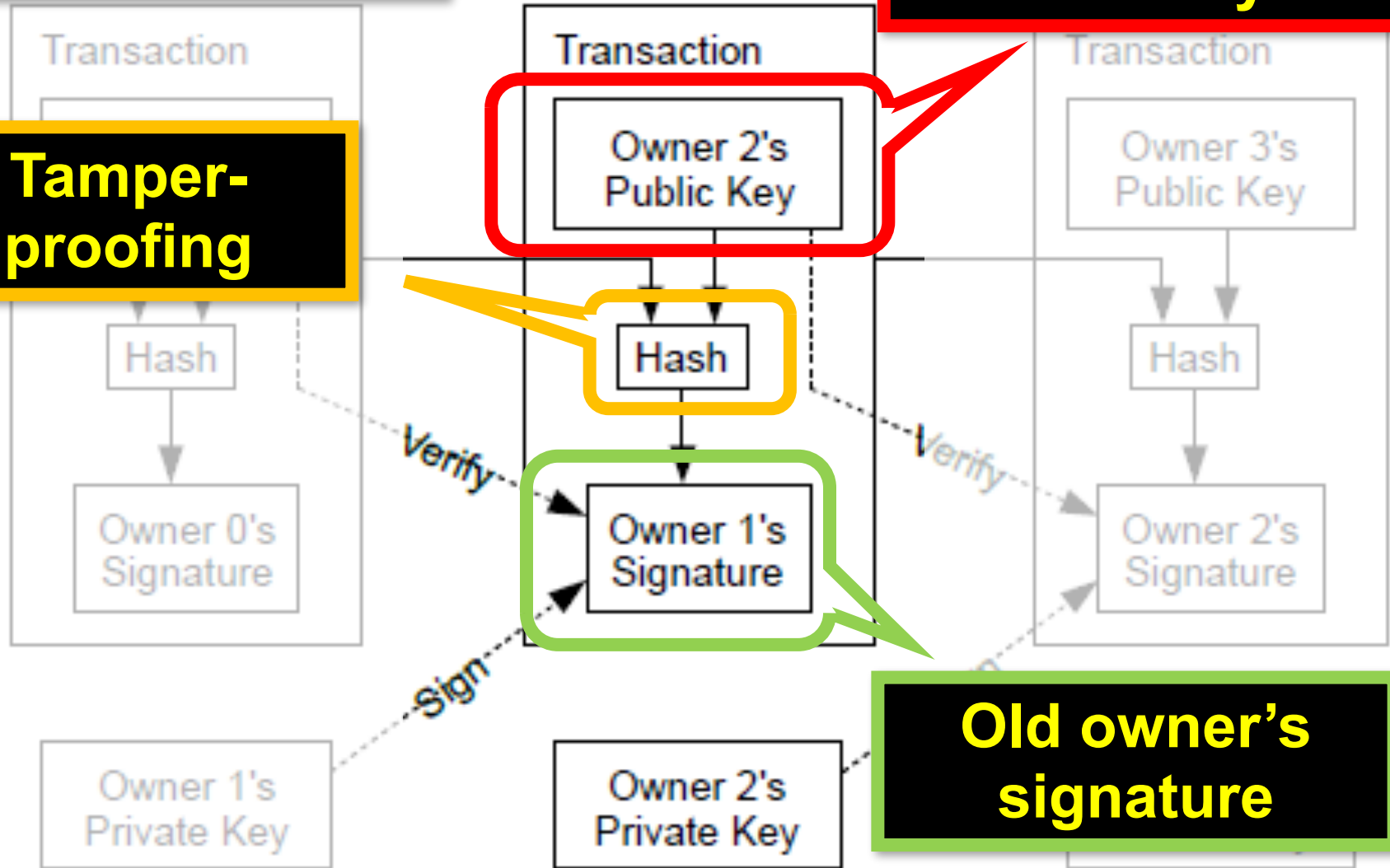
New owner's identity



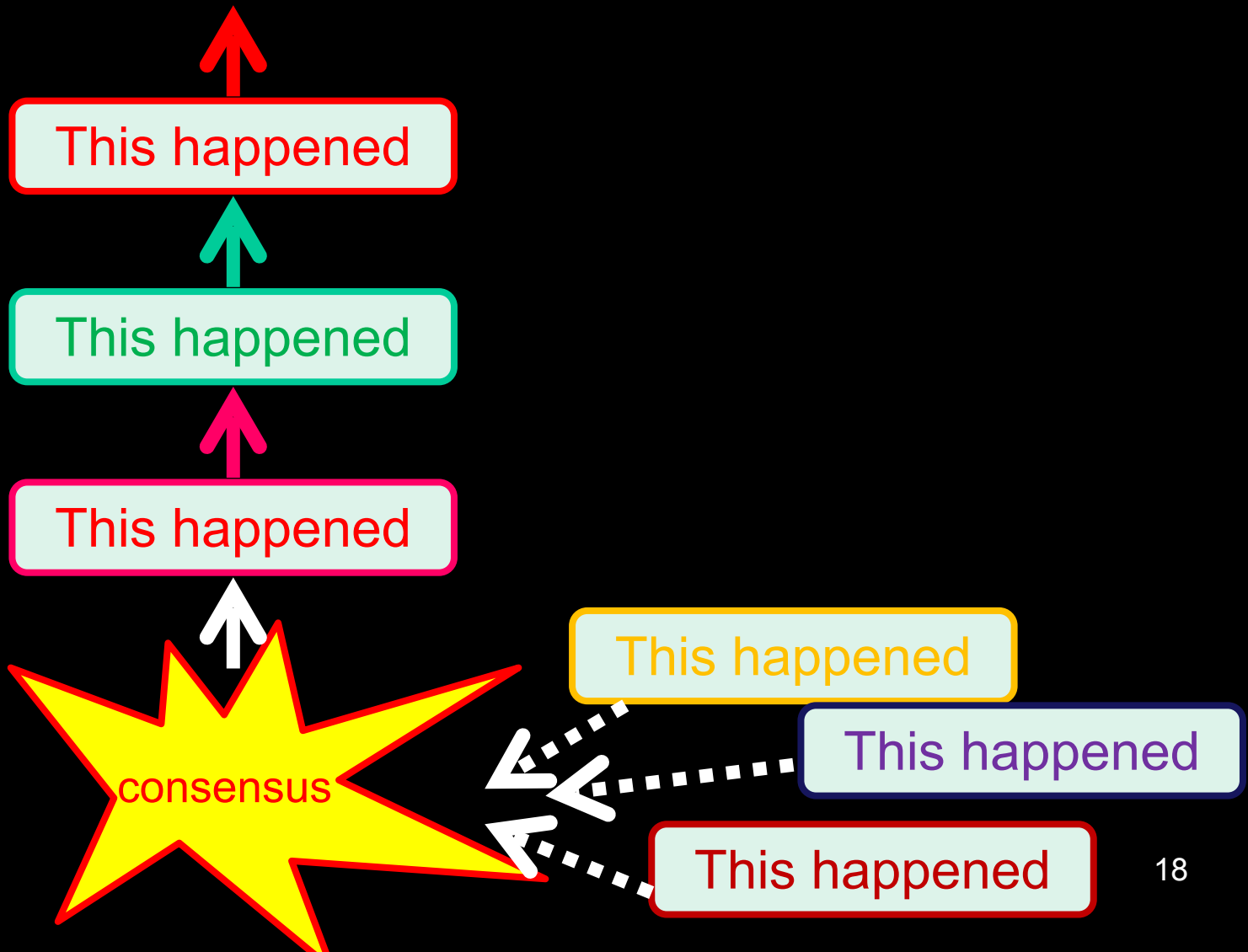
How it works

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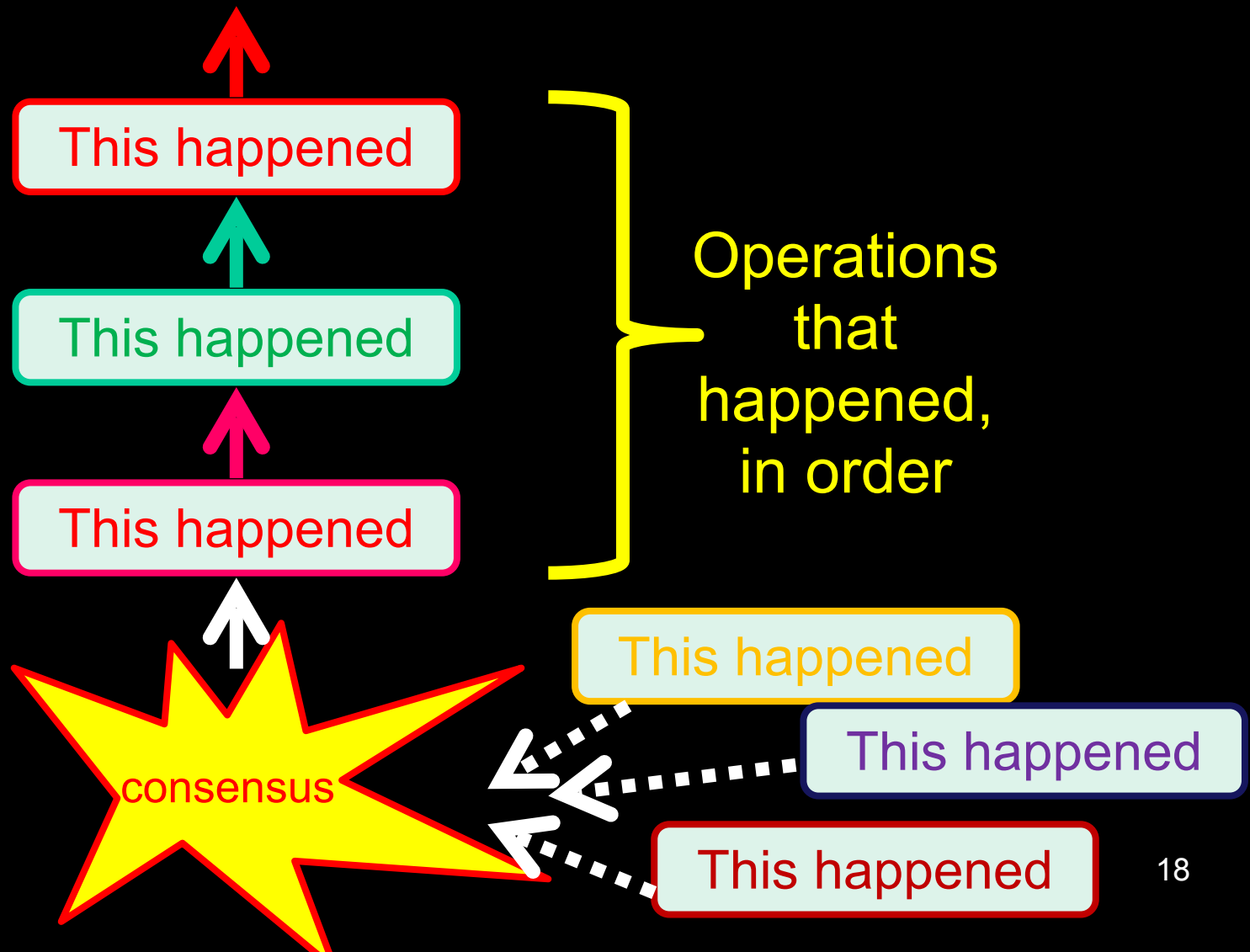
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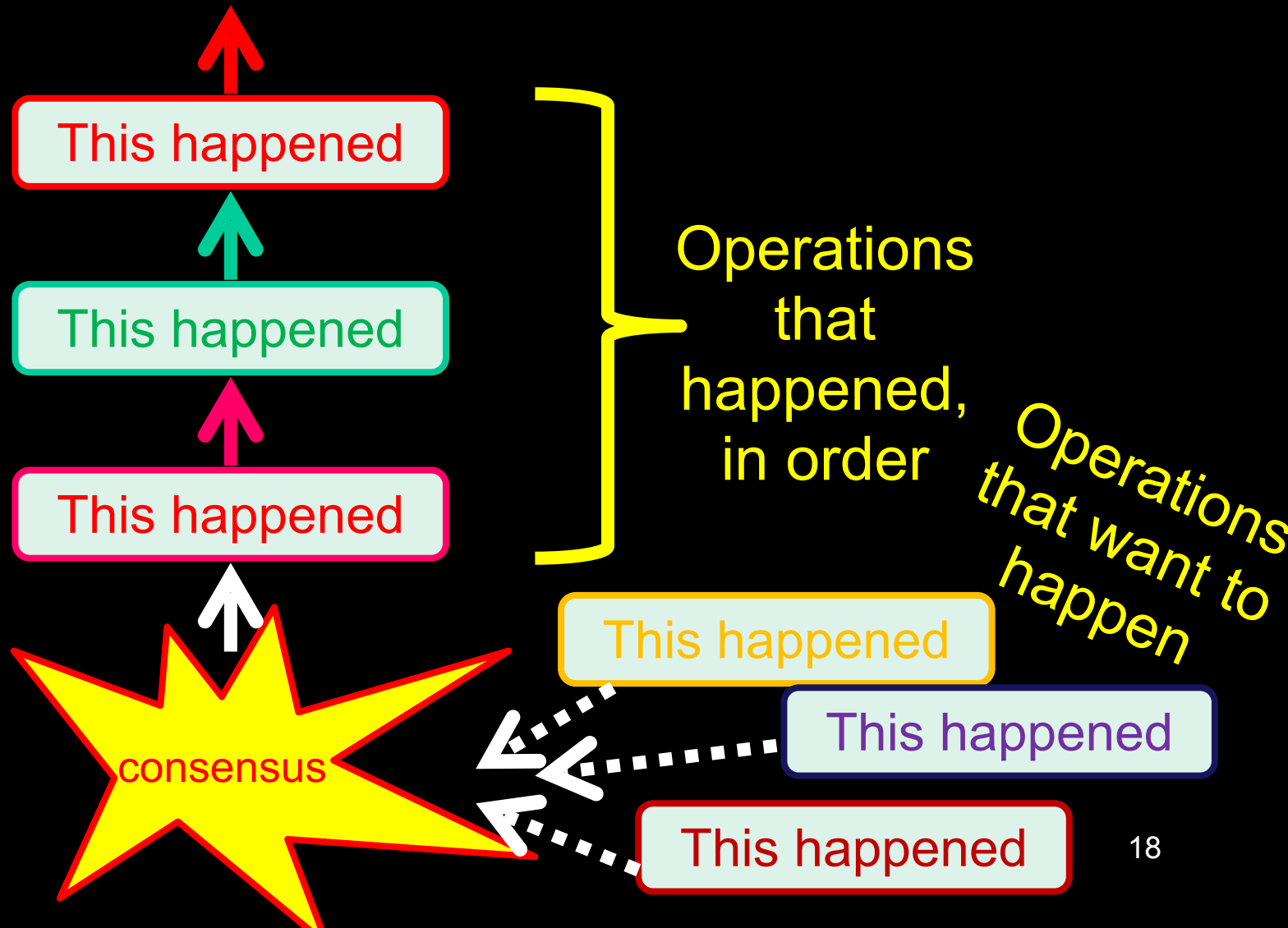
Blockchain Construction (simplified)



Blockchain Construction (simplified)



Blockchain Construction (simplified)



Clients



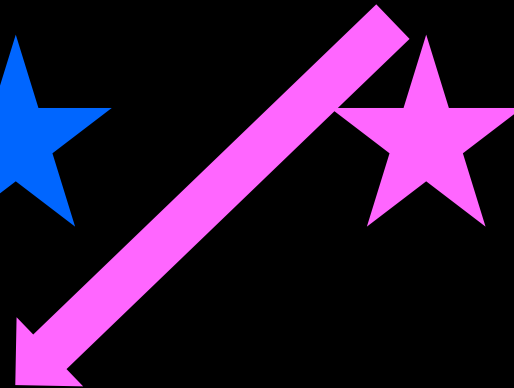
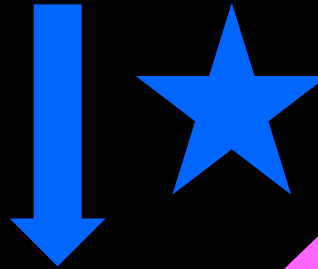
Clients



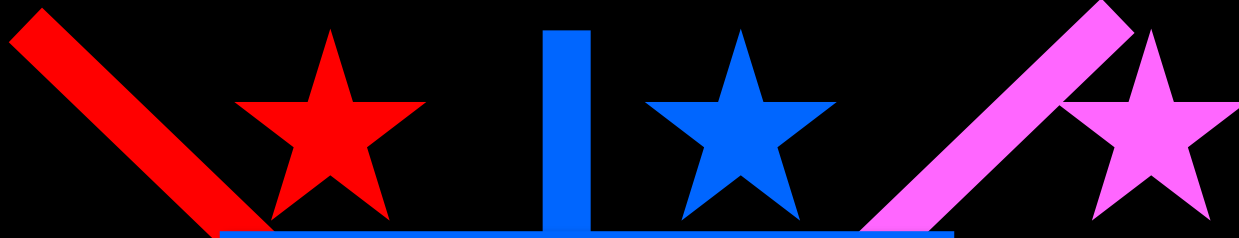
Miners ...



Clients



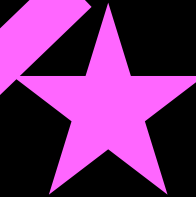
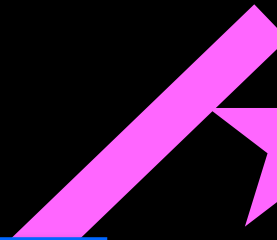
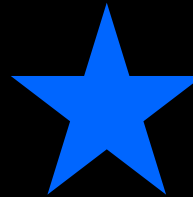
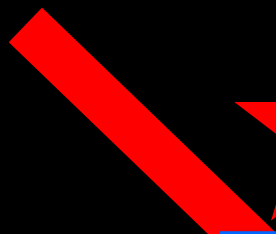
Clients



send transactions ...



Clients

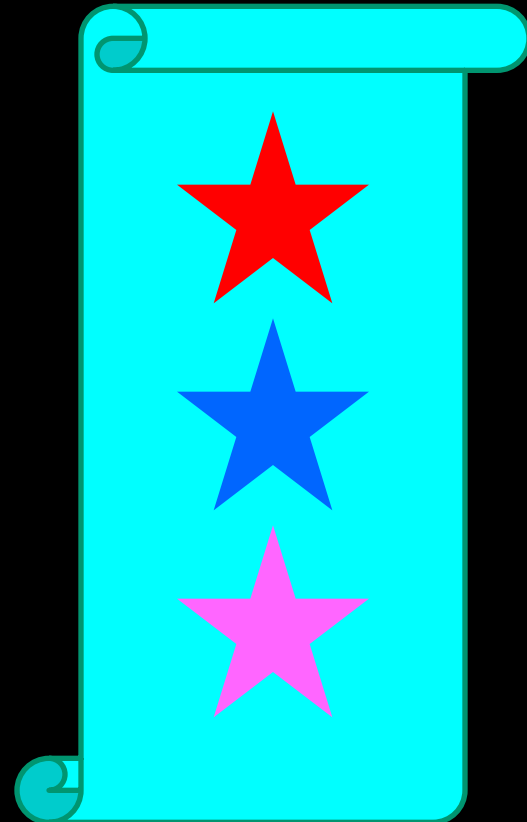


send transactions ...

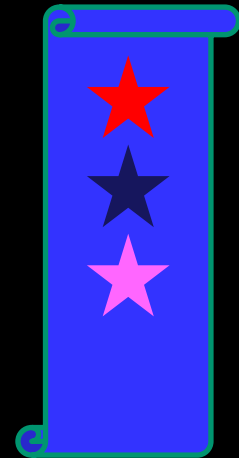
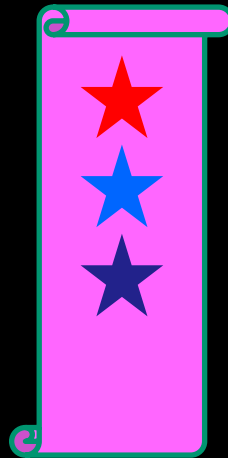
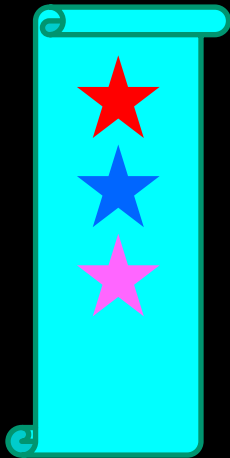
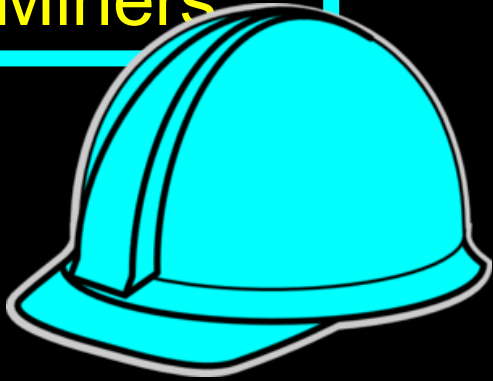


to miners.

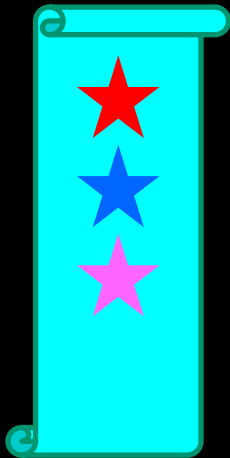
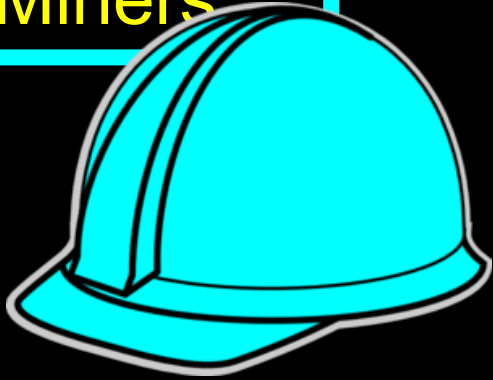
Miner batches transactions in blocks



Miners



Miners



do consensus to pick one block ...

Every node keeps a copy of every transaction

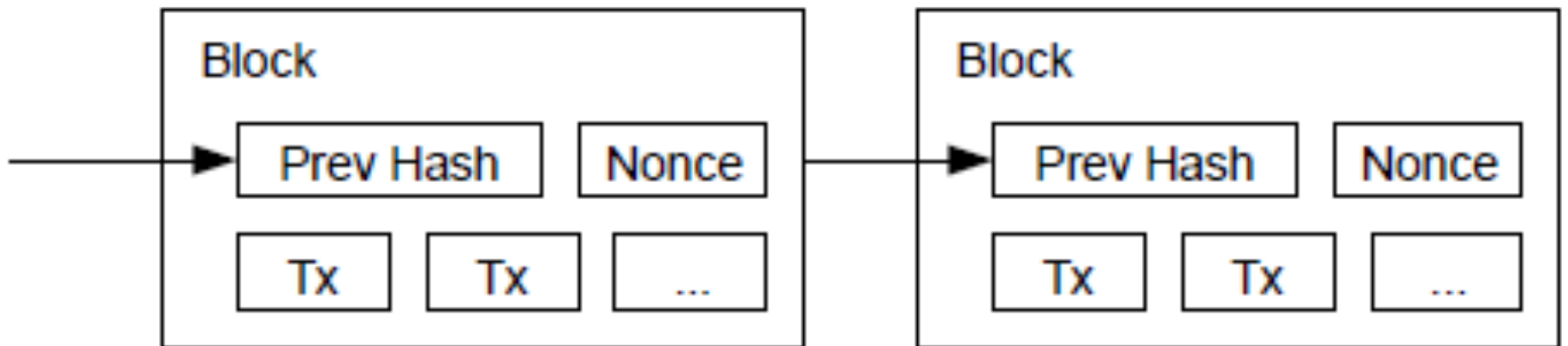
Widely considered reckless at the time

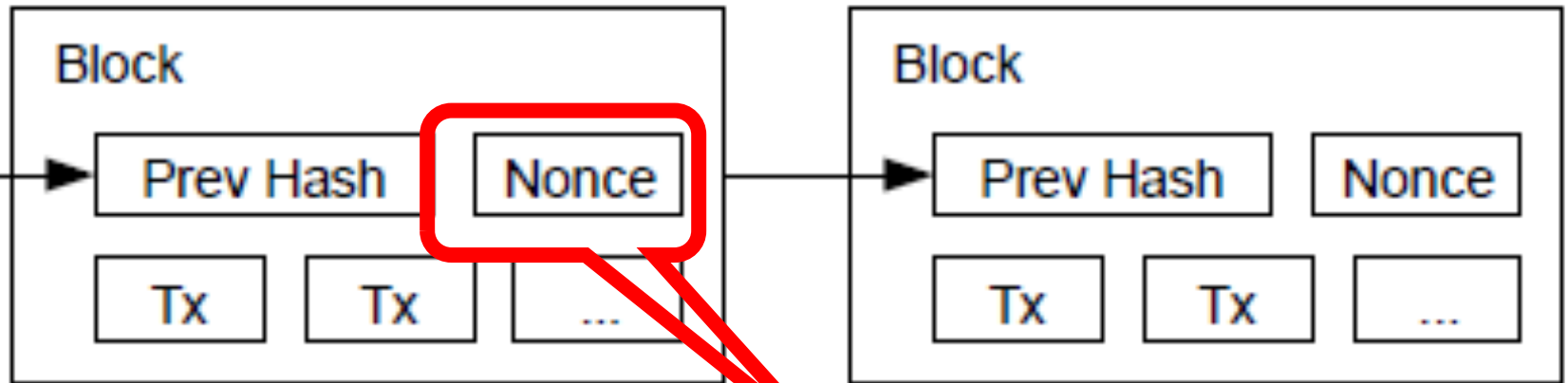
The diagram shows a series of overlapping green rectangles, each representing a node in a distributed ledger system. Each node contains a copy of the entire transaction history, starting from the initial balance and including subsequent transactions. This illustrates how every node maintains a full, identical copy of the ledger.

Date	Description	Increase	Decrease	Balance
Jan. 1, 20X3	Balance forward			\$ 50,000
Jan. 3, 20X3	Received from customer	10,000		60,000
Jan. 5, 20X3	Paid for office supplies		5,000	65,000
Jan. 7, 20X3	Paid salary		3,000	58,000
Jan. 9, 20X3	Received from customer	1,000		59,000
Jan. 10, 20X3	Paid tax		2,000	57,000
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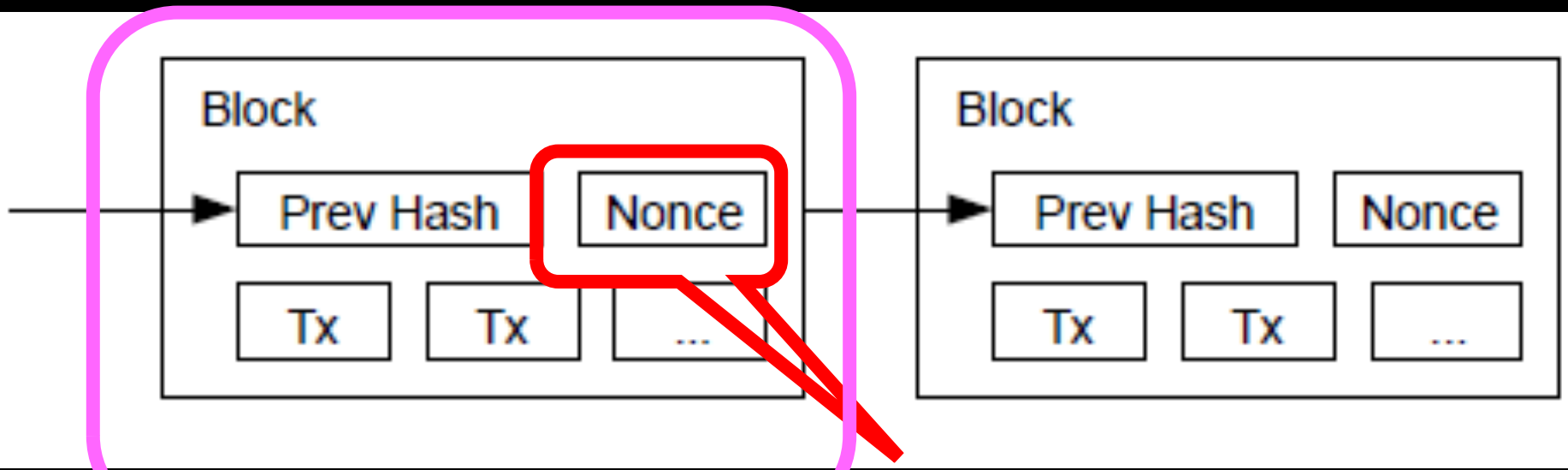
Every node keeps a copy of every transaction

Widely considered reckless at the time





**Find a value to
put here ...**



**Find a value to
put here ...**

**To give this hash
k leading 0s**

Nakamoto Consensus in One Line

SHA256(h | T | K | nonce) < D

Nakamoto Consensus in One Line

SHA256(h | T | K | nonce) < D

standard
hash
function

Nakamoto Consensus

hash of block
on longest
branch

$$\text{SHA256}(h \mid T \mid K \mid \text{nonce}) < D$$

standard
hash
function

Nakamoto Consensus

hash of block
on longest
branch

$$\text{SHA256}(h \parallel T \parallel K \parallel \text{nonce}) < D$$

standard
hash
function

transactions

Nakamoto Consensus

hash of block
on longest
branch

public
key

$$\text{SHA256}(h | T | K | \text{nonce}) < D$$

standard
hash
function

transactions

Nakamoto Consensus

hash of block in One Line

on longest
branch

public
key

difficulty

The diagram shows the hash formula: $\text{SHA256}(h | T | K | \text{nonce}) < D$. Each component is enclosed in a red box, and red lines point from descriptive labels to these boxes. The labels are: 'hash of block on longest branch' pointing to 'h', 'public key' pointing to 'K', 'difficulty' pointing to 'D', 'standard hash function' pointing to 'SHA256', and 'transactions' pointing to 'T'.

$$\text{SHA256}(h | T | K | \text{nonce}) < D$$

standard
hash
function

transactions

Nakamoto Consensus

hash of block in One Line

on longest
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SHA256(h T K nonce) < D

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function

transactions

Find this!

Nakamoto Consensus in One Line

SHA256(h | T | K | nonce) < D

**No formal proof of collision-resistance
Possible quantum attacks?
Ethereum uses similar, not same hash**

Nakamoto Consensus in One Line

$$\text{SHA256}(\text{h} \mid \text{T} \mid \text{K} \mid \text{nonce}) < \text{D}$$


Tamper-proofing

Nakamoto Consensus in One Line

$$\text{SHA256}(h \mid \boxed{T} \mid K \mid \text{nonce}) < D$$

Actually hash of txn “Merkel tree” root
Constant size
Too expensive to hash txns themselves

Nakamoto Consensus in One Line

$$\text{SHA256}(h \mid T \mid \text{K} \mid \text{nonce}) < D$$


Pay “coinbase” reward to this address

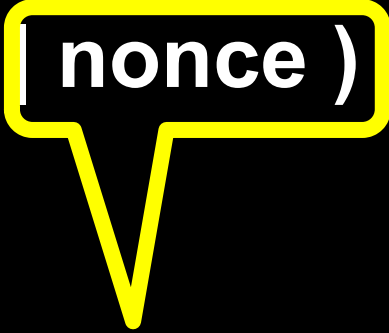
Nakamoto Consensus in One Line

$$\text{SHA256}(h \mid T \mid K \mid \text{nonce}) < \text{D}$$


Too easy? frequent forks, finality slow
Too hard? slow progress, low throughput
Adjusted dynamically: ~1 block / 10 min

Nakamoto Consensus in One Line

$\text{SHA256}(h \mid T \mid K \mid \text{nonce}) < D$



Keep trying one after another ...

Chain Property



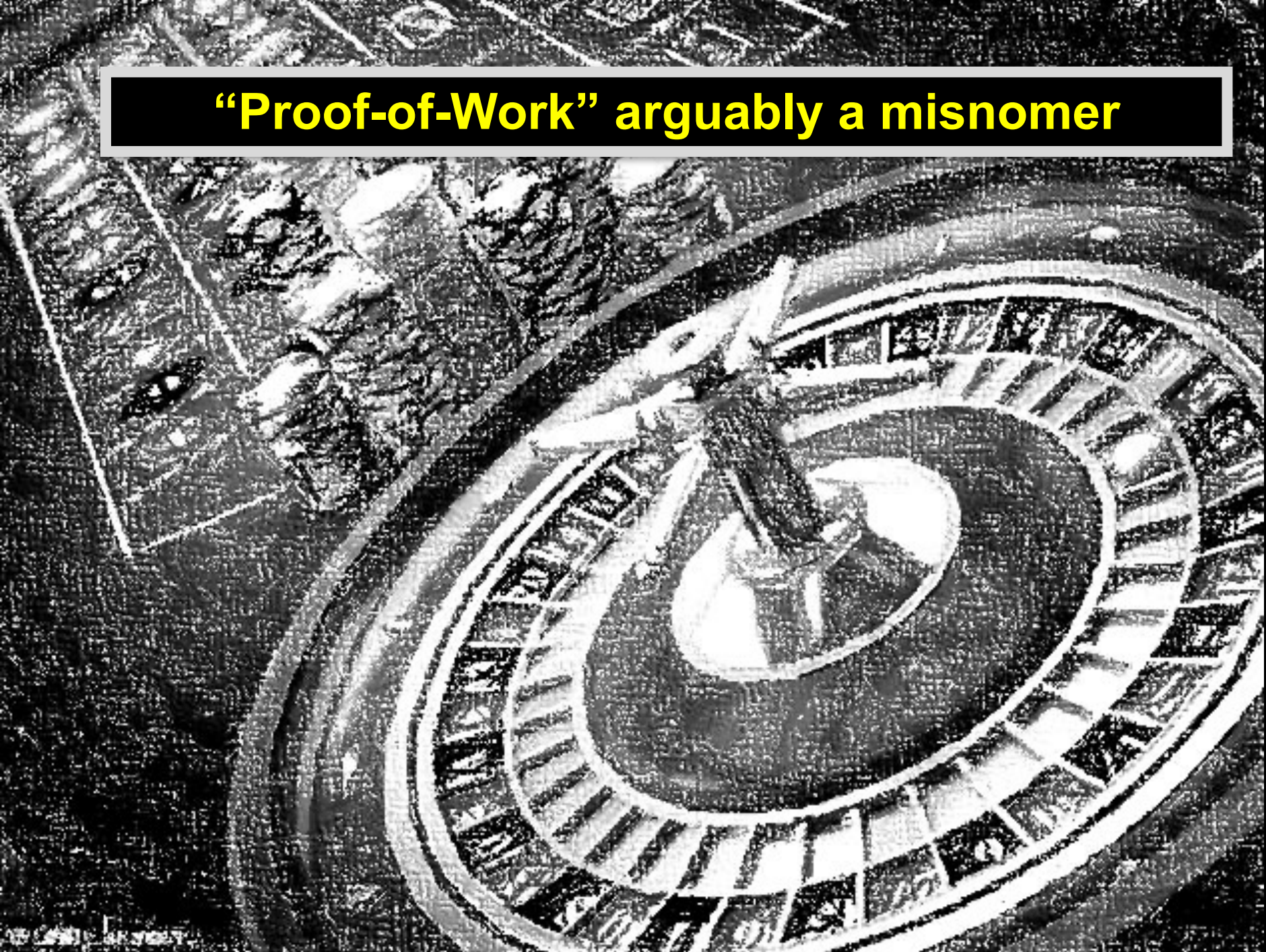
SAFETY

Chain Property

Control x% of miners, control x% of blocks

SAFETY

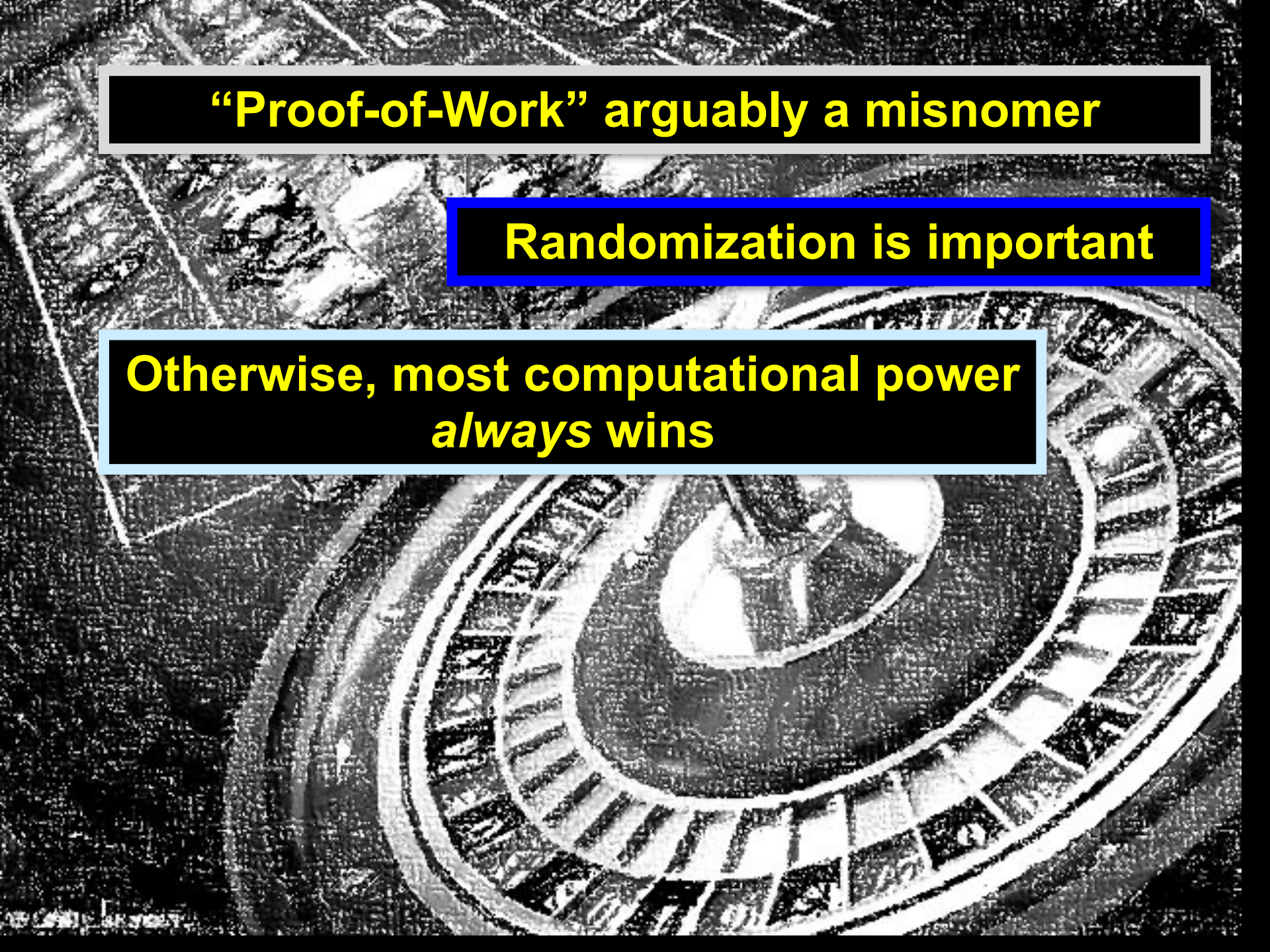
“Proof-of-Work” arguably a misnomer





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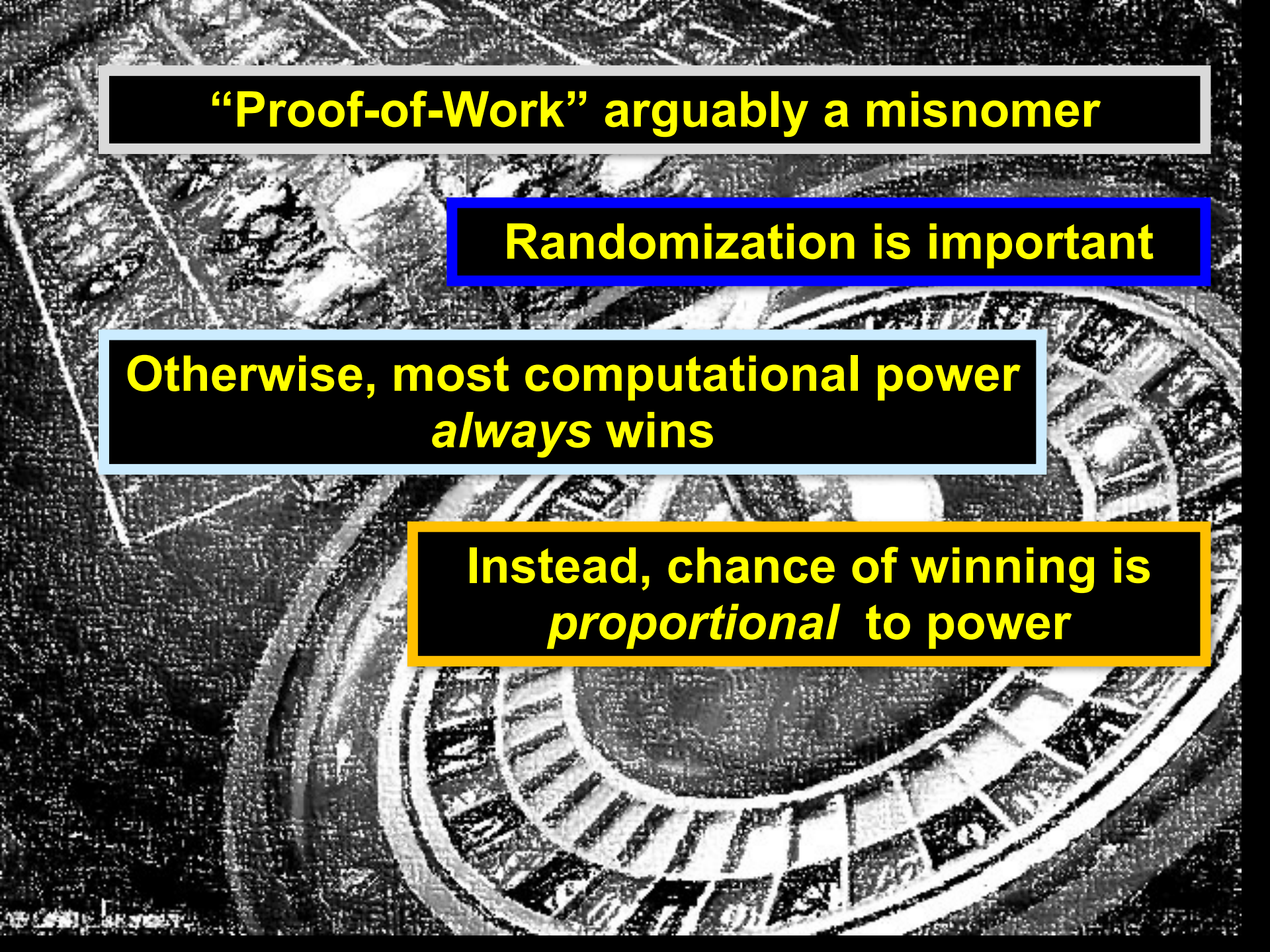
Randomization is important



“Proof-of-Work” arguably a misnomer

Randomization is important

**Otherwise, most computational power
always wins**

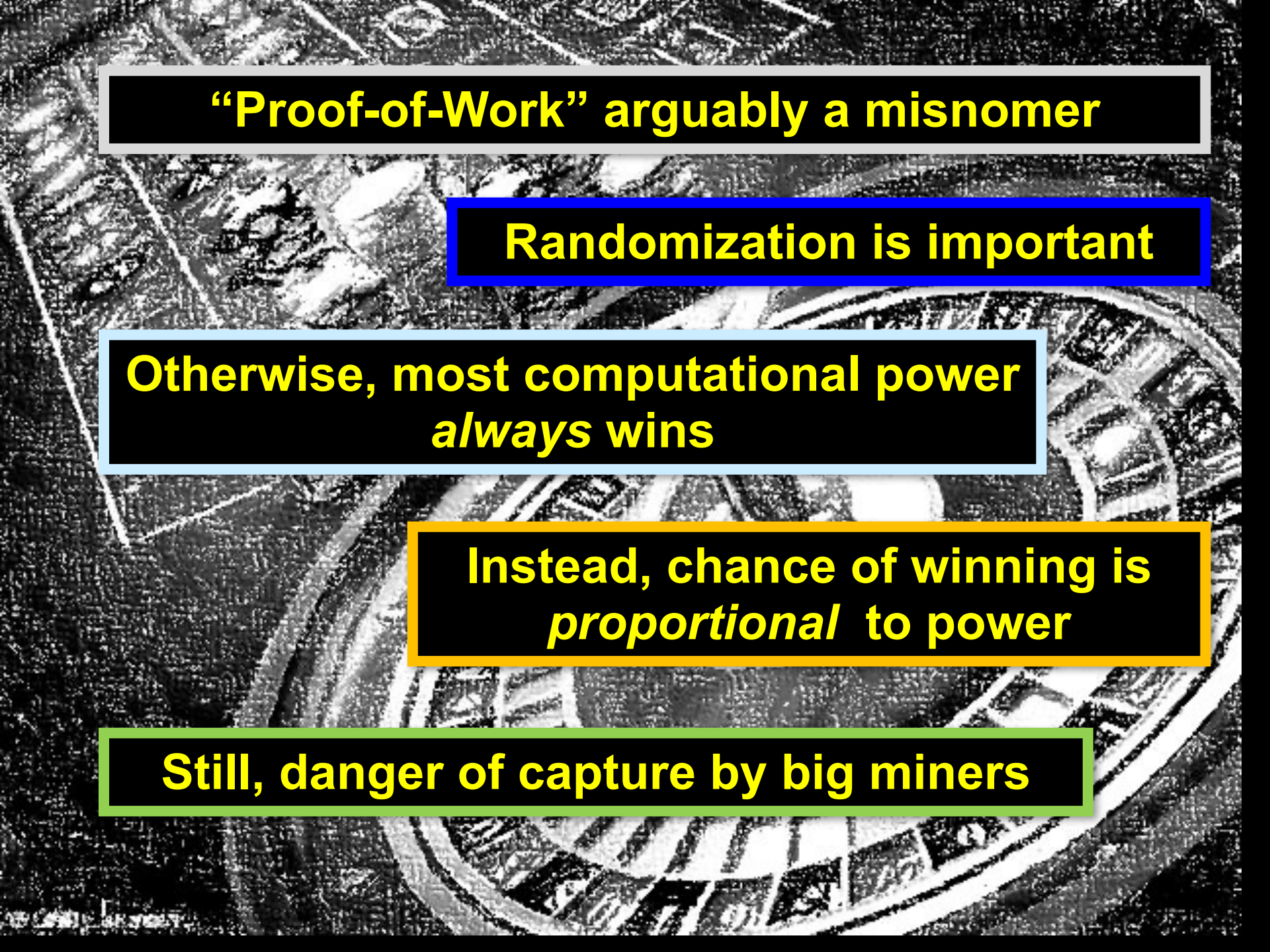


“Proof-of-Work” arguably a misnomer

Randomization is important

**Otherwise, most computational power
always wins**

**Instead, chance of winning is
proportional to power**



“Proof-of-Work” arguably a misnomer

Randomization is important

**Otherwise, most computational power
always wins**

**Instead, chance of winning is
proportional to power**

Still, danger of capture by big miners

Longest Chain Rule

in it. If a majority of CPU power is controlled by honest nodes, the honest chain will grow the fastest and outpace any competing chains. To modify a past block, an attacker would have to redo the proof of work of the block and all blocks after it and then catch up with and surpass the

**Honest miners build on
longest chain ...**

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**Honest miners build on
longest chain ...**

**Dishonest miners would have to out-
compute all honest miners**

Limited Throughput is Feature, not Bug

To compensate for increasing hardware speed and varying interest in running nodes over time, the proof-of-work difficulty is determined by a moving average targeting an average number of blocks per hour. If they're generated too fast, the difficulty increases.

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By varying PoW difficulty

Limited scalability becomes a problem as Bitcoin becomes successful

Calculation

p = probability an honest node finds the next block

Back of the envelope calculation

q = probability the attacker will ever catch up from z blocks behind

$$q_z = \begin{cases} 1 & \text{if } p \leq q \\ (q/p)^z & \text{if } p > q \end{cases}$$

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How likely dishonest miner can overtake honest miner to reverse transaction?

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How likely dishonest miner can overtake honest miner to reverse transaction?

Exponentially small

Calculation naïve but probably mostly right

The Bitcoin Backbone Protocol: Analysis and Applications*

Juan A. Garay
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akiayias@inf.ed.ac.uk

Nikos Leonardos^{†§}
National and Kapodistrian University of Athens.
nikos.leonardos@gmail.com

June 23, 2017

Abstract

Bitcoin is the first and most popular decentralized cryptocurrency to date. In this work, we extract and analyze the core of the Bitcoin protocol, which we term the *Bitcoin backbone*, and prove two of its fundamental properties which we call *common prefix* and *chain quality* in the static setting where the number of players remains fixed. Our proofs hinge on appropriate and novel assumptions on the “hashing power” of the adversary relative to network synchronicity; we show our results to be tight under high synchronization.

Next, we propose and analyze applications that can be built “on top” of the protocol, specifically focusing on Byzantine agreement (BA) and on transaction ordering. Regarding BA, we observe that Nakamoto-style BA is not possible in the Bitcoin action ledger. Regarding transaction ordering, we observe that a simple alternative which is bounded by $1/3$. The public ledger is a cryptographically secure ledger.

The Bitcoin Backbone Protocol: Analysis and Applications*

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Here is a more complete calculation ...

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more precise bounds on hashing power

Crime doesn't Pay

The incentive may help encourage nodes to stay honest. If a greedy attacker is able to assemble more CPU power than all the honest nodes, he would have to choose between using it to defraud people by stealing back his payments, or using it to generate new coins. He ought to

**Suppose dishonest party acquires
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Or collect all the rewards?

Vandalism destroys coin values!

The steps to run the network are as follows:

- 1) New transactions are broadcast to all nodes.
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- 3) Each node works on finding a difficult proof-of-work for its block.
- 4) When a node finds a proof-of-work, it broadcasts the block to all nodes.
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On the Bitcoin P2P layer

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**Miners assemble transactions
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**Economy of scale: single
transaction too expensive**

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Cartels with access to cheap power and ASICs control most of hashing power

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**Result: high latency because need to wait until
your transaction deep enough in chain**

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Double spending filter

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Break ties arbitrarily

Where's my Money?

By convention, the first transaction in a block is a special transaction that starts a new coin owned by the creator of the block. This adds an incentive for nodes to support the network, and provides a way to initially distribute coins into circulation, since there is no central authority to issue them.

You mine a block, you get paid

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How new bitcoins are generated

“Coinbase” transaction

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The incentive can also be funded with transaction fees. If the output value of a transaction is less than its input value, the difference is a transaction fee that is added to the incentive value of the block containing the transaction. Once a predetermined number of coins have entered circulation, the incentive can transition entirely to transaction fees and the complete inflation

Customers can include transaction fee

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Customers can include transaction fee

Higher fees buy lower latency?

Deflationary

the block containing the transaction. Once a predetermined number of coins have entered circulation, the incentive can transition entirely to transaction fees and be completely inflation free.

Limit on number of BTC ever minted

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Fear of inflation?

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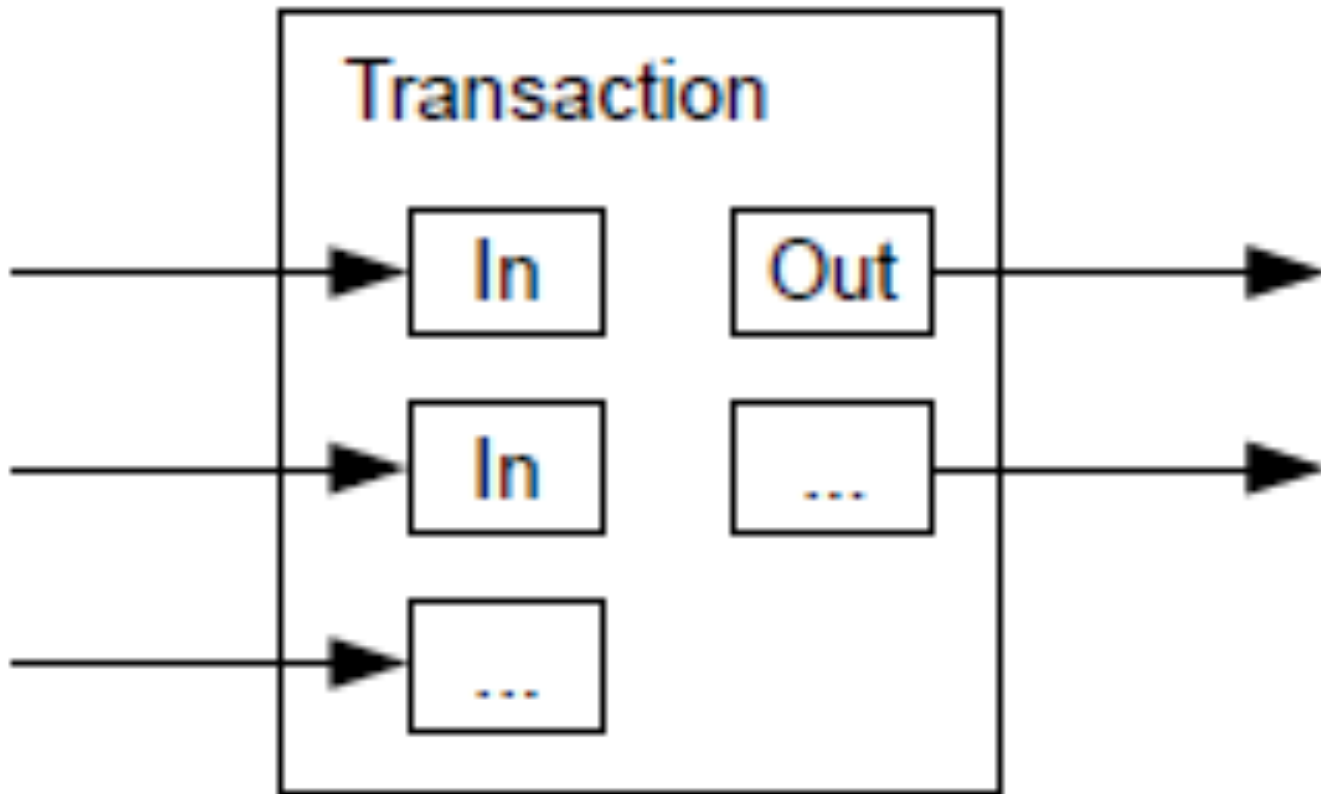
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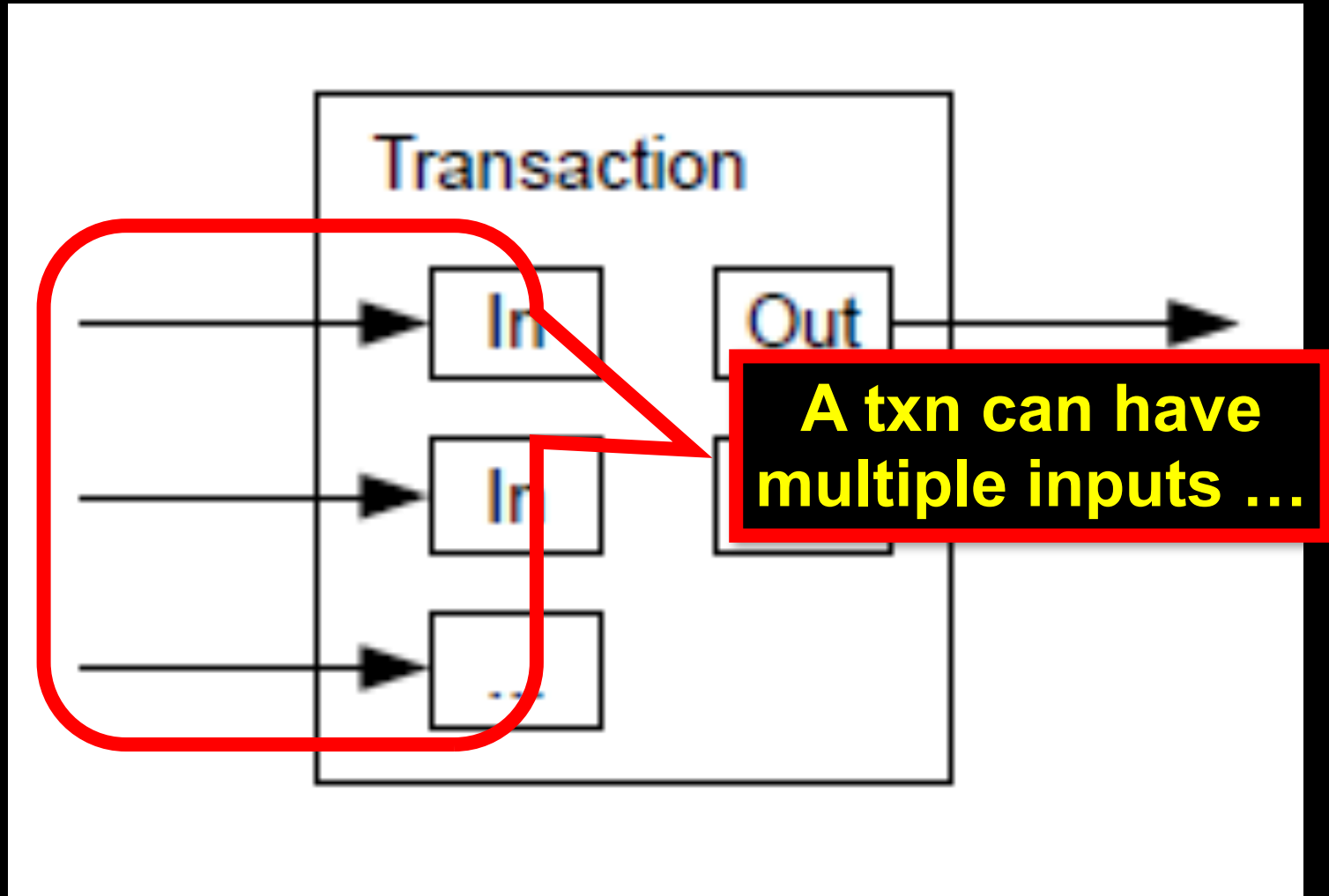
But not deflation?

Deflation implies inflated fees

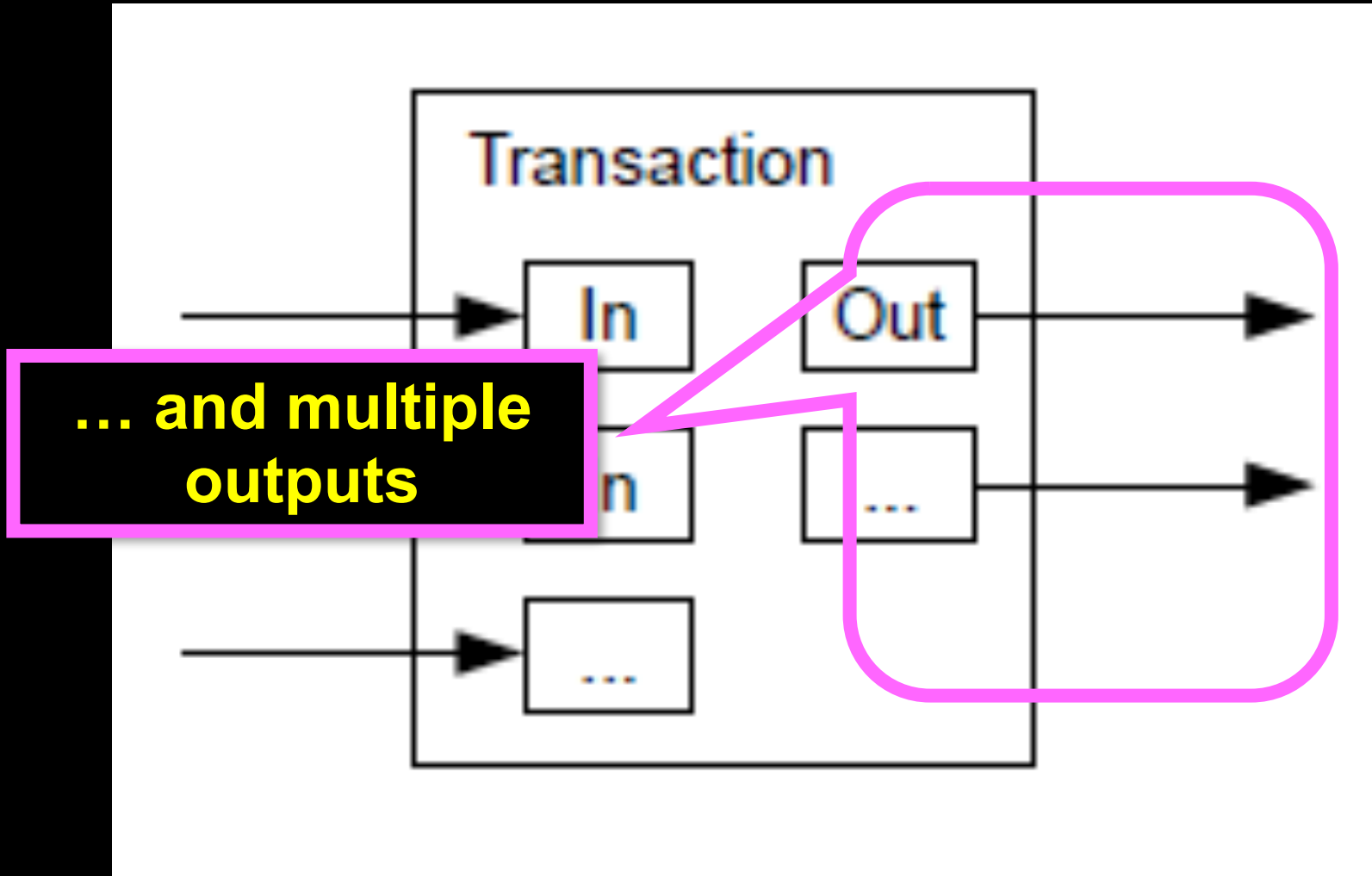
UTXO (unspent transaction output) Model



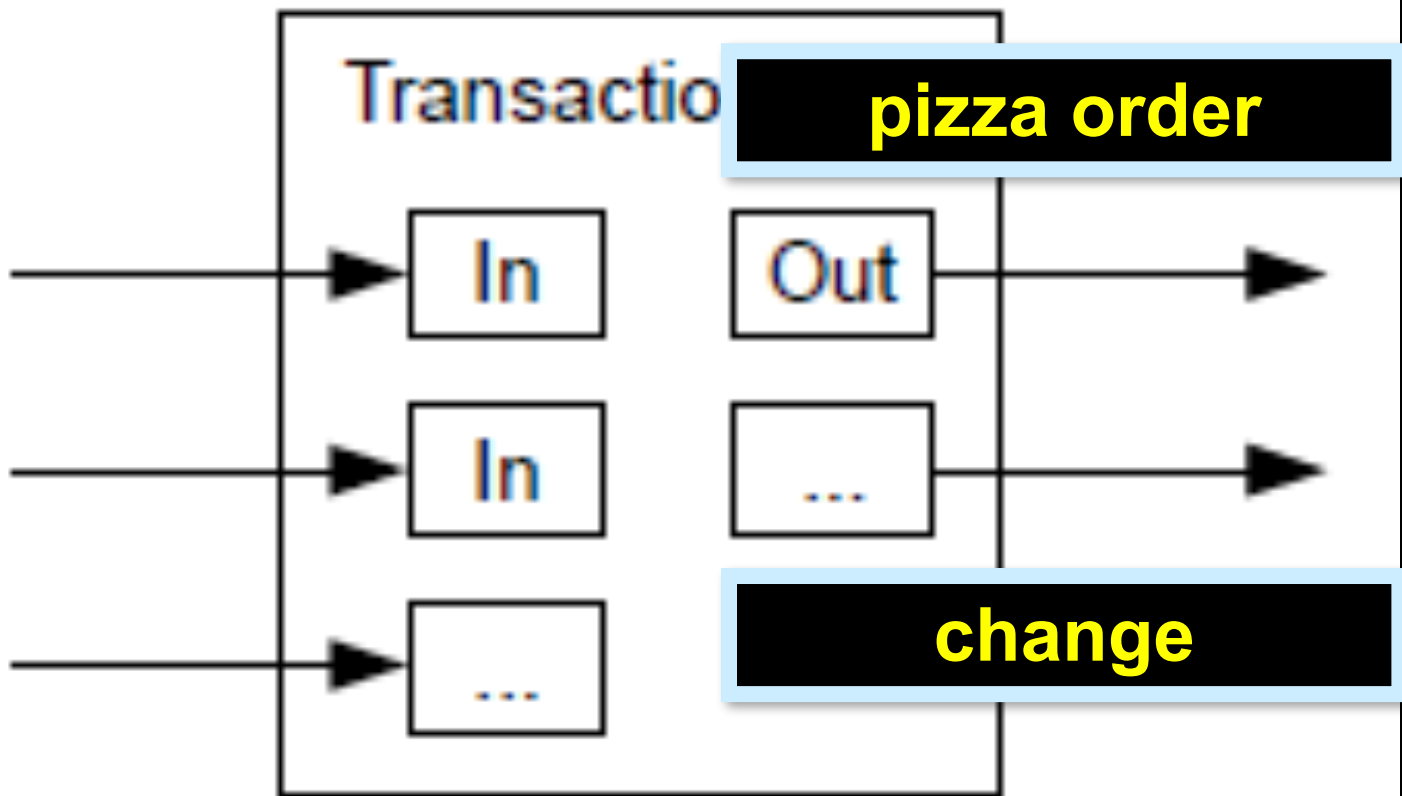
UTXO Model



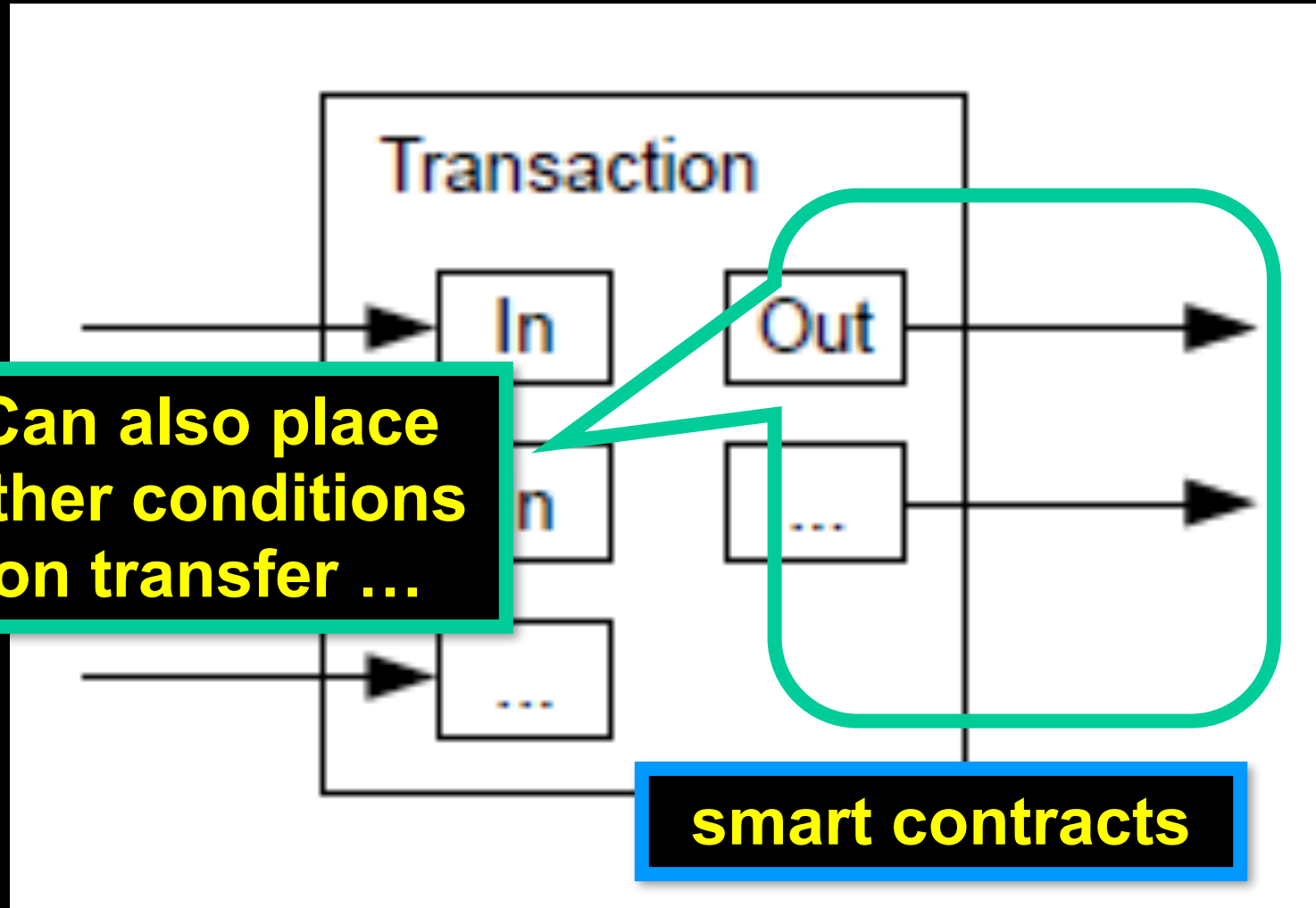
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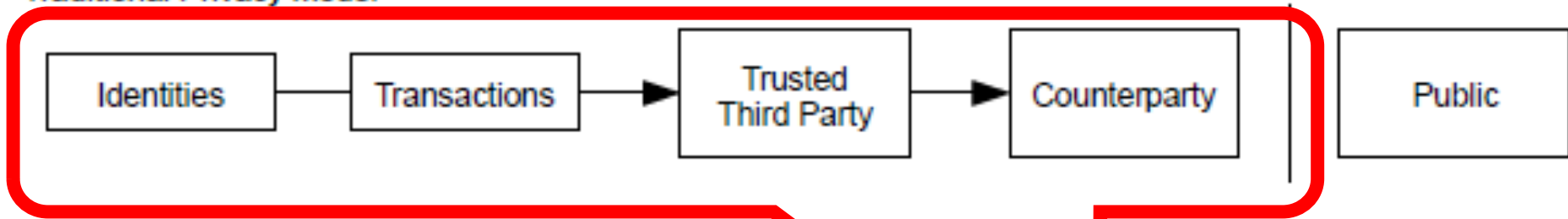


UTXO Model

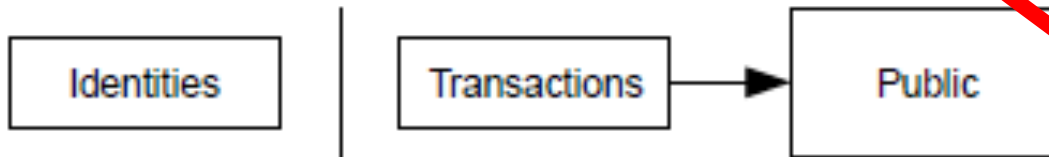


Privacy

Traditional Privacy Model



New Privacy Model

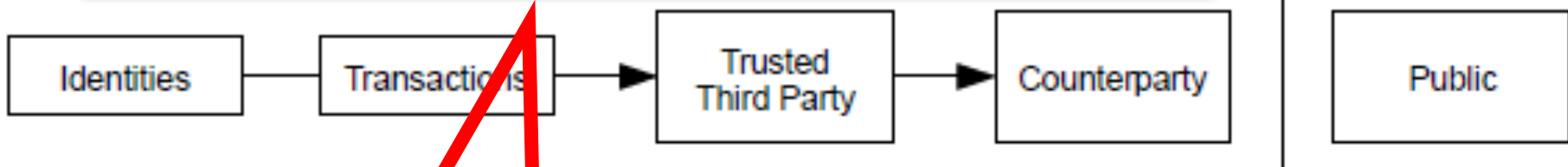


Old-school: Trusted 3rd party keeps IDs and transactions secret

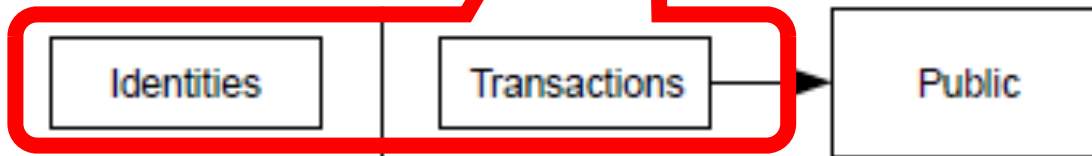
Privacy

Bitcoin: all transactions visible

Traditi



New Privacy Model



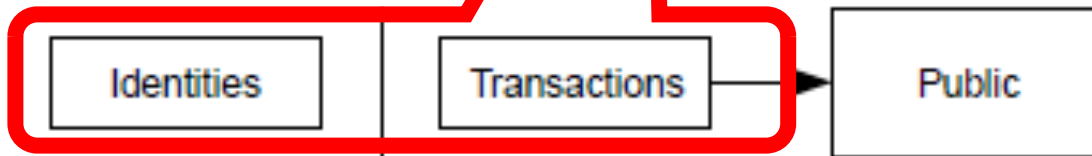
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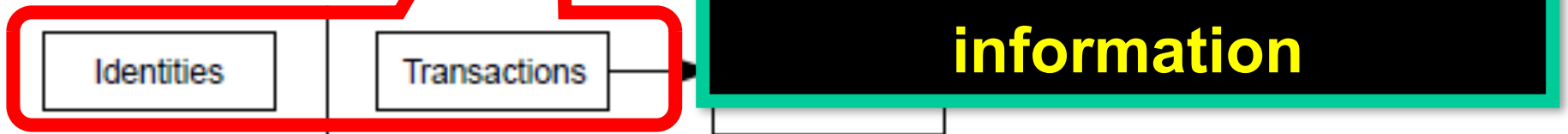
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New Privacy Model



“pseudonymous”

“Decentralized Trust Infrastructure”



Centralized



Decentralized

Centralized Trust



Centralized Trust

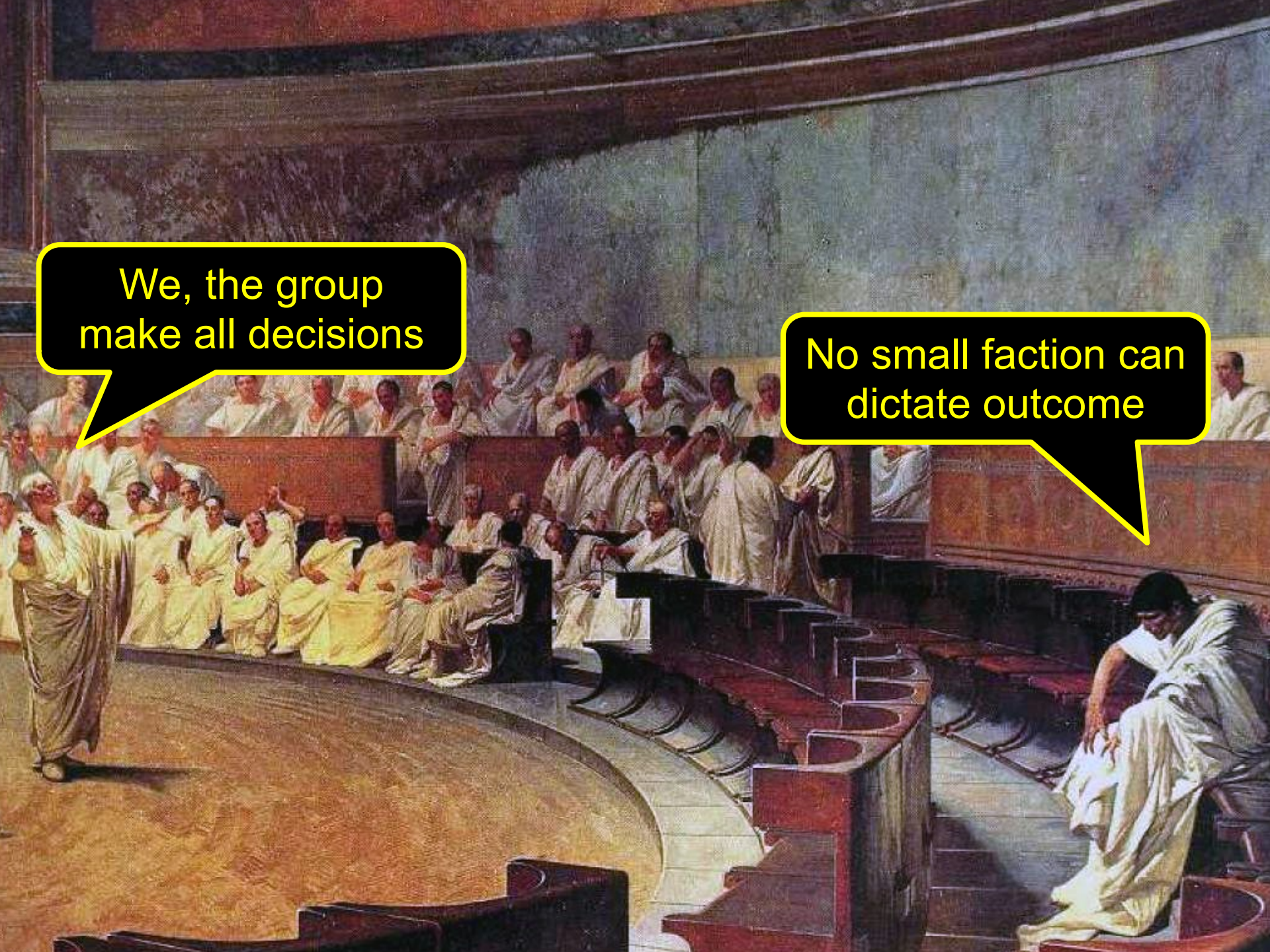
Only I can make decisions





A classical painting depicting a Roman Senate in session. The scene is set in a large, semi-circular room with tiered seating. Numerous men in white togas are seated, some looking towards the front, others in conversation. In the foreground, a man in a white toga stands on the left, gesturing with his right arm towards the seated group. On the right, another man in a white toga is seated, looking down. The background features a large, textured wall. A yellow speech bubble is overlaid on the left side of the image.

We, the group
make all decisions



We, the group
make all decisions

No small faction can
dictate outcome

Who Votes?



Who Votes?

Adult Men: 1792, 1848

Who Votes?

Adult Men: 1792, 1848

Adult Women: 1944

Who Votes?

Adult Men: 1792, 1848

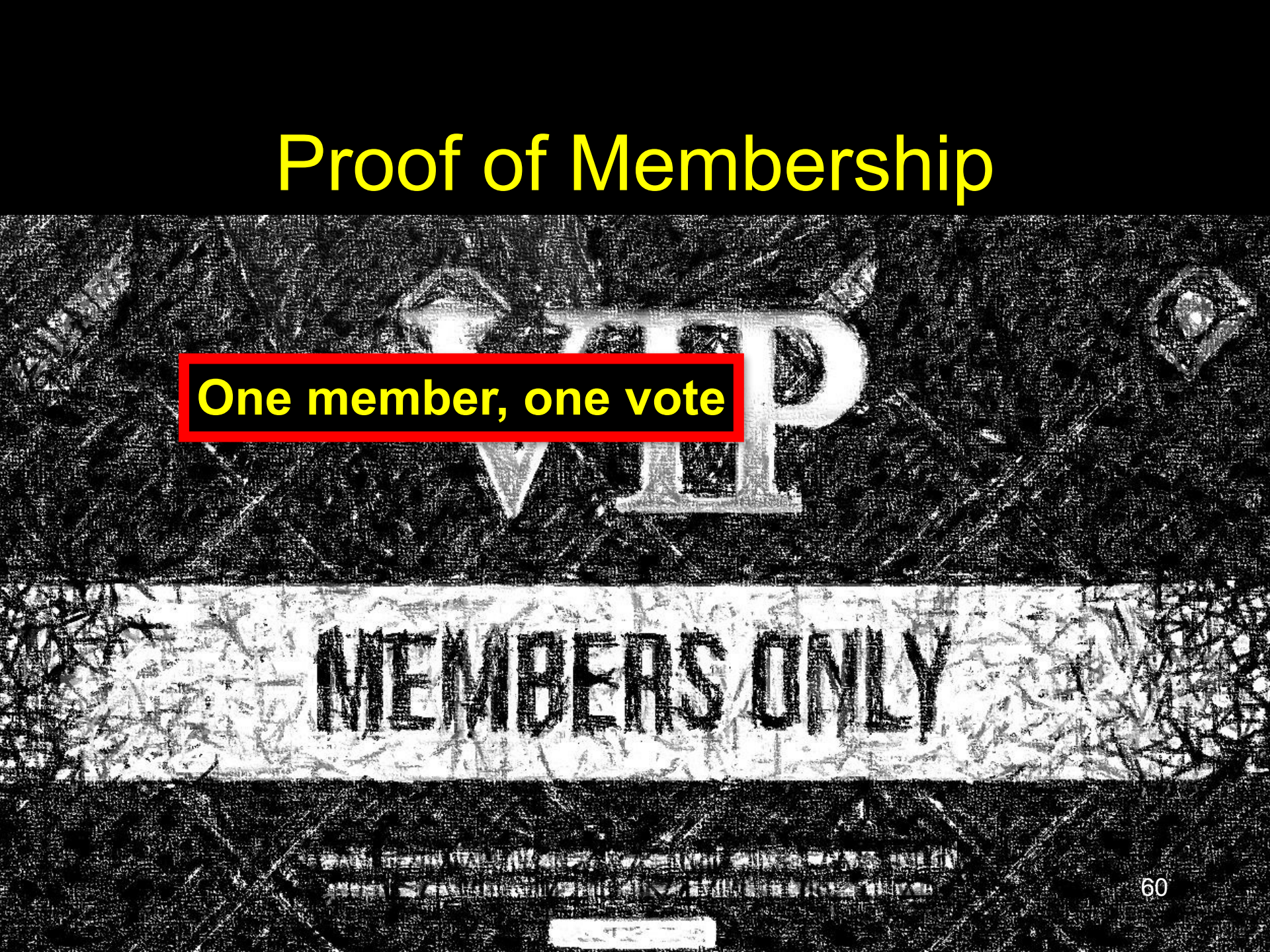
Adult Women: 1944

Only citizens ...

Proof of Membership

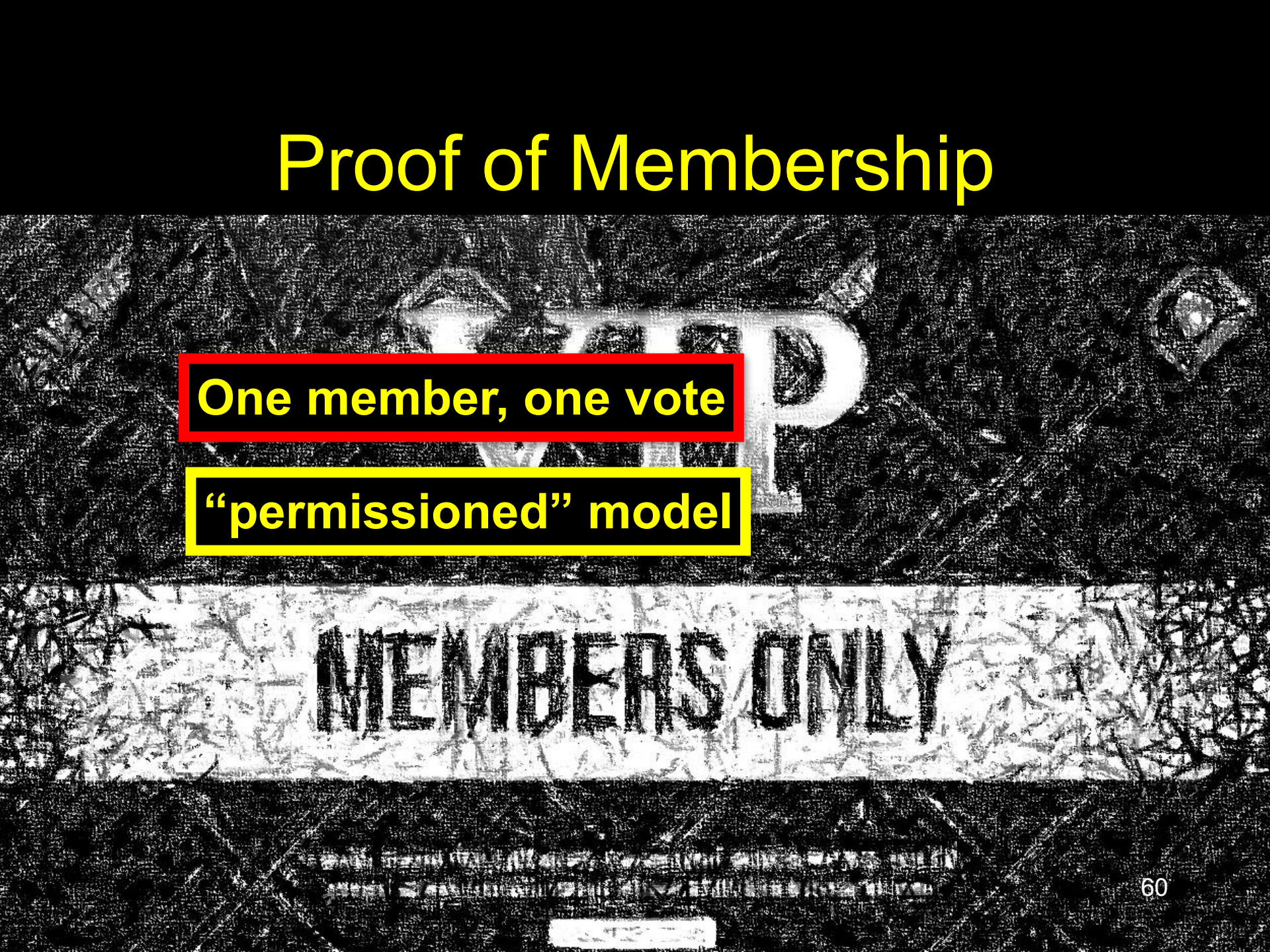


Proof of Membership

A black and white photograph of a metal door. The door has a large, stylized 'VAP' logo in the center. Below the logo is a horizontal band with the text 'MEMBERS ONLY' in a bold, sans-serif font. The door is surrounded by a dark, textured background.

One member, one vote

Proof of Membership



One member, one vote

“permissioned” model

MEMBERS ONLY

Proof of Membership

One member, one vote

“permissioned” model

Adversary controls $< 1/3$ votes

Proof of Membership

One member, one vote

“permissioned” model

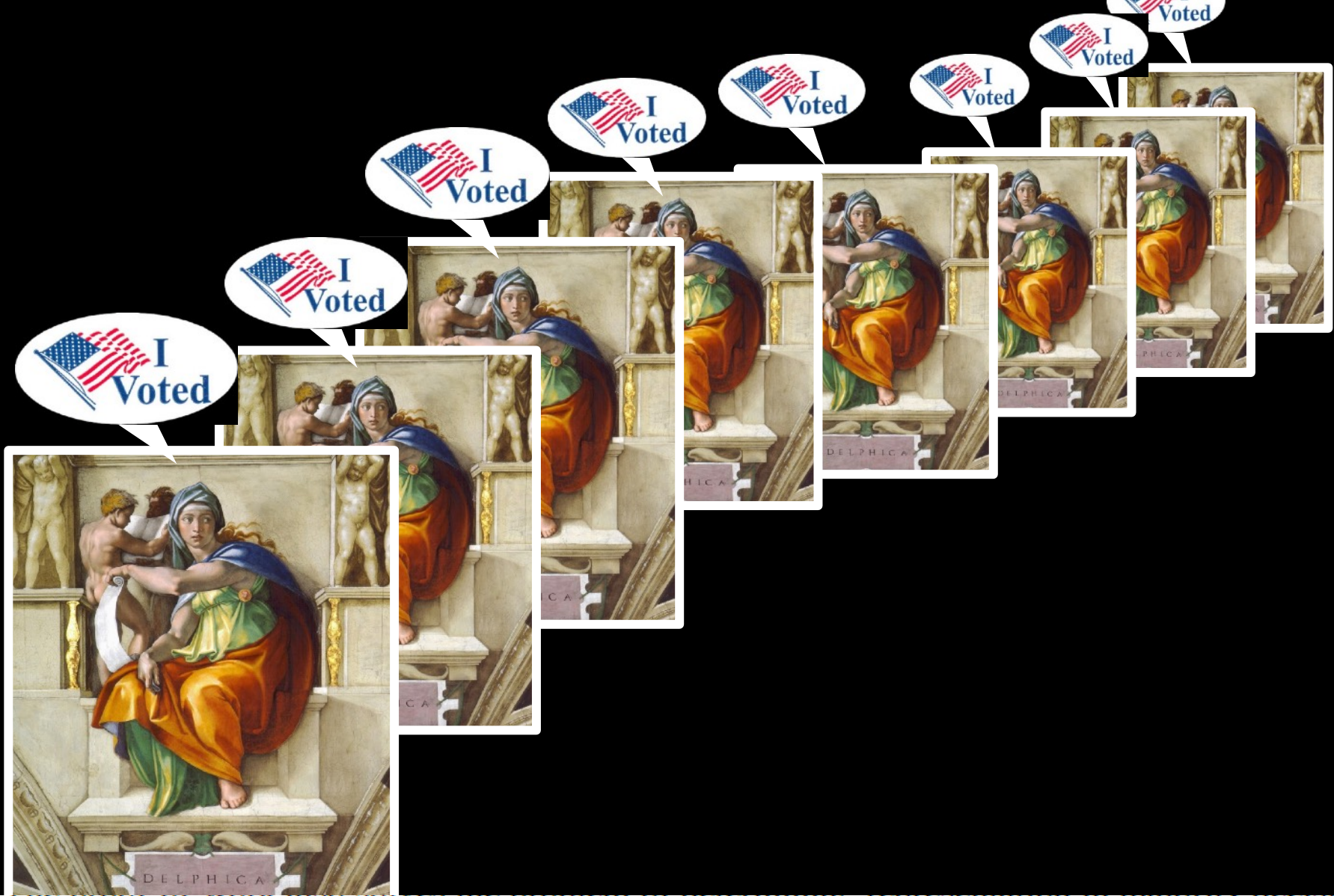
Adversary controls $< 1/3$ votes

Classical distributed systems



DELPHICA

The proof of work also solves the problem of determining representation in majority decision making. If the majority were based on one-IP-address-one-vote, it could be subverted by anyone able to allocate many IPs. Proof-of-work is essentially one-CPU-one-vote. The majority decision is represented by the longest chain, which has the greatest use of a fixed resource.



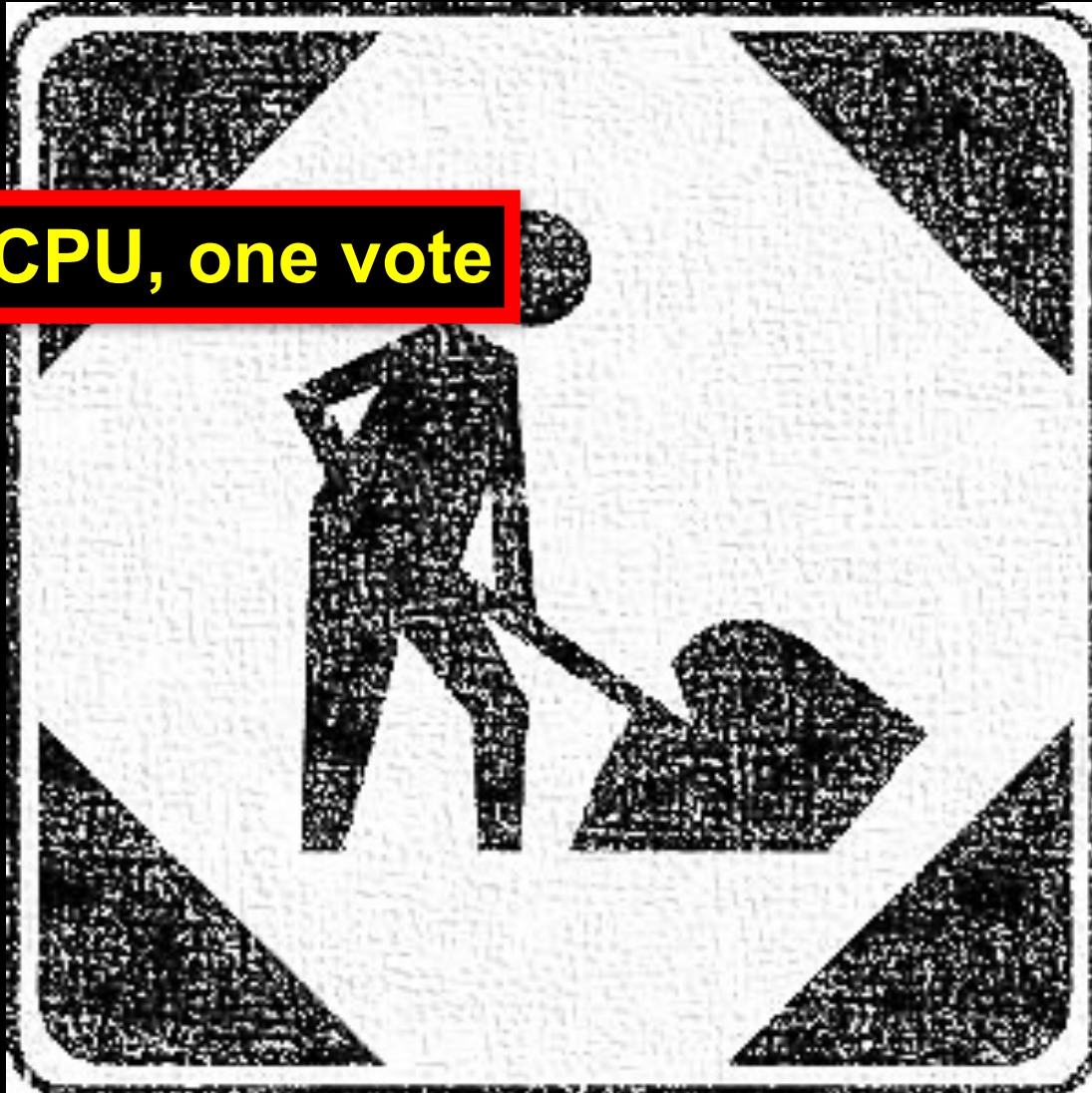
The proof of work also solves the problem of determining representation in majority decision making. If the majority were based on one-IP-address-one-vote, it could be subverted by anyone able to allocate many IPs. Proof-of-work is essentially one-CPU-one-vote. The majority decision is represented by the longest chain, which has the greatest number of work effort invested.

Proof of Work



Proof of Work

One CPU, one vote



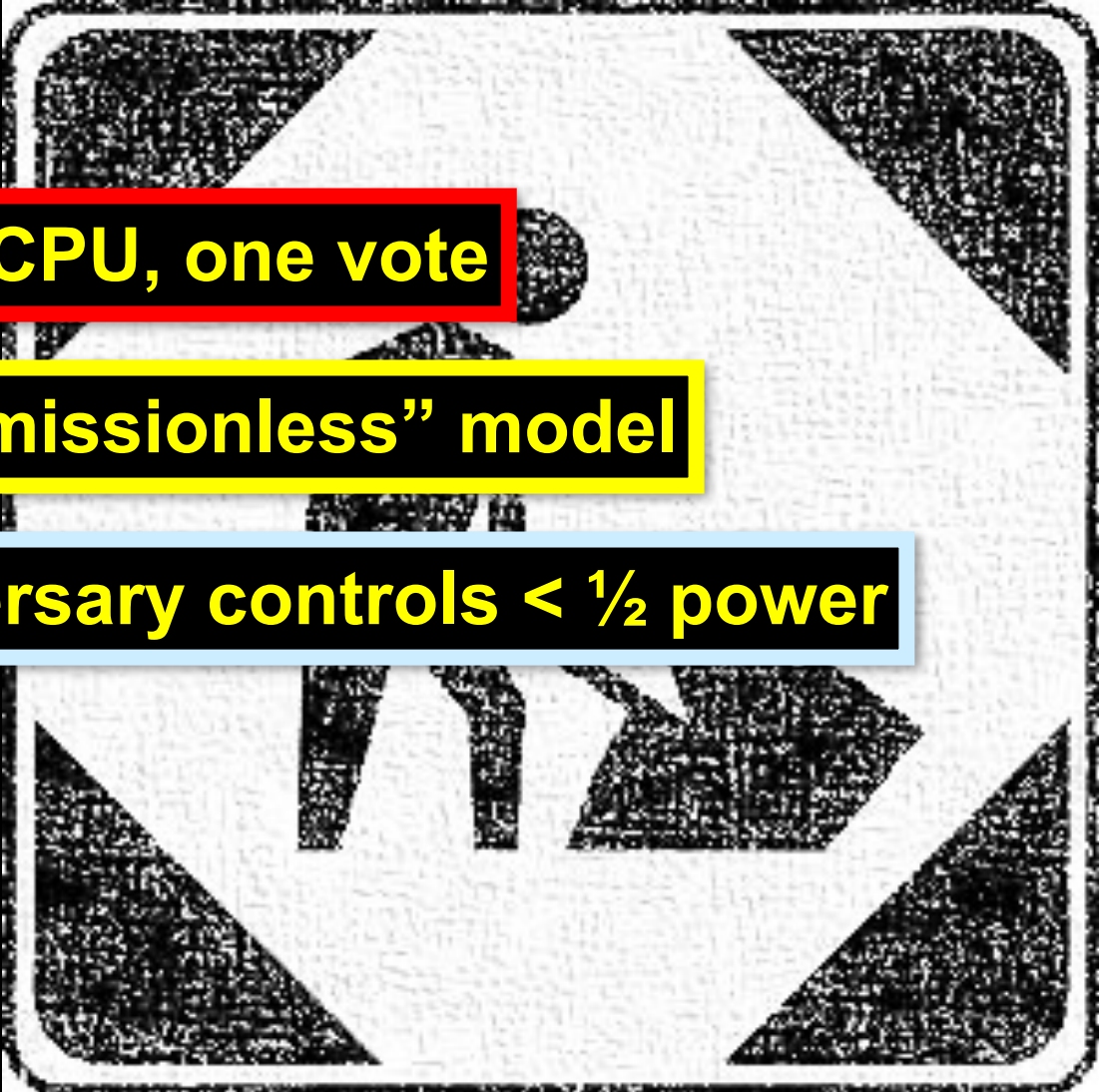
Proof of Work

One CPU, one vote

“permissionless” model



Proof of Work



One CPU, one vote

“permissionless” model

Adversary controls $< \frac{1}{2}$ power

Proof of Work

One CPU, one vote

“permissionless” model

Adversary controls $< \frac{1}{2}$ power

Nakamoto's breakthrough

Bitcoin Adversary

“The system is secure as long as honest nodes collectively control more CPU power than any cooperating group of attacker nodes.”

S. Nakamoto

Popular Culture

Silicon Valley Season 5 Finale Episode – 51 % Attack

by Rohit Kukreja | May 21, 2018 | [Bitcoin](#), [Cryptocurrency](#), [Cryptocurrency News](#)



Decentralization in Bitcoin and Ethereum Networks

Adem Efe Gencer^{1,2}, Soumya Basu^{1,2}, Ittay Eyal^{1,3}, Robbert van Renesse^{1,2},
and Emin Gün Sirer^{1,2}

¹ Initiative for Cryptocurrencies and Contracts (IC3)

² Computer Science Department, Cornell University

³ Electrical Engineering Department, Technion

Abstract. Blockchain-based cryptocurrencies have demonstrated how to securely implement traditionally centralized systems, such as currencies, in a decentralized fashion. However, there have been few measurement studies on the level of decentralization they achieve in practice. We present a measurement study on various decentralization metrics of two of the leading cryptocurrencies with the largest market capitalization and user base, Bitcoin and Ethereum. We investigate the extent of decentralization by measuring the network resources of nodes and the interconnection among them, the protocol requirements affecting the operation of nodes, and the robustness of the two systems against attacks. In particular, we adapted existing internet measurement techniques to use the Falcon Relay Network for measuring network resources. We used the Falcon Relay Network to measure the network resources of nodes. In our data, we did not find any significant correlation between the network resources of nodes and the extent of decentralization.

Fake Decentralization

Decentralization in Bitcoin and Ethereum Networks

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Sanjiv Siroer^{1,2}

Robbert van Renesse^{1,2}

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Bitcoin (& Ethereum) are pretty centralized

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Top 4 Bitcoin Miners > 53% power

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Department, Technion

Top 4 Bitcoin Miners > 53% power

Top 3 Ethereum miners > 61% power

15 Bitcoin, 11 Ethereum miners > 90%

PoW Encourages Centralization

TECH INDUSTRY

This coal power plant is being reopened for blockchain mining

The now shuttered coal-fired power station on Australia's east coast will offer cheap power prices to blockchain operators.

BY CLAIRE REILLY | APRIL 11, 2018 12:27 AM PDT



PoW Encourages Centralization



Wasteful

Wasteful

Bitcoin mining uses as much energy as mining for gold, study finds

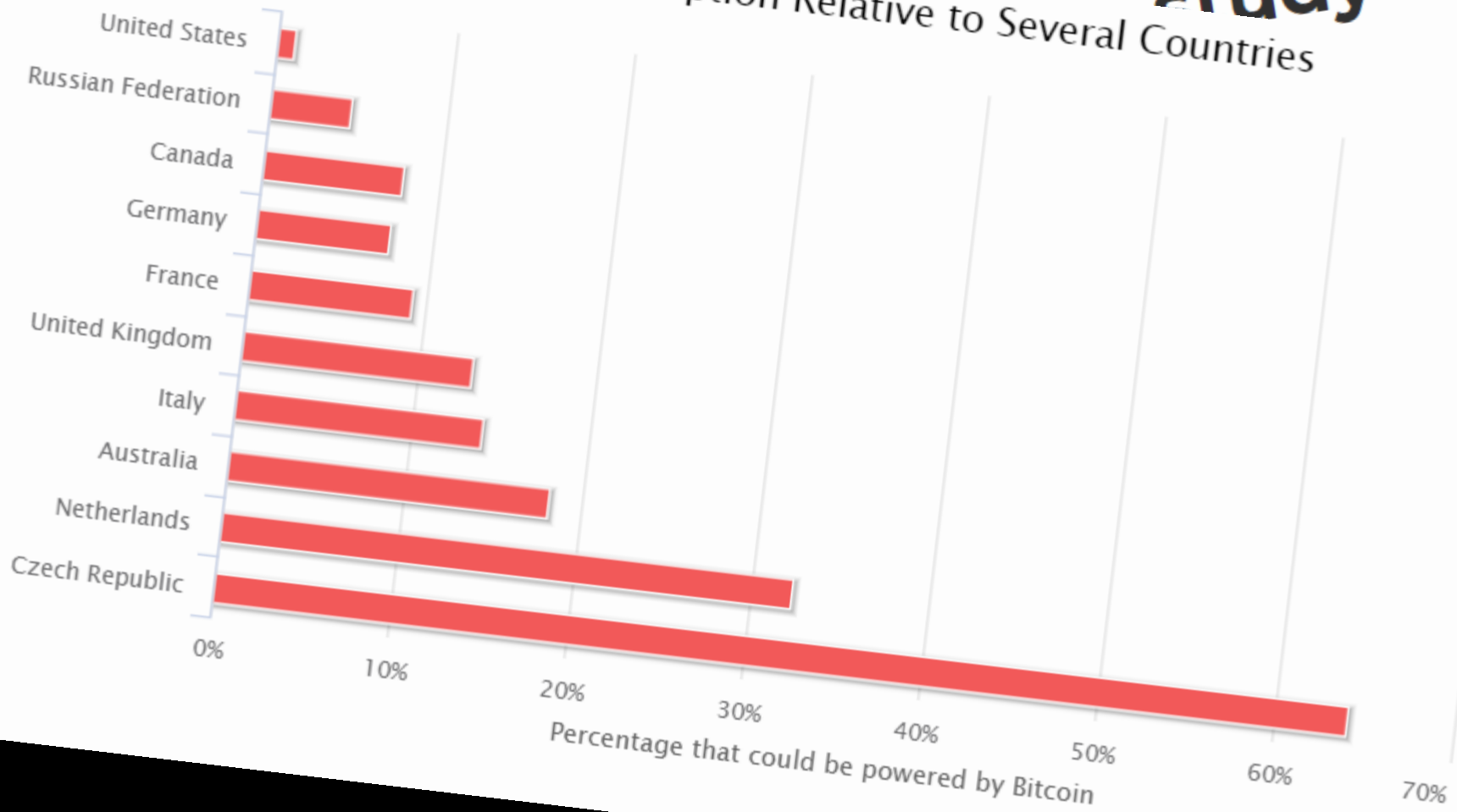
What does it mean for the future of the cryptocurrency movement and its impact on the environment?

REUBEN JACKSON 08 November, 2018

Wasteful

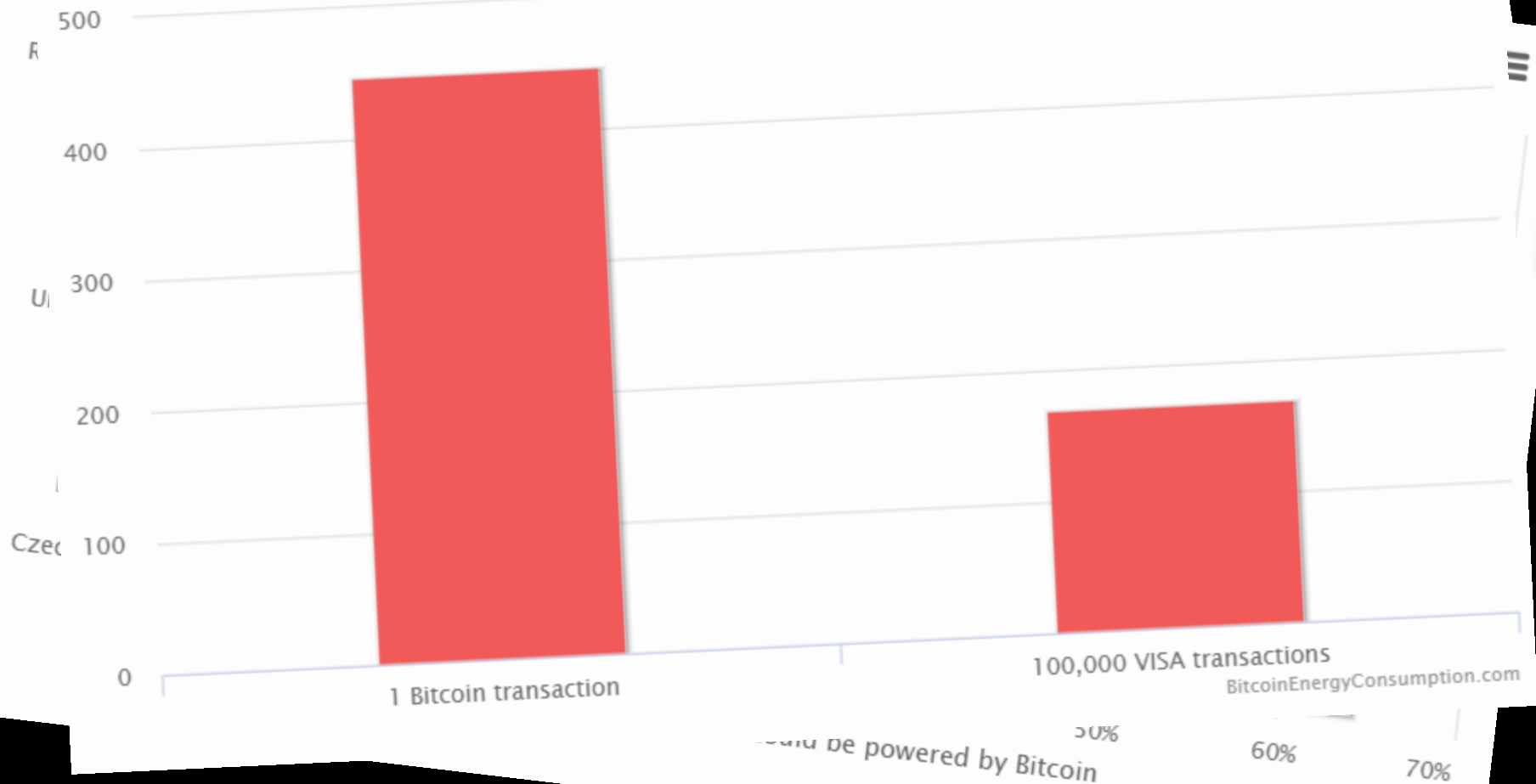
as much study

Bitcoin Energy Consumption Relative to Several Countries

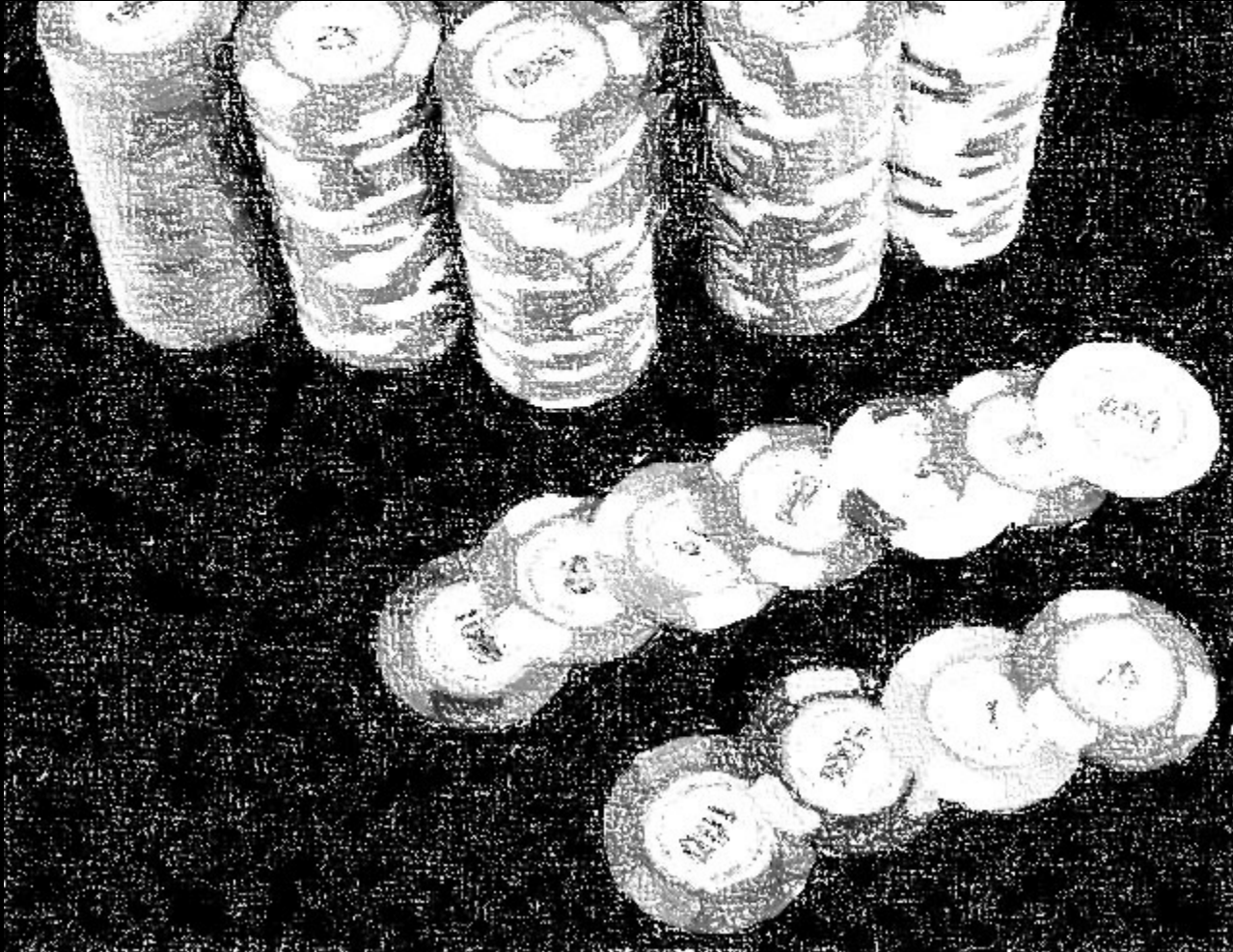


Wasteful

Bitcoin network versus VISA network average consumption



Proof of Stake



Proof of Stake



One coin, one vote

Proof of Stake

The background of the slide features a high-contrast, black and white image of numerous coins. Some coins are stacked vertically into tall columns, while others are scattered across the surface. The image has a grainy, textured appearance.

One coin, one vote

Voters have something to lose

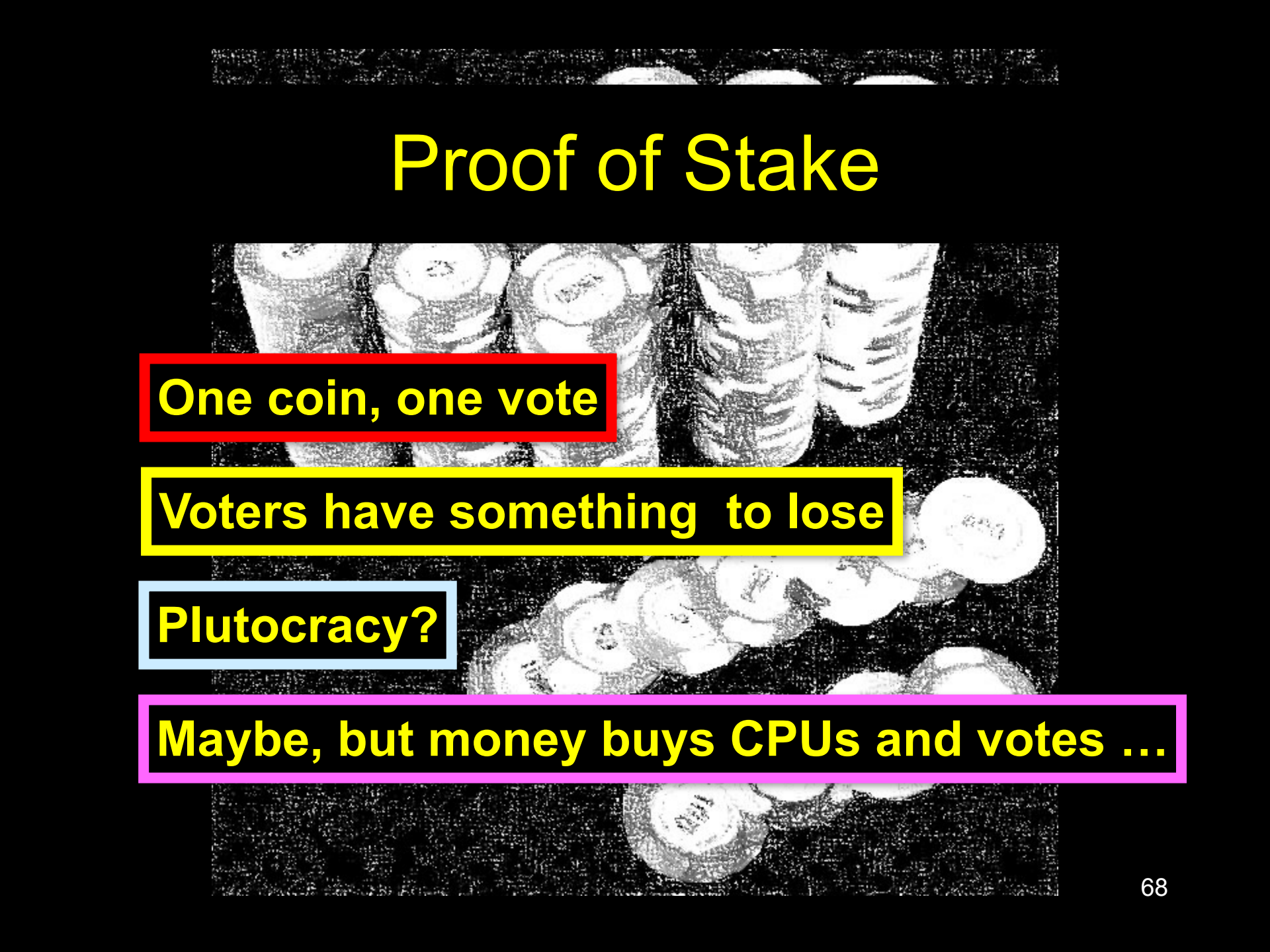
Proof of Stake

The background of the slide is a high-contrast, black and white image showing several tall stacks of coins on the left and a pile of scattered coins on the right. The coins appear to be of different denominations, with some showing distinct markings.

One coin, one vote

Voters have something to lose

Plutocracy?

The background of the slide is a high-contrast, black and white image showing several tall stacks of coins, likely Euro coins, arranged in a row. The coins are stacked on a dark surface, and the lighting creates strong highlights and shadows, emphasizing the texture and metallic nature of the coins.

Proof of Stake

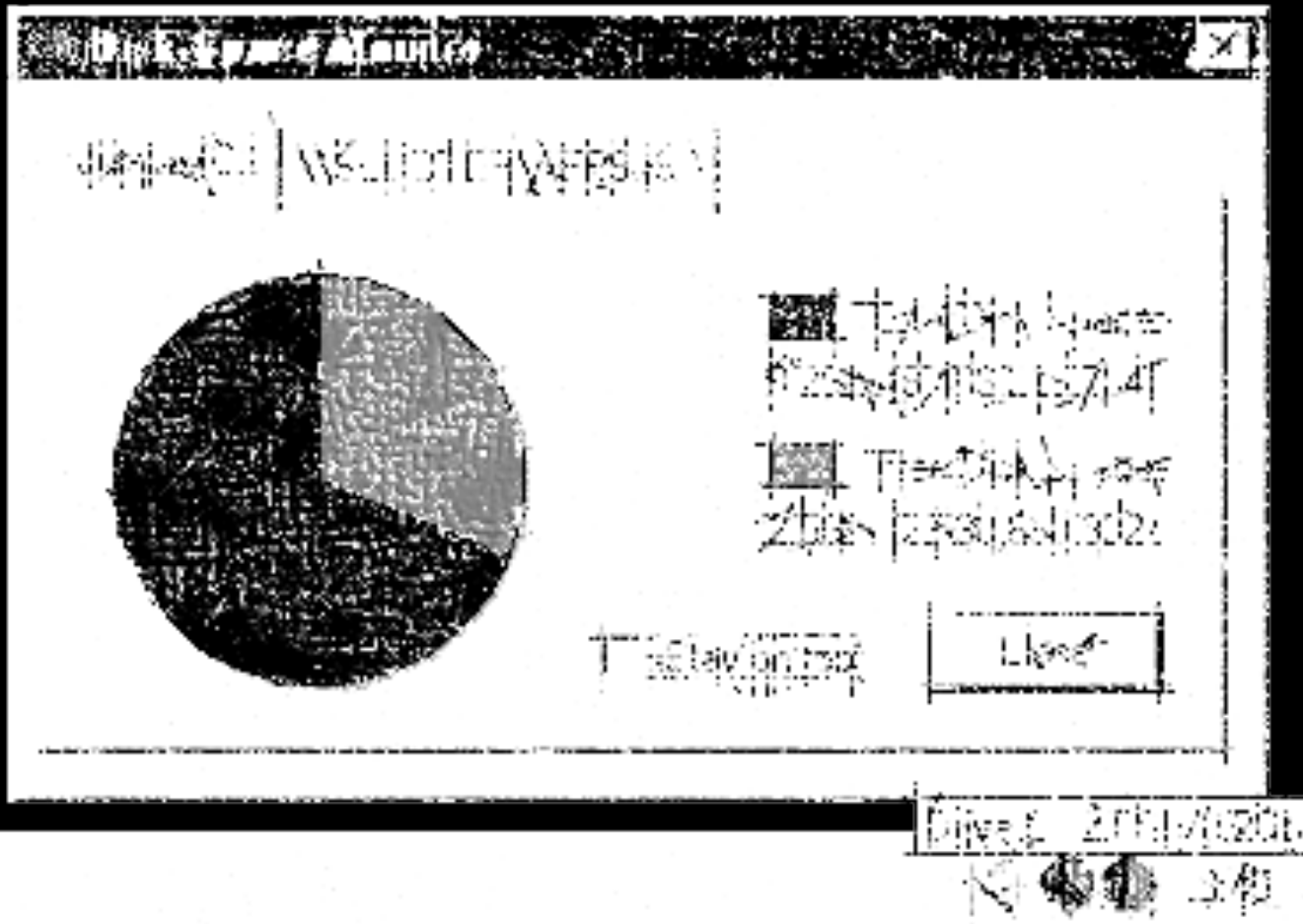
One coin, one vote

Voters have something to lose

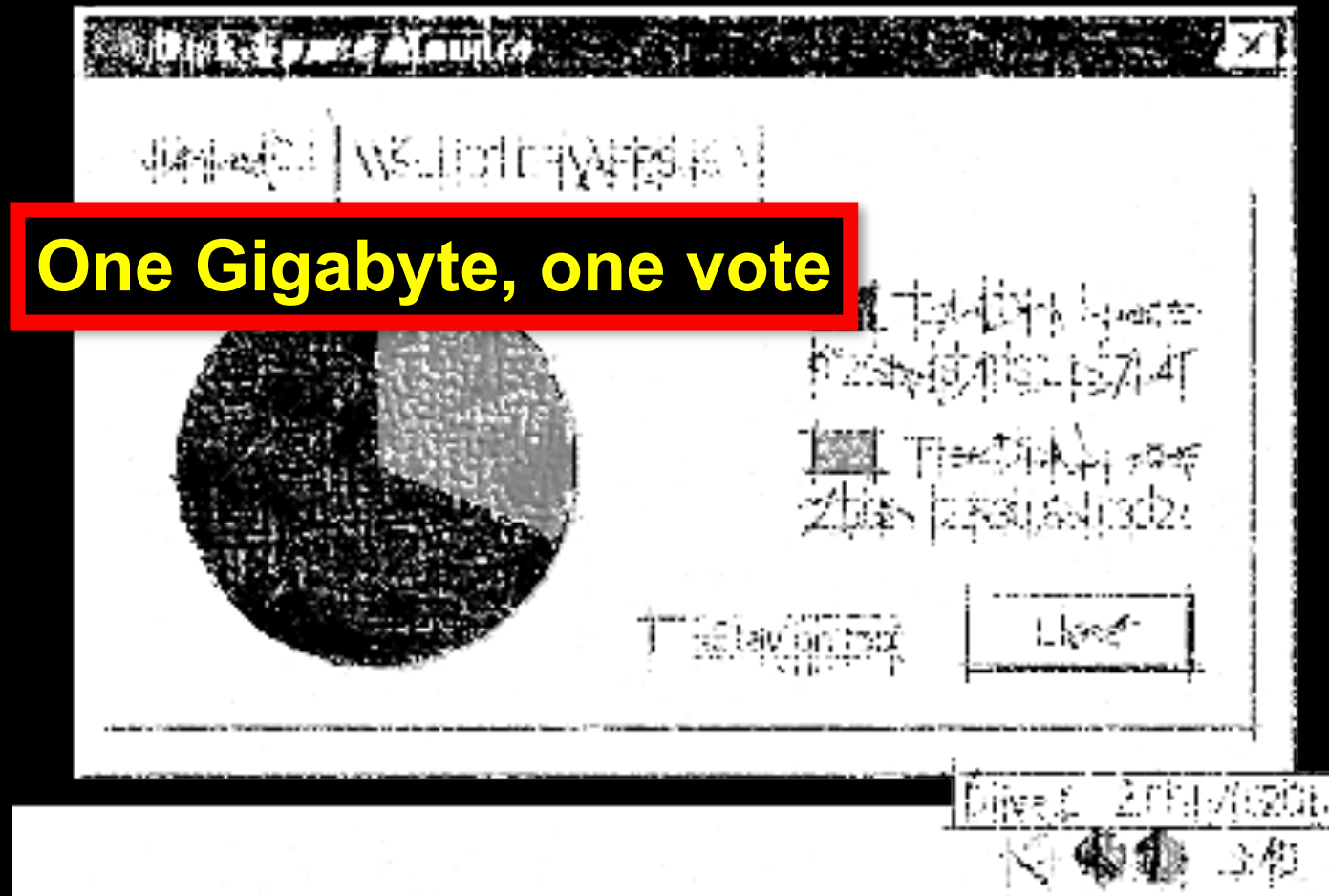
Plutocracy?

Maybe, but money buys CPUs and votes ...

Proof of Space



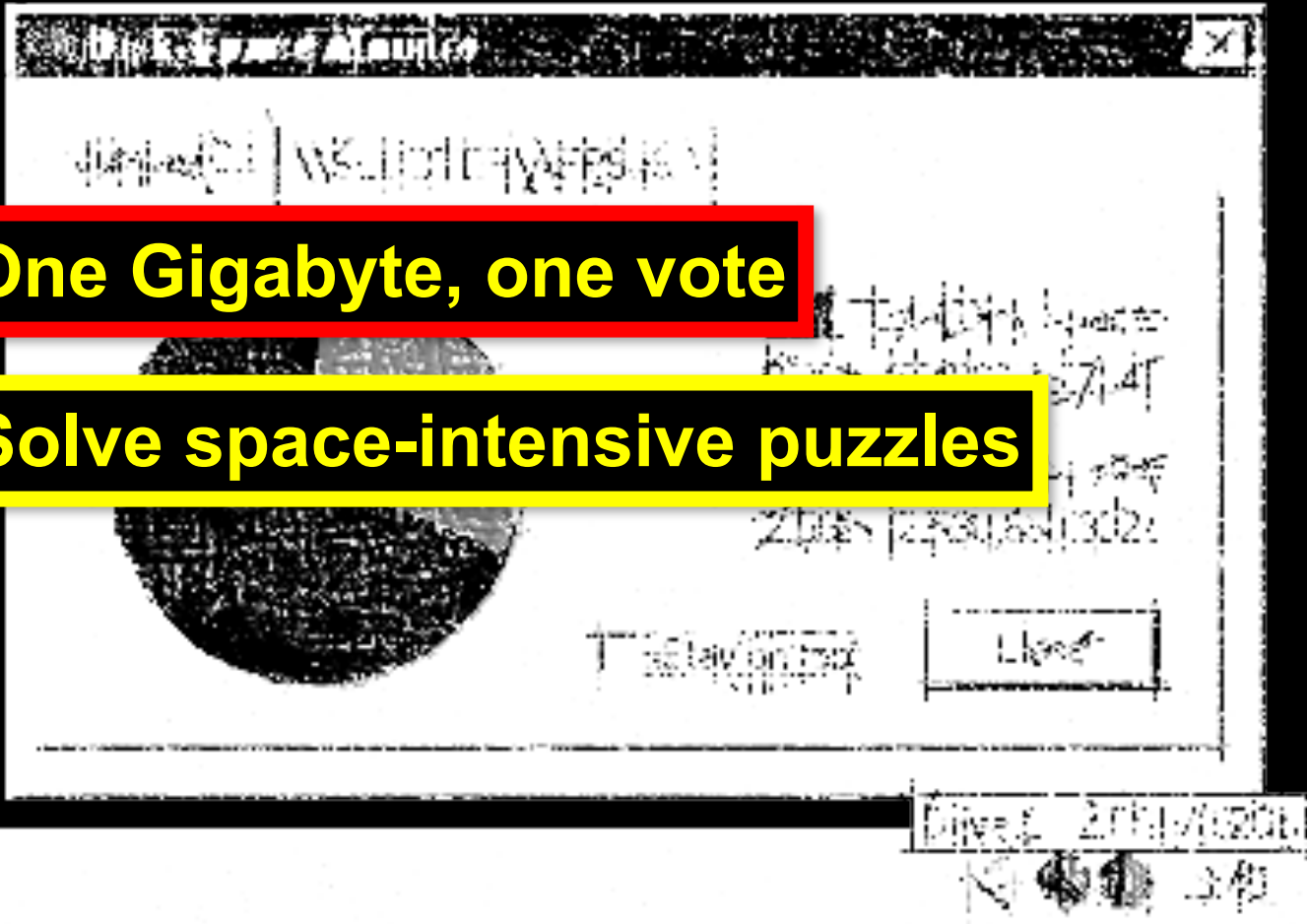
Proof of Space



Proof of Space

One Gigabyte, one vote

Solve space-intensive puzzles

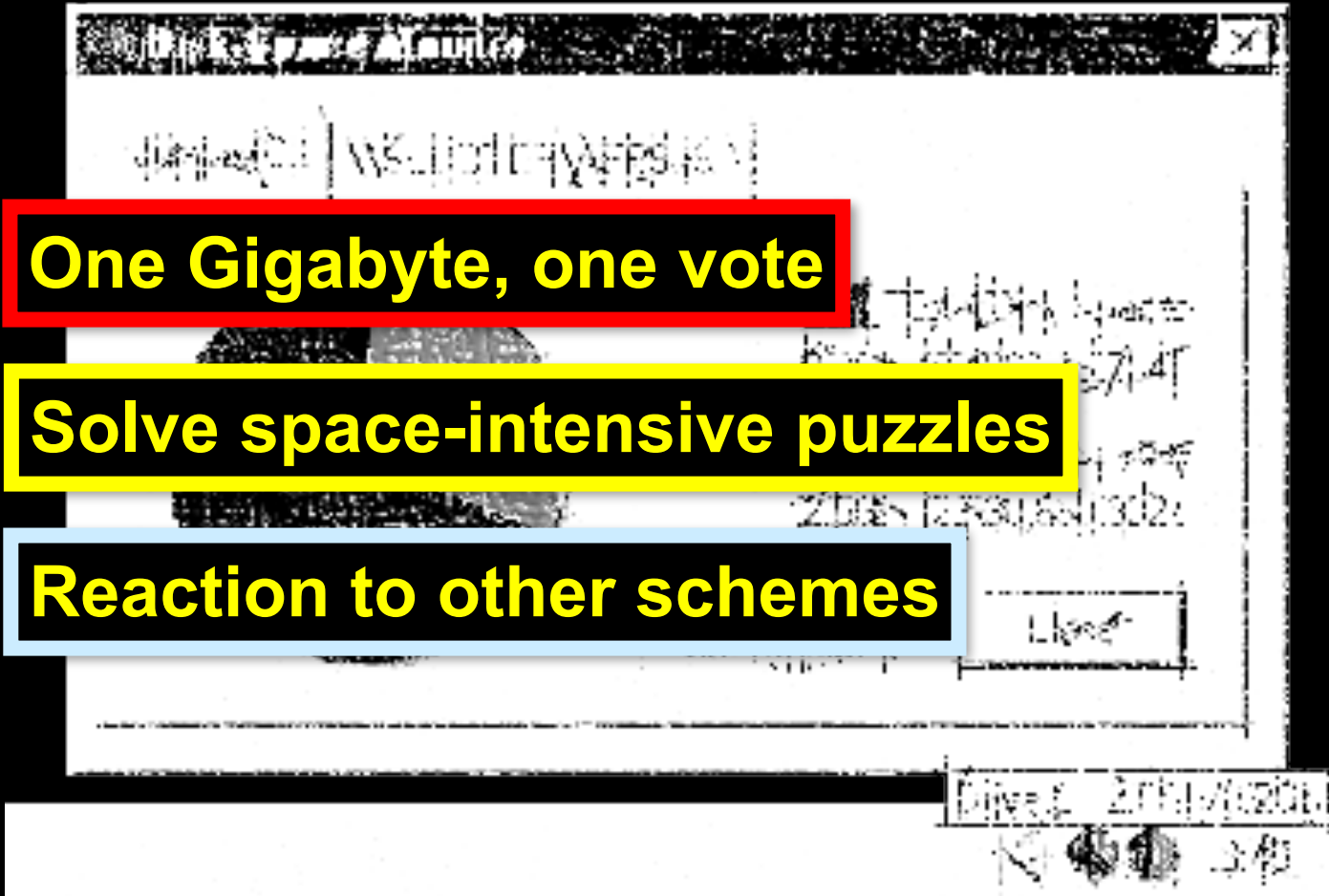


Proof of Space

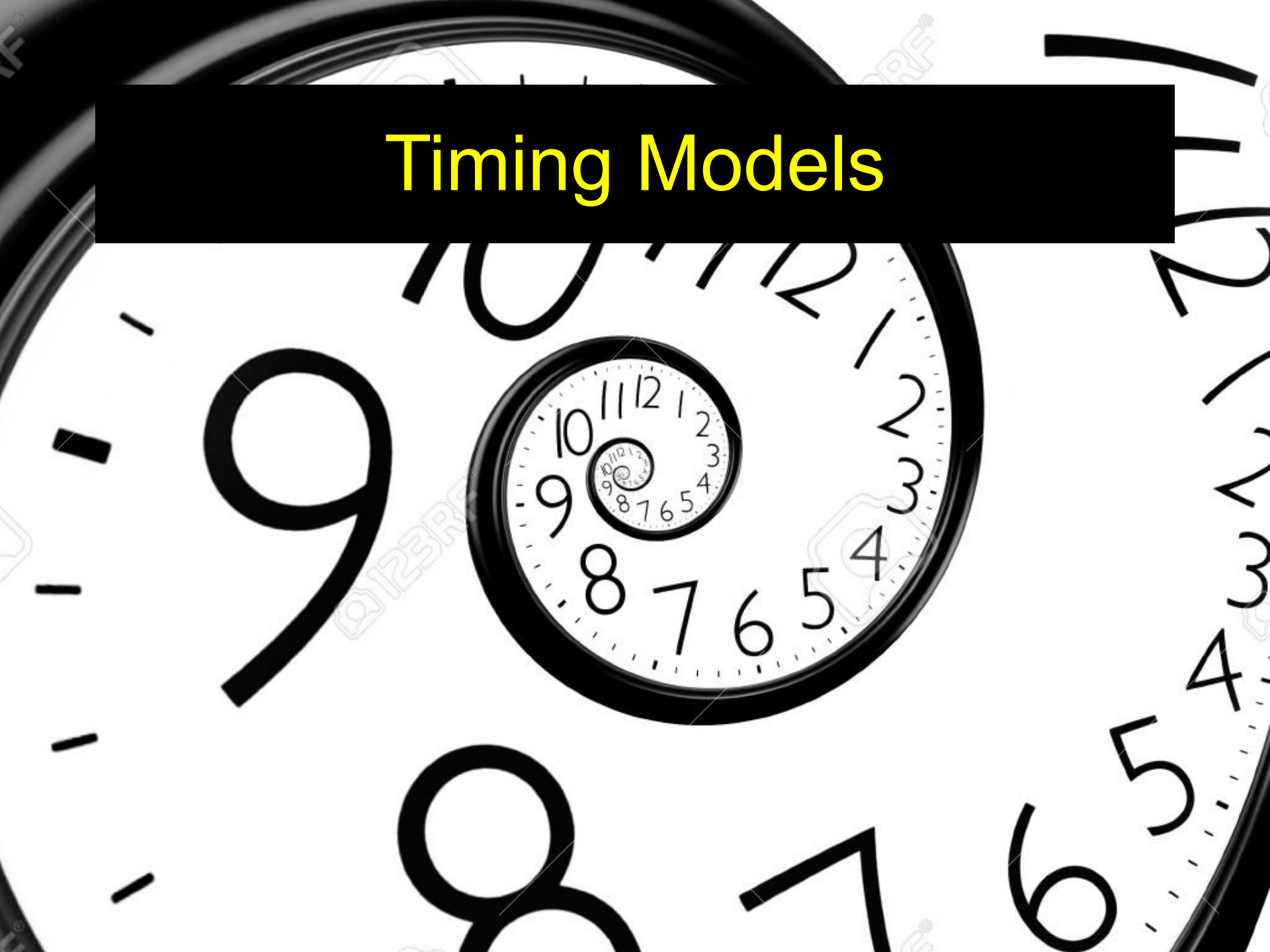
One Gigabyte, one vote

Solve space-intensive puzzles

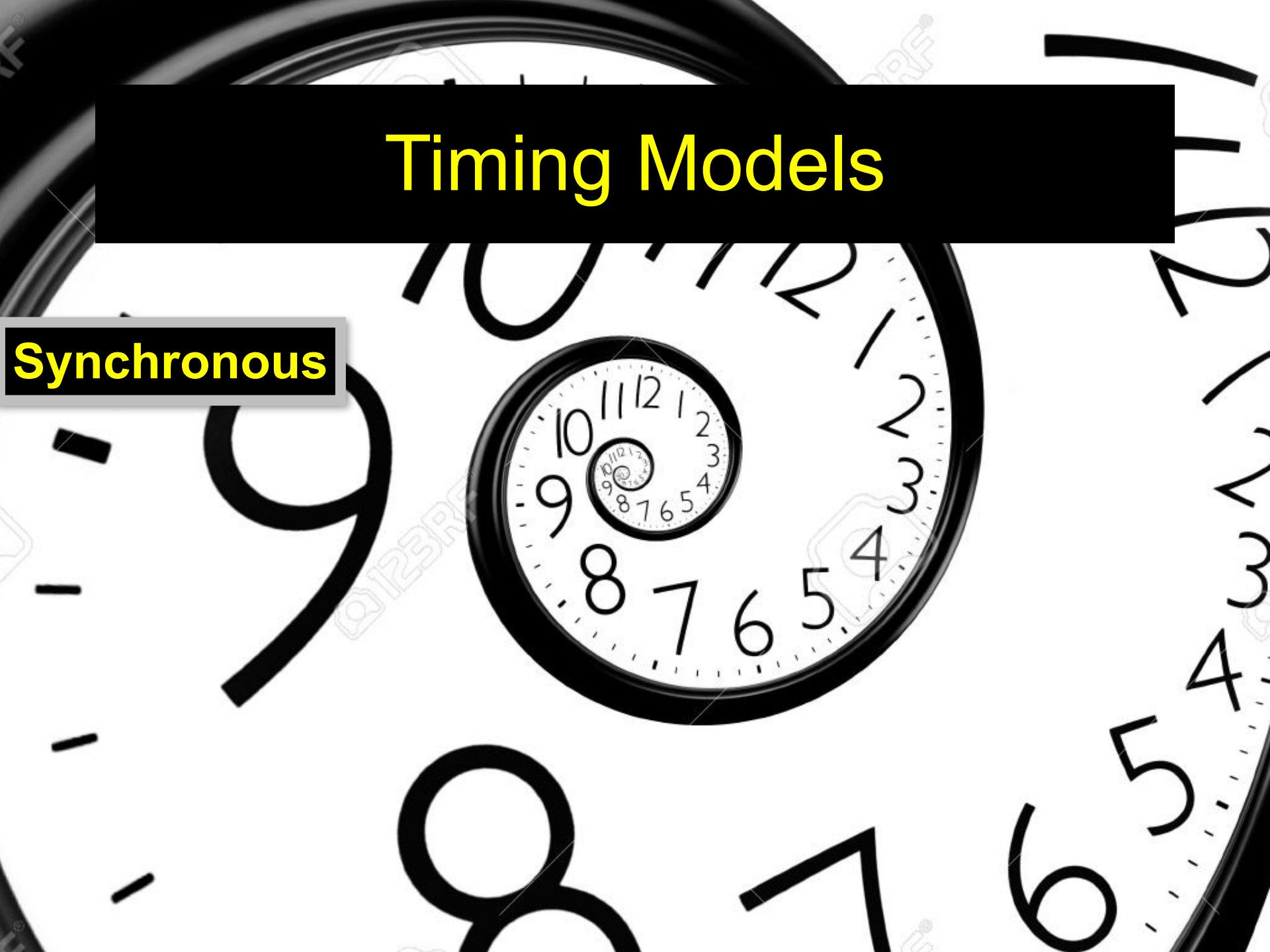
Reaction to other schemes



Timing Models



Timing Models

The background of the slide is a large, stylized clock face. The clock face is white with black numbers and hands. In the center of the clock face, there is a spiral pattern of smaller clock faces, each with its own set of numbers and hands, creating a recursive or fractal-like effect.

Synchronous

Timing Models

Synchronous

Adversary delays messages $< \text{known } \Delta$

Timing Models

Synchronous

Adversary delays messages $< \text{known } \Delta$

Bitcoin lives here

Timing Models

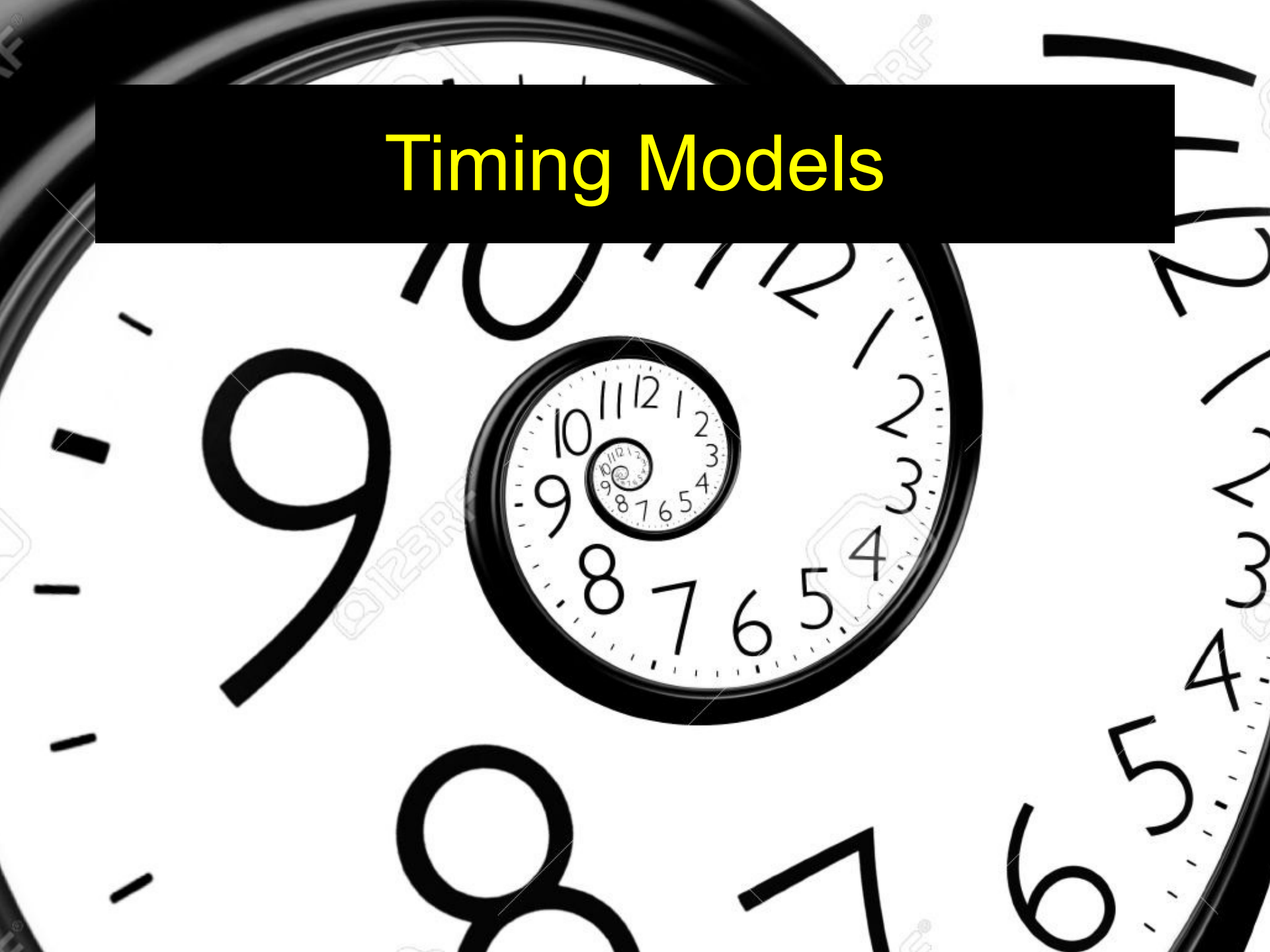
Synchronous

Adversary delays messages $< \text{known } \Delta$

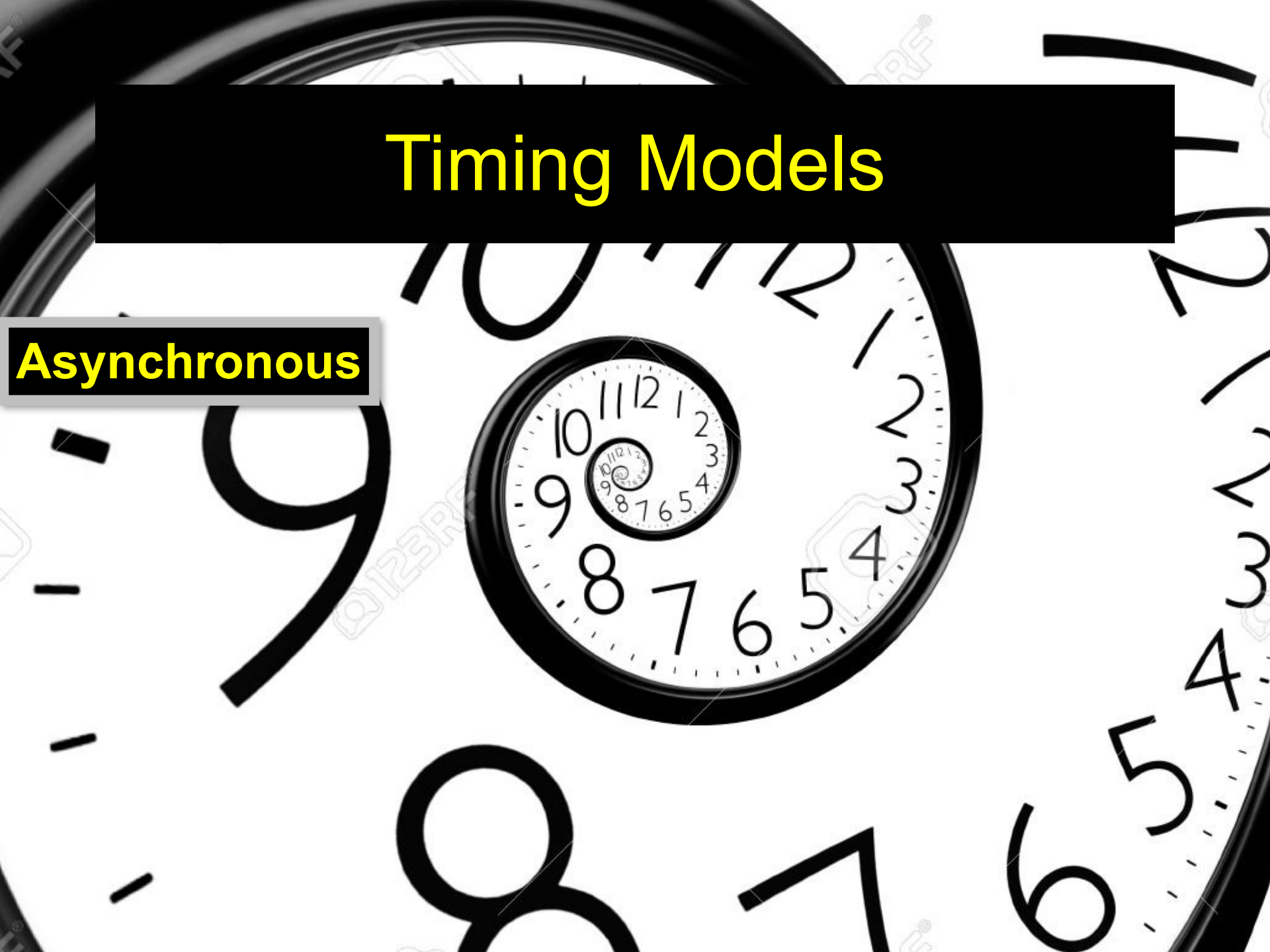
Bitcoin lives here

Needed for safety, not just liveness

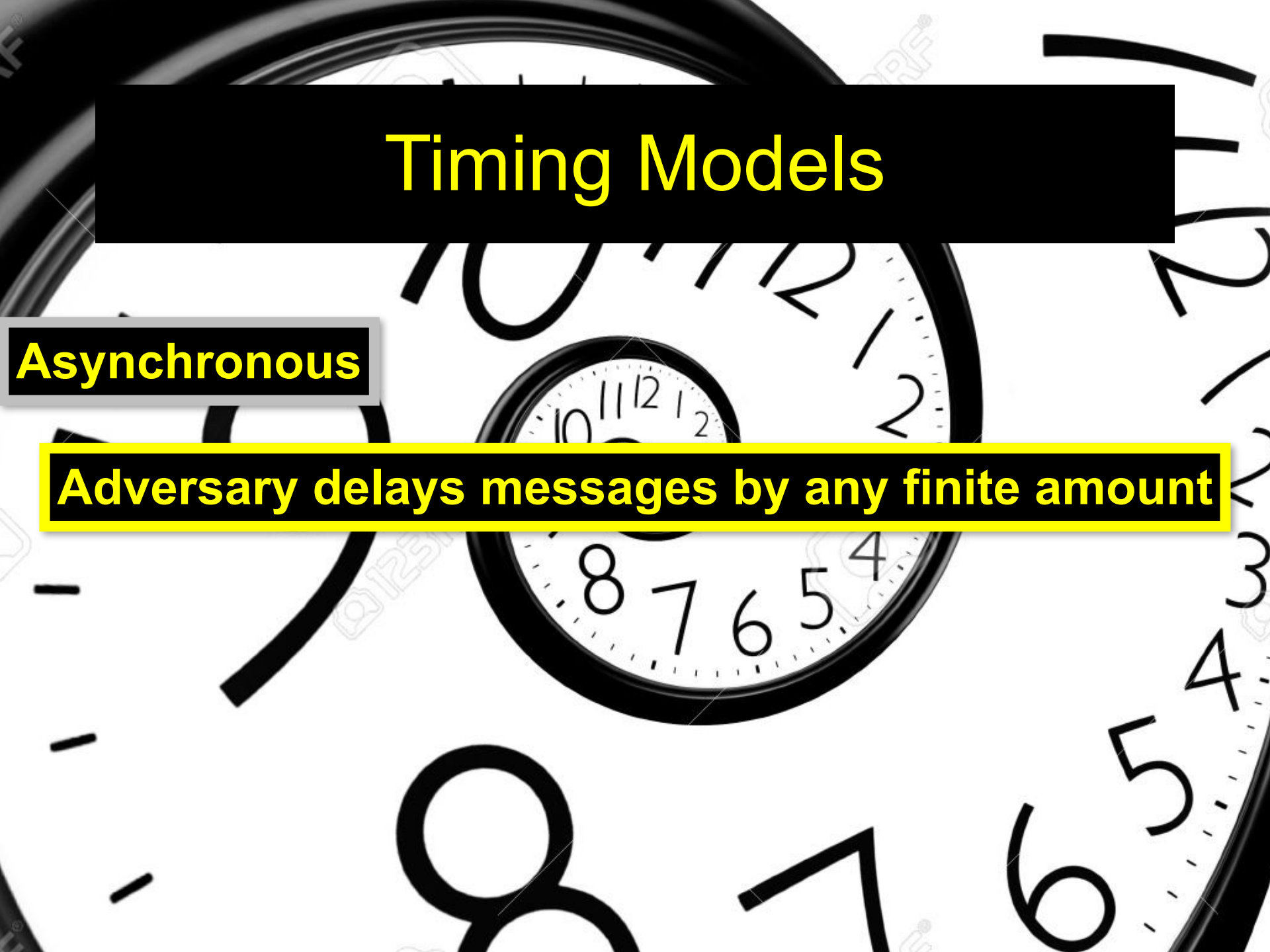
Timing Models



Timing Models

The background of the slide is a large, stylized clock face. The numbers 1 through 12 are visible, though some are partially cut off. In the center of this large clock is a smaller, spiral-shaped clock face, which also has numbers 1 through 12. The overall theme is time and timing.

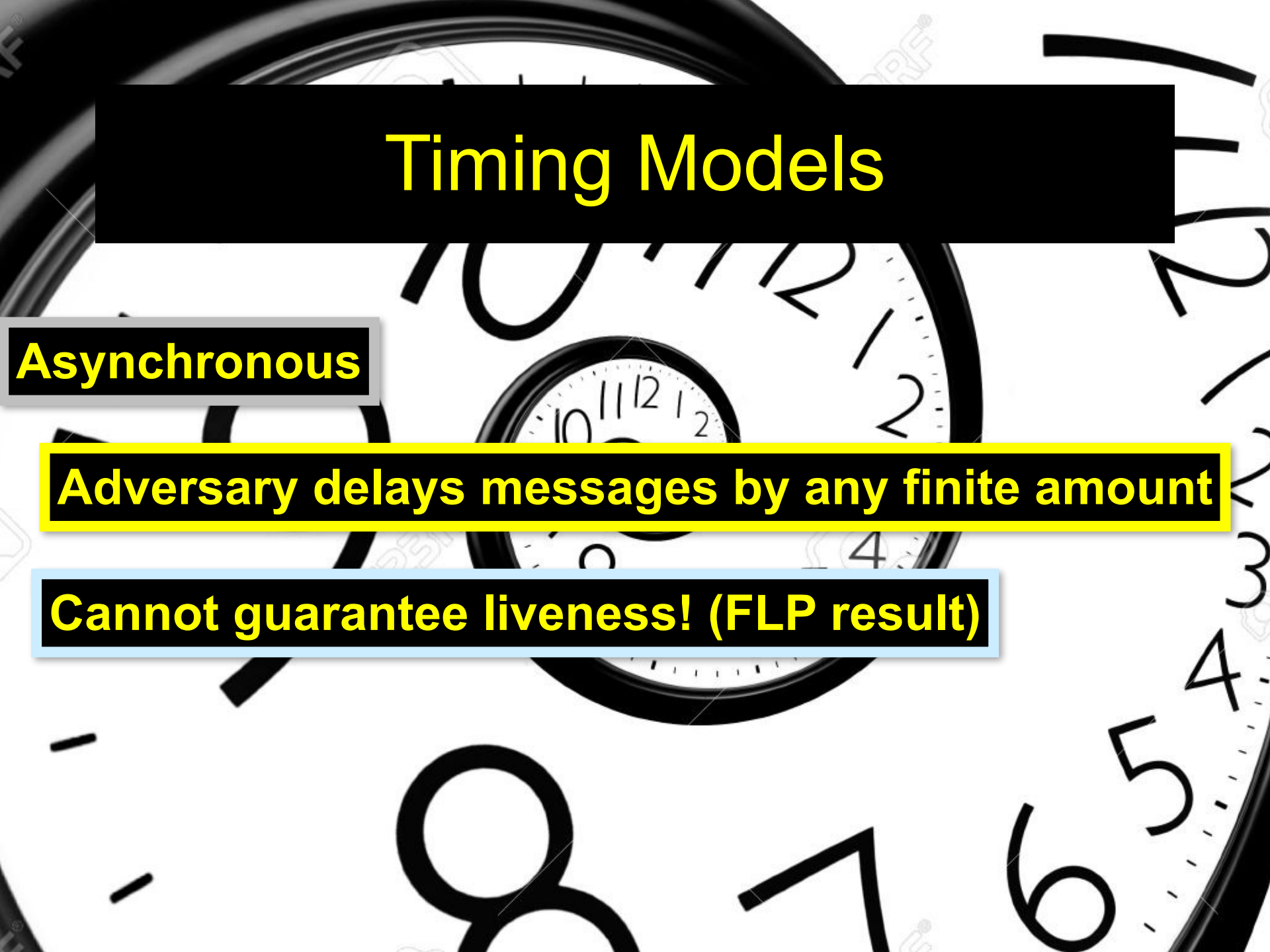
Asynchronous

The background of the slide is a close-up, slightly blurred image of a clock face. The clock has a white face with black numbers and hands. The numbers 1 through 12 are visible, though some are partially cut off by the edges of the frame. The clock hands are black and thin. The overall effect is a sense of time and timing.

Timing Models

Asynchronous

Adversary delays messages by any finite amount



Timing Models

Asynchronous

Adversary delays messages by any finite amount

Cannot guarantee liveness! (FLP result)

Timing Models

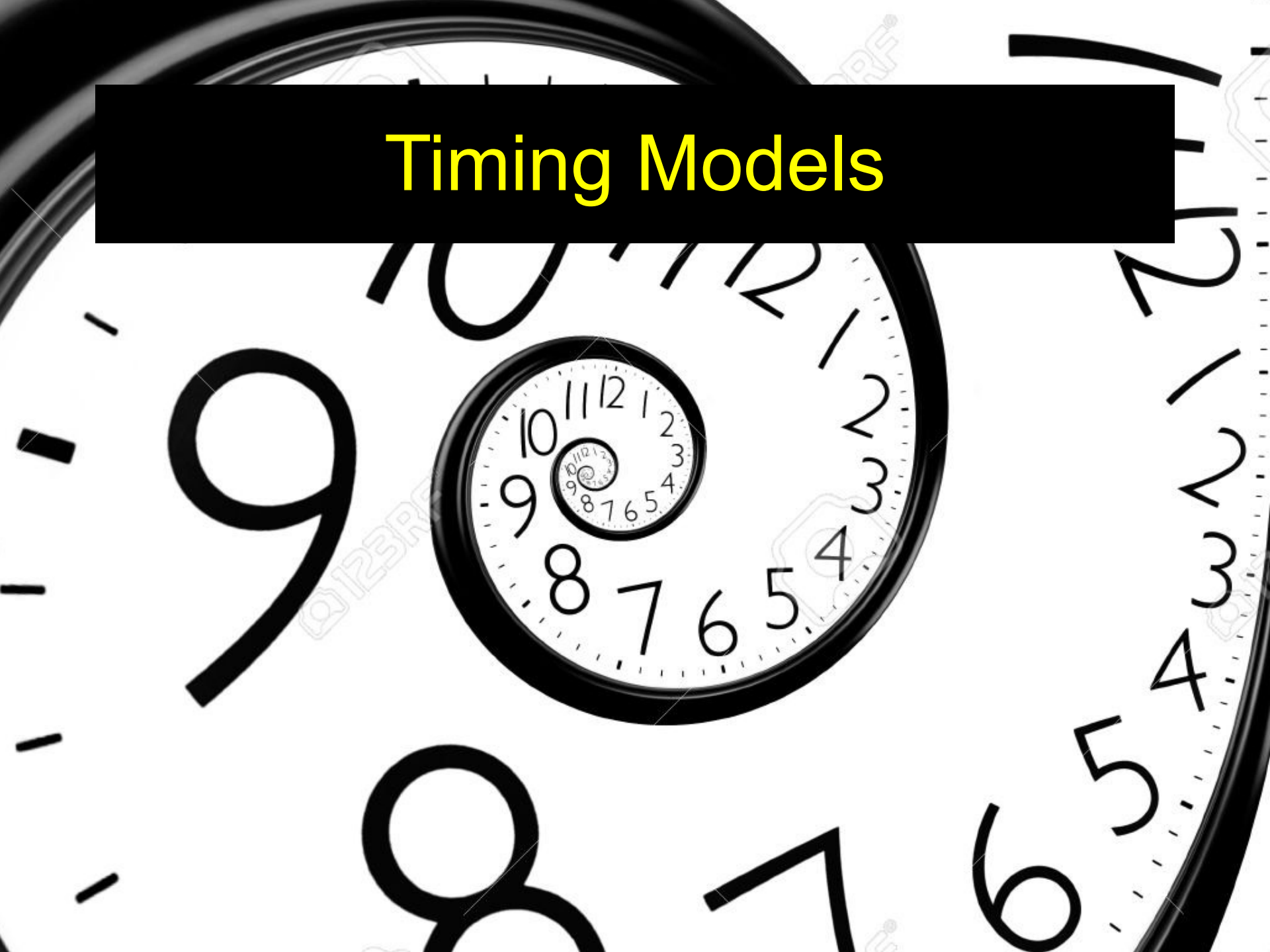
Asynchronous

Adversary delays messages by any finite amount

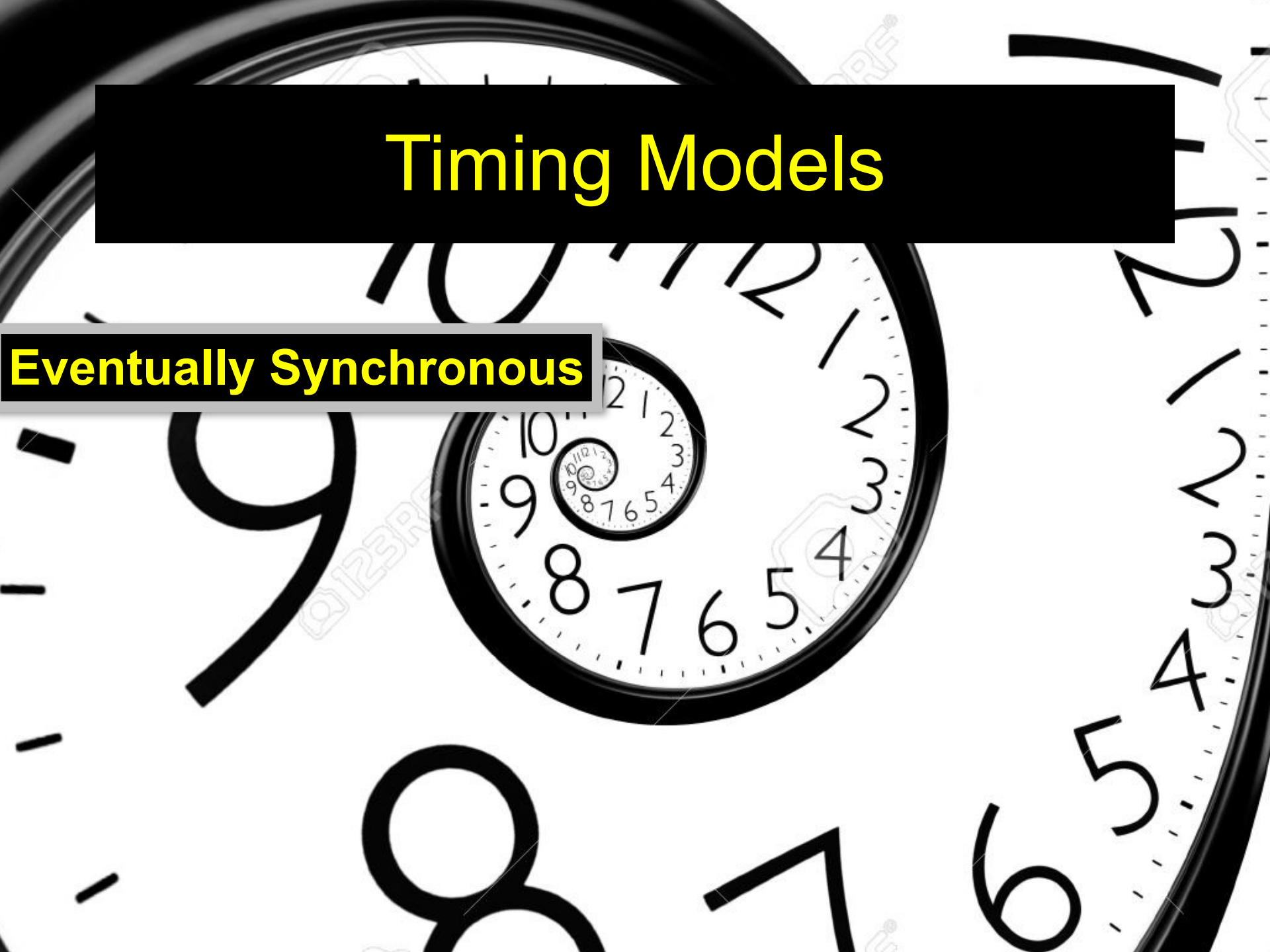
Cannot guarantee liveness! (FLP result)

Randomization for expected termination

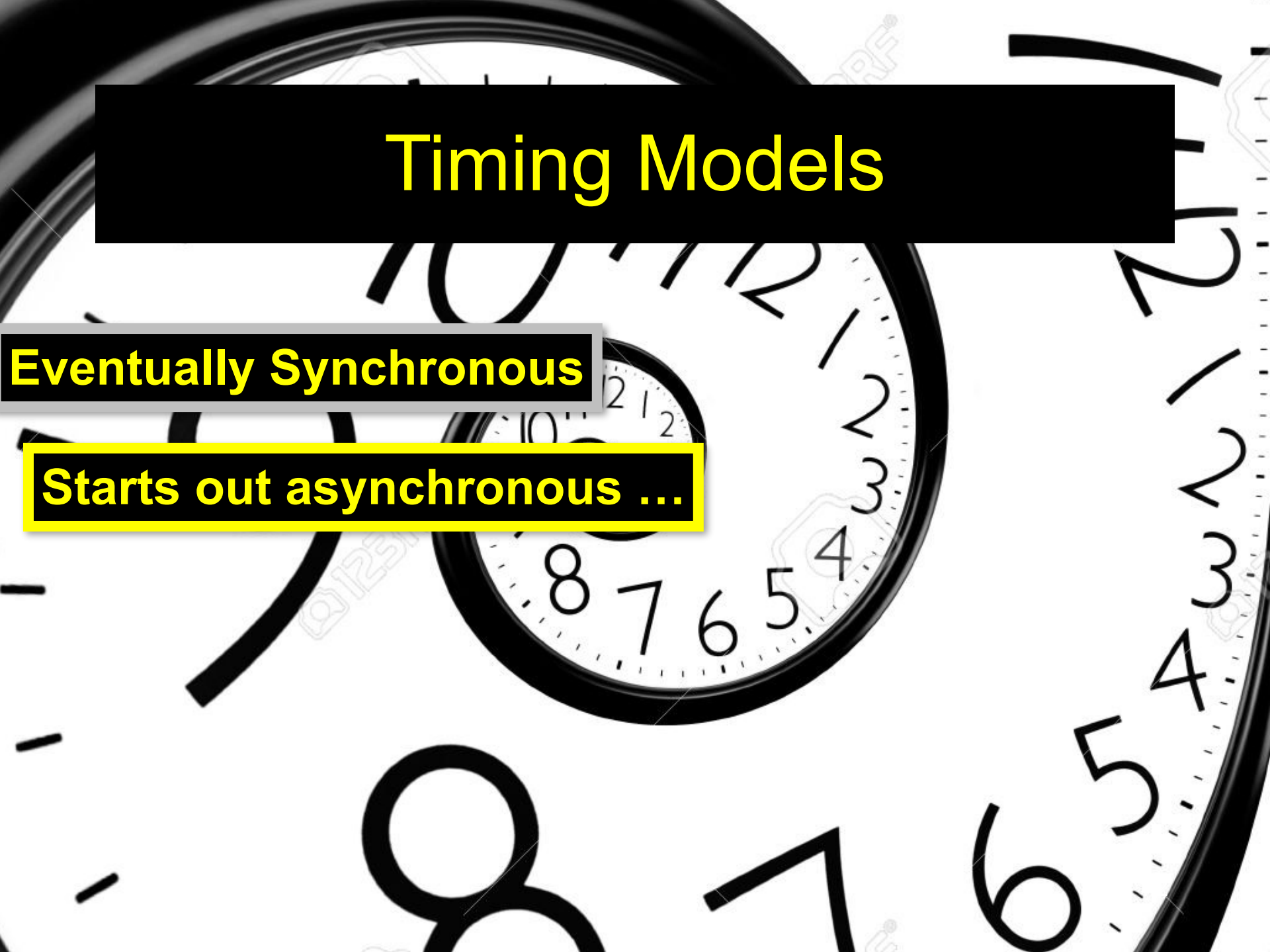
Timing Models



Timing Models

The background of the slide features a large, stylized clock face. The clock has a black frame and white face with black numerals. A spiral inset is visible, showing a smaller clock face within a circular path. The overall theme is time and timing.

Eventually Synchronous

The background of the slide features several overlapping clock faces. The clocks have black frames and white faces with black numerals and hands. They are arranged in a way that creates a sense of depth and movement, with some clocks partially obscured by others. The overall theme is time and timing.

Timing Models

Eventually Synchronous

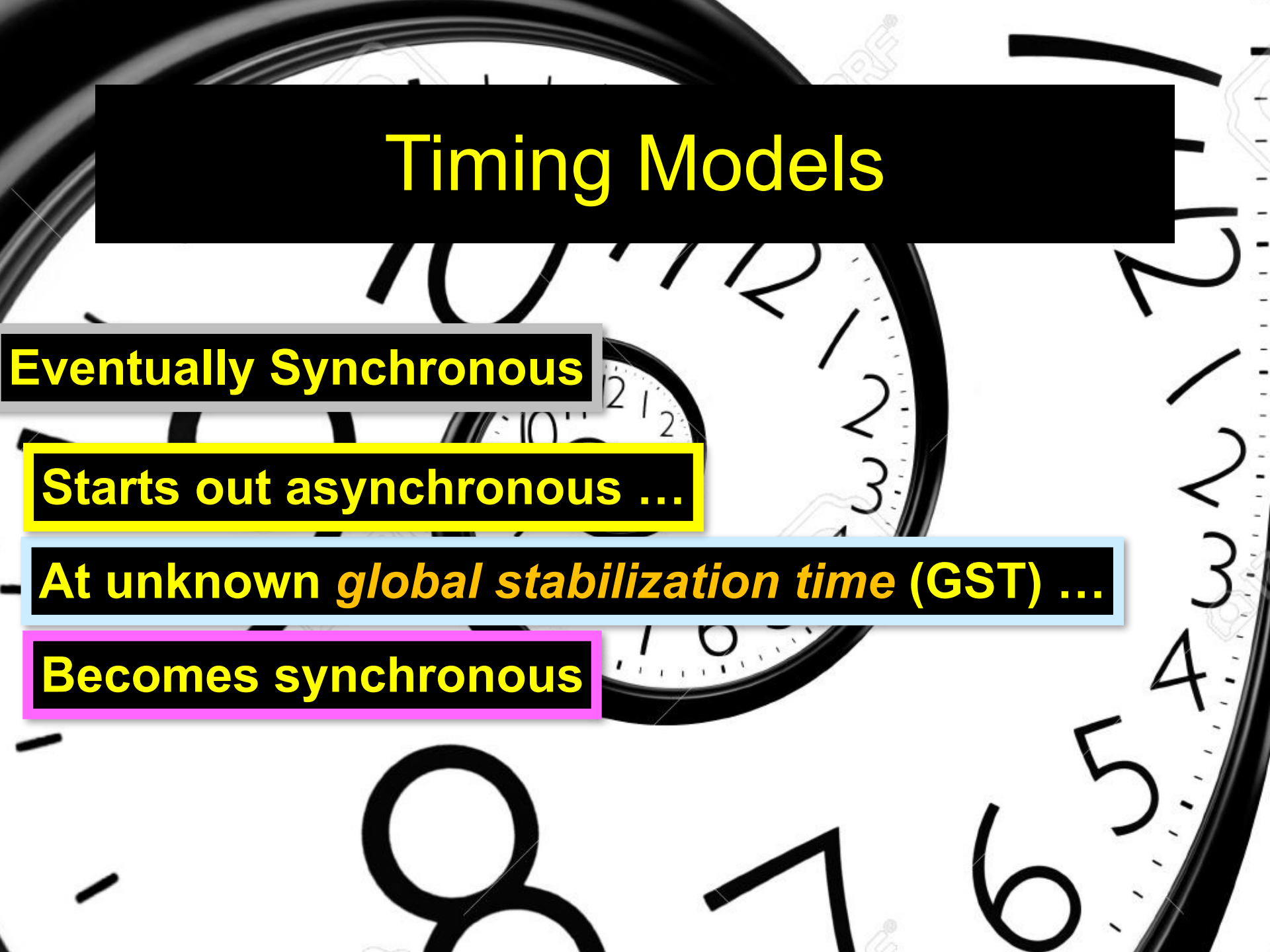
Starts out asynchronous ...

Timing Models

Eventually Synchronous

Starts out asynchronous ...

At unknown *global stabilization time* (GST) ...



Timing Models

Eventually Synchronous

Starts out asynchronous ...

At unknown *global stabilization time* (GST) ...

Becomes synchronous

Timing Models

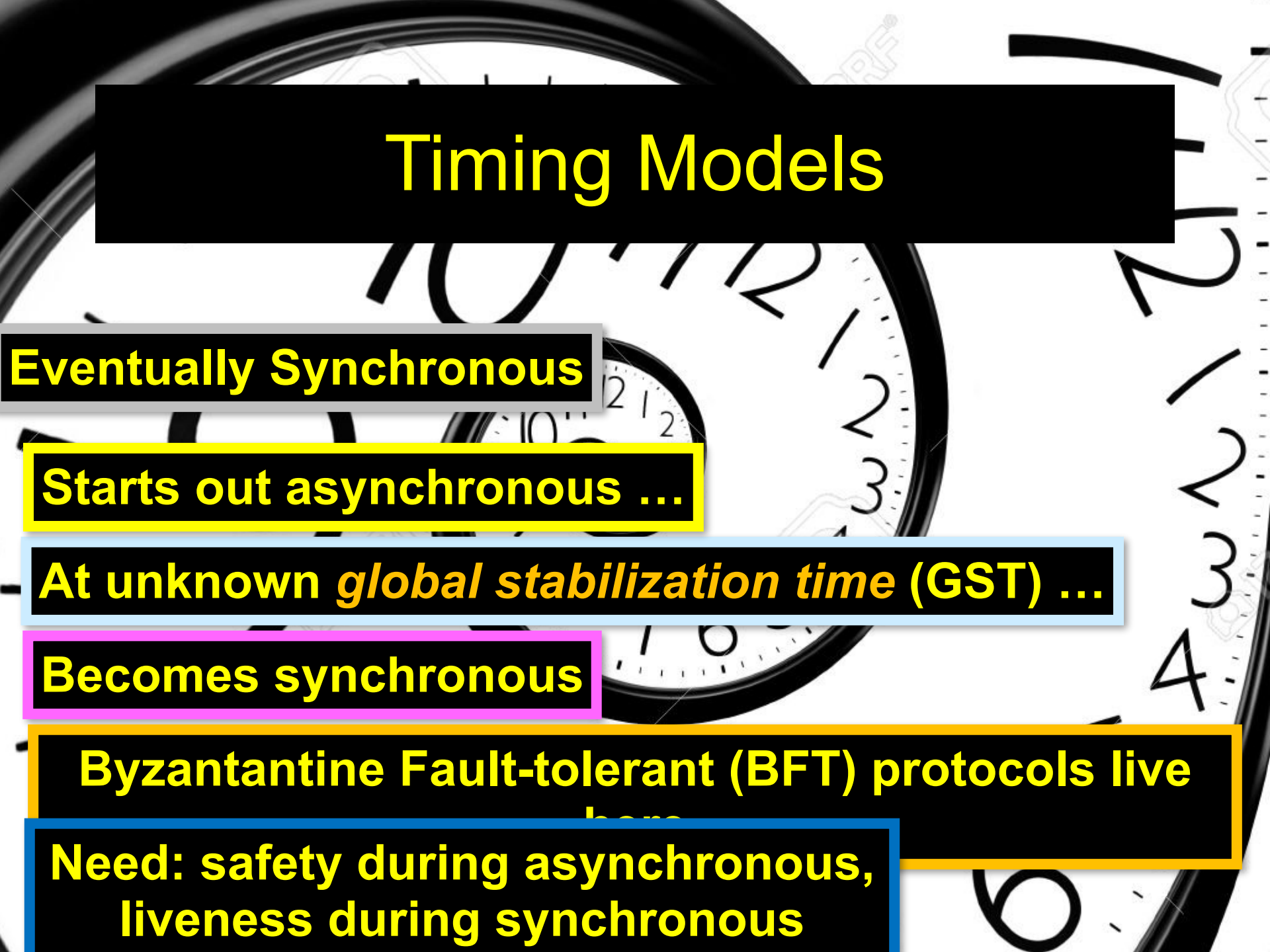
Eventually Synchronous

Starts out asynchronous ...

At unknown *global stabilization time* (GST) ...

Becomes synchronous

Byzantine Fault-tolerant (BFT) protocols live here



Timing Models

Eventually Synchronous

Starts out asynchronous ...

At unknown *global stabilization time* (GST) ...

Becomes synchronous

Byzantine Fault-tolerant (BFT) protocols live

**Need: safety during asynchronous,
liveness during synchronous**